

definitions

important notes

stakeholder any group or individual with an interest in a business such as workers, managers, suppliers, the local community, customers and owners

scarcity the limited nature of society's resources

economics the study of how society manages its scarce resources

THE TEN PRINCIPLES OF ECONOMICS

The word 'economy' comes from the Greek word *oikonomos*, which means for 'one who manages a household'. At first, this origin might seem peculiar. But, in fact, businesses and economies have much in common.

A business faces many decisions. It must decide how many people to employ, what each of those people will do to contribute to the business, how much each employee gets in return, who should receive a bonus scheme and who should not, how to increase output per worker, how best to manage costs, when to invest, how much to invest and where best to get the funds to invest from, what products to produce, what products not to produce, when to stop producing products, when to expand, when to contract, how best to manage the sales process and customer relations, whether to be environmentally friendly or whether to give the impression of being so, how to deal with competitors and whether to charge a high price or a low price (or one in between). In short, business must allocate scarce resources among competing uses, taking into account a range of stakeholder wants and needs. **A stakeholder is any group or individual with an interest in a business.**

Like businesses, a society faces many decisions. A society must decide what jobs will be done and who will do them. It needs some people to grow food, other people to make clothing and still others to design computer software. Once society has allocated people (as well as land, buildings and machines) to various jobs, it must also allocate the output of goods and services that they produce. It must decide who will eat caviar and who will eat potatoes. It must decide who will drive a Ferrari and who will take the bus.

The management of society's resources is important because resources are scarce. **Scarcity means that society has limited resources and therefore cannot produce all the goods and services people wish to have.** Just as a household cannot give every member everything he or she wants, a society cannot give every individual the highest standard of living to which he or she might aspire.

Economics is the study of how society manages its scarce resources. In most societies, resources are allocated not by a single central planner but through the combined actions of millions of households and firms. Economists therefore study how people make decisions: how much they work, what they buy, how much they save and how they invest their savings. Economists also study how people and people in businesses interact with one another. For instance, they examine how the multitude of buyers and sellers of a good together determine the price at which the good is sold and the quantity that is sold. Economists analyse forces and trends that affect the economy as a whole, including the growth in average income, the fraction of the population that cannot find work and the rate at which prices are rising. Many of the concepts that economists use are also directly applicable and of relevance to businesses. In a sense, economics can be described as the science of decision making and since businesses all around the world have to make millions of decisions every day then having some understanding of the process of decision making and the consequences that might arise from such decision making is important.

Although the study of economics has many facets, the field is unified by several central ideas, all of which are of direct relevance to business. In the rest of this chapter we look at *Ten Principles of Economics*. Don't worry if you don't understand them all at first, or if you don't find them completely convincing. In the coming chapters we will explore these ideas more fully. The ten principles are introduced here just to give you an overview of what economics is all about. You can think of this chapter as a 'preview of coming attractions'.

HOW PEOPLE AND BUSINESSES MAKE DECISIONS

We use the term 'the economy' on a regular basis but have you ever stopped to think about what the term really means? Whether we are talking about the economy of a group of countries such as the European Union or the Middle East, or the economy of

one particular country, such as South Africa or the United Kingdom, or of the whole world, an economy is just a group of people interacting with one another as they go about their lives. This interaction is invariably through a process of exchange. Whether it be an individual buying a morning newspaper, a business buying several hundred tonnes of steel for a construction project or a government funding a higher education institution, the interaction consists of millions of individuals all making decisions and together we describe these interactions as 'the economy'. Because the behaviour of an economy reflects the behaviour of the individuals, both acting on their own and as part of businesses who make up the economy, we start our study of economics with four principles of individual decision making.

CASE STUDY

Coal or Sun – or do you not care provided you have electricity?

For many people in Europe, the Middle East and parts of Africa, the idea of switching on a light, charging a mobile phone with the flick of a switch, plugging in the power cable for a laptop, taking a shower with hot running water, doing the washing and thousands of other daily tasks is taken largely for granted. There are plenty of people and businesses in the region, however, who recognize how fortunate they are to have access to supplies of electricity 24 hours a day, 365 days a year. South Africa is one country which suffers from shortages of electricity and the government is under pressure to increase capacity.

The question is, how should this capacity increase? South Africa is relatively rich in coal supplies. The country is the fourth largest coal exporter in the world and produces around 244 million tonnes of coal a year. Over 90 per cent of the electricity generated in South Africa comes from coal fired power stations. Despite this rich endowment of a valuable resource, the country's people and businesses suffer from regular power shortages. The demand for electricity has risen at a faster rate than the ability of the country to generate it and as a result electricity prices are high and could lead to slower economic growth.

Given the endowment of coal supplies, one way to increase electricity supplies would be to build new coal fired power stations and also upgrade existing older stations. The benefits of such a programme would be an increase in the amount of electricity available to businesses and people in the country, but the downside is that coal fired power stations are alleged to be a major contributor to carbon emissions and climate change. Other options open to South Africa include investing more in solar and wind energy – so called 'clean energy'.

Indeed, the World Bank agreed in October 2011 to help the South African utility company, Eskom, fund the construction of a 100 megawatt wind and solar power plant in Uppington in the Northern Cape province. The funding will amount to \$250 million. This sounds like a positive step forward but critics argue that the plant will not increase the supply of electricity sufficiently to have much effect on the power shortage the country experiences.

The World Bank had previously been criticized for helping to fund another power generation project for Eskom, the construction of new coal fired power stations worth around \$3.75 billion. Critics argued that this would just exacerbate Eskom's carbon footprint but the company countered by saying it had to adopt a balanced approach to power generation and would have to not only upgrade existing coal fired stations to become more efficient (in terms of reducing carbon emissions and producing more electricity with lower inputs and at lower cost) but to build new stations to meet the growing demand of the country.



The sun is a natural resource in South Africa which can be utilized to provide energy as in this case of a solar power field designed to capture the energy of the sun and convert it to electricity.

The case here highlights the trade-offs facing businesses in making decisions. In economics, there are rarely ‘right answers’ – businesses have to make decisions and face trade-offs. On the one hand, demand for electricity in South Africa means it is only ‘right’ and ‘sensible’ to exploit its natural resources and provide for the needs of its people. In doing so it has to recognize the negative impact of using coal as a means of generating electricity. Solar and wind may be cleaner sources of energy but can enough investment be generated in these sources, quickly, to meet the more immediate demands of the nation for power which will help to boost economic growth and help it deal with the pockets of chronic poverty that exist? There is no right answer to the predicament that Eskom finds itself in – it has to chart a path to balance out the costs and benefits to try and maximize the benefits but reduce the costs to a minimum and that involves recognizing and accepting the trade-offs that exist.

Principle 1: Decision Making Involves Trade-Offs

The first lesson about making decisions is summarized in an adage popular with economists: ‘There is no such thing as a free lunch’. To get the benefits of one thing that we like, we usually have to give up the benefits of another thing that we also like, or accept that we might have to give up something else and incur a cost of some sort (cost here being used in its widest sense not just a monetary one). Making decisions requires trading off one goal or the benefits against another.

Consider a business manager who must decide how to allocate her most valuable resource – her time. She can spend all of her time reflecting on strategy in her office; she can spend all of her time walking around the business premises talking to staff; or she can divide her time between the two fields. For every hour she spends reflecting and strategizing, she gives up an hour she could have used talking to staff. And for every hour she spends doing either, she gives up an hour that she could have spent talking to customers, being out promoting the business, working with suppliers to improve efficiency or networking with colleagues at conferences.

Or consider employees of this business deciding how to spend the income they receive from working at the business. They can buy food, clothing or a family holiday. Or they can save some of the family income for retirement or perhaps to help the children buy a house or a flat when they are grown up. When they choose to spend an extra euro on one of these goods, they have one less euro to spend on some other good.

When people are grouped into societies, they face different kinds of trade-offs. The classic trade-off is between ‘guns and butter’ – these two products just represent defence and consumer goods in general. The more we spend on national defence (guns) to protect our country from foreign aggressors (the benefit), the less we can spend on consumer goods (butter) which brings the benefit of raising our standard of living at home. Also important in modern society is the trade-off between a clean environment and a high level of income. Laws that require firms to reduce pollution raise the cost of producing goods and services but bring the benefit to society as a whole (and possibly to the firm in the form of good publicity). Because of the higher costs, these firms end up earning smaller profits, paying lower wages, charging higher prices, or some combination of these three. Thus, while pollution regulations give us the benefit of a cleaner environment and the improved levels of health that come with it, they have the cost of reducing the incomes of the firms’ owners, workers and customers.

Another trade-off society faces is between efficiency and equity. **Efficiency** can be looked at in three different ways. In essence, **efficiency is about getting the most we can**

efficiency the property of society getting the most it can from its scarce resources

from scarce resources. More specifically we can look at it in four ways related to business:

- **Technical efficiency** – a business can improve its technical efficiency if it could find a way of using its existing resources to produce more. It may be that it could use machinery instead of people that do the same job but do it much faster without having to take a break!
- **Productive efficiency** – a business can improve productive efficiency by producing output at the lowest cost possible. If it can find a way of producing its products cheaper, for example, by sourcing cheaper raw materials, then it can improve its productive efficiency.
- **Allocative efficiency** – this looks at efficiency from the perspective of consumers. Are the products being produced by businesses actually wanted and valued by consumers (both individual consumers and business consumers)? Efficiency occurs where the goods and services being produced match the demand by consumers. Allocative efficiency occurs where the cost of resources used to produce the products is equal to the value placed on the product by consumers represented by the price they are willing to pay.
- **Social efficiency** – when businesses produce products they incur costs – raw materials, wages, rents, interest payments, insurance, plant and equipment and so on – the private costs. However, there are also costs which businesses may not take into account such as the pollution they generate in production. These are costs borne by society as a whole. Social efficiency occurs where the private and social cost of production is equal to the private and social benefits derived from their consumption.

the resources are allocated optimally to meet the ends and needs of society

an economic state where resources are allocated to maximize total societal welfare, occurring when the marginal social benefit (MSB) equals the marginal social cost (MSC).

Equity means that the benefits of those resources are distributed fairly among society's members. In other words, efficiency as an overall concept, refers to the size of the economic cake, and equity refers to how the cake is divided. Often, when government policies are being designed, these two goals conflict; in addition, when businesses make decisions conflicts arise between equity and efficiency.

equity the property of distributing economic prosperity fairly among the members of society

Pitfall Prevention If you are writing about efficiency, be careful to distinguish which measure of efficiency you mean – each may have different implications and consequences for businesses.

Consider, for instance, policies aimed at achieving a more equal distribution of economic well-being. Some of these policies, such as the social security system or unemployment insurance, try to help those members of society who are most in need. Others, such as the individual income tax, ask the financially successful to contribute more than others to support the government. Although these policies have the benefit of achieving greater equity, they have a cost in terms of reduced overall efficiency. When the government redistributes income from the rich to the poor, it reduces the reward for working hard; as a result, people work less and produce fewer goods and services. In other words, when the government tries to cut the economic cake into more equal slices, the cake gets smaller.

Equally, when a business decides to outsource some of its production abroad, workers in the domestic economy may find that they lose their jobs. For the business as a whole, the decision may increase productive efficiency but workers would consider the decision unfair and the consequences in terms of morale and motivation could affect productivity elsewhere in the business in a detrimental way.

Recognizing that people and businesses face trade-offs does not by itself tell us what decisions they will or should make. A chief executive officer (CEO) should not abandon time set aside for reflection and thinking just because doing so would increase the time available to talk to workers. Society should not stop protecting the environment just

Equity is the value of ownership in an asset or company, calculated as total assets minus total liabilities (debts).

because environmental regulations reduce our material standard of living. The poor should not be ignored just because helping them distorts work incentives. Nevertheless, acknowledging life's trade-offs is important because people and businesses are likely to make good decisions only if they understand the options that they have available and can quantify them in some way to enable them to make informed decisions.

Quick Quiz Does the adage 'there is no such thing as a free lunch' simply refer to the fact that someone has to have paid for the lunch to be provided and served? Or does the recipient of a 'free lunch' also incur a cost?

Principle 2: The Cost of Something Is What You Give Up to Get It

Because people and businesses face trade-offs, making decisions requires comparing the costs and benefits of alternative courses of action. In many cases, however, the cost of some action is not as obvious as it might first appear.

Consider, for example, the decision by a business to cease production of a product that is not selling very well any more. The benefit is that resources can be made available to invest in other parts of the business that are more successful. But what is the cost? To answer this question, you might be tempted to add up the money the business has to pay in redundancy to workers who may no longer be needed, to close down plant, get rid of defunct machinery and equipment. Yet this total does not truly represent what the business gives up when it ceases production of a product.

The first problem with this answer is that it ignores many wider issues that the business might face as a result of its decision. How do competitors view the decision? Will they seek to use it as an example of the decline of the business? What about customers – will they be disappointed that the product has disappeared? A number of businesses have found themselves under pressure to bring back much loved products that may not have been financially viable and have incurred disappointment from customers and possible loss of loyalty as a result. Then there is the attitude amongst workers – is this closure the first of others, does it send negative signals to the rest of the workforce and result in a decline in motivation and increases in staff turnover as workers seek to get out before they are pushed out?

The second problem is that it does not include the lost revenue from sales of the product. It may be that sales were low and not that it was not viable to continue with production. Assuming that sales were not zero there was some revenue being generated and this will now be lost. That has to be taken into consideration.

The decision, therefore, has costs far greater than pure money costs. The cost of loss of goodwill, worker and customer loyalty and bad publicity also has to be taken into consideration in assessing the costs of the decision and these may not be immediately obvious and sometimes not easy to work out.

opportunity cost whatever must be given up to obtain some item – the value of the benefits foregone (sacrificed)

The opportunity cost of an item is what you give up to get that item. When making any decision, such as whether to close down production of a product, decision makers should be aware of the opportunity costs that accompany each possible action.

Quick Quiz A bakery is planning to invest €20 000 in a new machine which will increase its production of croissants by 10 per cent. The financial director of the company points out that account needs to be taken of what that €20 000 could earn if invested in another way, such as in an interest-bearing account at a bank, as part of the cost calculations. Why do you think she insisted that this was taken into consideration and why is it classed as a cost?

If the bakery invests €20,000 in the machine, that money cannot be used elsewhere. For example, it could have been deposited in a bank to earn interest. By choosing the machine, the bakery gives up the potential earnings from the bank investment.

So, the lost interest is the next best alternative benefit that is sacrificed.

Principle 3: Rational People and Businesses Think at the Margin

Decisions in life are rarely black and white but usually involve shades of grey. The decision by a company worker of whether to put in the extra hour at the end of the working day is not a case of working all day or not but about that extra time to finish a project before going home. A publisher may have to make a decision about whether to print an extra thousand copies of a textbook when it is not certain that all these extra books will sell. Many workers have to make decisions about whether to spend a bit more time with their families or whether to shut themselves away and get on with the bit of work that needs finishing for a deadline the next week. Economists use the term **marginal changes** to describe small incremental adjustments to an existing plan of action. Keep in mind that 'margin' means 'edge', so marginal changes are adjustments around the edges of what you are doing.

marginal changes small incremental adjustments to a plan of action

In many situations, people make the best decisions by thinking at the margin. Suppose, for instance, that you asked a friend for advice about how many years to stay in education. If he were to compare for you the lifestyle of a person with a PhD to that of someone who finished secondary school with no qualifications, you might complain that this comparison is not helpful for your decision. Perhaps you have already been at university for a few years but you're getting a little tired of studying and not having enough money and so you're deciding whether or not to stay on for that last year. To make this decision, you need to know the additional benefits that an extra year in education would offer (higher wages throughout your life and the sheer joy of learning) and the additional costs that you would incur (another year of tuition fees and another year of foregone wages). By comparing these *marginal benefits* and *marginal costs*, you can evaluate whether the extra year is worthwhile.

As another example, consider an airline company deciding how much to charge passengers who fly standby. Suppose that flying a 200-seat aeroplane from London to Warsaw costs the airline €100 000. In this case, the average cost of each seat is €100 000/200, which is €500. One might be tempted to conclude that the airline should never sell a ticket for less than €500. In fact, however, the airline can raise its profits by thinking at the margin. Imagine that a plane is about to take off with ten empty seats, and a standby passenger is waiting at the gate willing to pay €300 for a seat. Should the airline sell it to him/her? Of course it should. If the plane has empty seats, the cost of adding one more passenger is minuscule. Although the *average* cost of flying a passenger is €500, the *marginal* cost is merely the cost of the airline meal that the extra passenger will consume (which may have gone to waste in any case) and possibly an extremely slight increase in the amount of aircraft fuel used. As long as the standby passenger pays more than the marginal cost, selling him or her a ticket is profitable.

As these examples show, individuals and firms can make better decisions by thinking at the margin. A rational decision maker takes an action if and only if the marginal benefit of the action exceeds the marginal cost. For businesses this principle is extremely important because there are more likely to be attempts to rationalize decision making than individuals do on a day-to-day basis.

Highlighting only the marginal parts

Marginal Benefit:

The €300 the standby passenger is willing to pay for the seat.

Marginal Cost:

The extra meal cost for the passenger.

A very small increase in fuel consumption.

Why it is profitable

If Marginal Benefit (€300) > Marginal Cost (meal + tiny fuel cost), the airline should sell the seat because it adds profit even though the average cost per seat is €500.

Principle 4: People and Businesses Respond to Incentives

Because people and businesses make decisions by comparing costs and benefits, their behaviour may change when the costs or benefits change. That is, people respond to incentives. When the price of an apple rises, for instance, people decide to eat more pears and fewer apples because the cost of buying an apple is higher. At the same time, apple orchards decide to hire more workers and harvest more apples, because the benefit of selling an apple is also higher. As we shall see, the effect of price on the behaviour of buyers and sellers in a market – in this case, the market for apples – is crucial for understanding how the economy works.

An incentive is something that motivates or encourages a person to do something.

Public policy makers should never forget about incentives, because many policies change the costs or benefits that people face and, therefore, alter behaviour. A tax on petrol, for instance, encourages people to drive smaller, more fuel-efficient cars. For vehicle manufacturers this change in demand has to be accommodated and shifts resources into the production of smaller, more fuel efficient cars and away from larger, more fuel expensive cars. It also encourages people to use public transport rather than drive and to live closer to where they work. This in turn has an effect on decisions about provision of public transport and what type of transport and affects businesses in the construction industry. If the tax were large enough, people would start driving electric cars, which requires investment by vehicle manufacturers in this technology and a host of other businesses who may be involved in setting up the infrastructure to facilitate the use of electric cars.

When policy makers fail to consider how their policies affect incentives, they often end up with results they did not intend. For example, consider public policy regarding motor vehicle safety. Today all cars sold in the European Union, South Africa and most countries in the Middle East have to have seat belts fitted by law (although actual seat belt use – especially by rear-seat passengers – varies widely, with official estimates ranging from about 30 per cent of car occupants in some member states of the EU to around 90 per cent in others, notably Sweden).

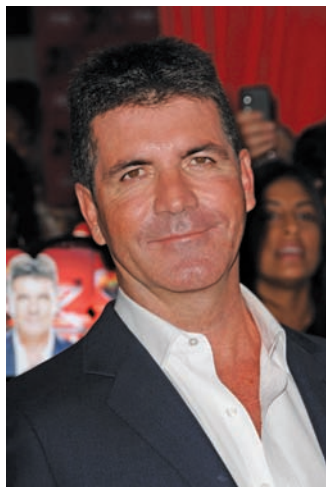
How does a seat belt law affect car safety? The direct effect is obvious: when a person wears a seat belt, the probability of surviving a major car accident rises. But that's not the end of the story, for the law also affects behaviour by altering incentives. The relevant behaviour here is the speed and care with which drivers operate their cars. Driving slowly and carefully is costly because it uses the driver's time and energy. When deciding how safely to drive, rational people compare the marginal benefit from safer driving to the marginal cost. They drive more slowly and carefully when the benefit of increased safety is high. It is no surprise, for instance, that people drive more slowly and carefully when roads are icy than when roads are clear.

There is a valid case for this argument. Improvements in car safety and road design reduce the perceived risk of driving, which lowers the marginal cost of risky behavior such as speeding.

At the same time, the marginal benefit of driving faster (e.g., saving time, convenience) remains the same or may even increase. As a result, some drivers may choose to drive faster because the marginal benefit now outweighs the marginal cost.

This idea is explained by the Peltzman Effect, which suggests that individuals adjust their behavior in response to perceived levels of risk.

However, this effect is only partial, and overall improvements in safety still lead to a reduction in accidents and fatalities.



Music, TV and television producer and executive Simon Cowell understood opportunity cost and incentives. He decided to leave school before completing his post-16 education and is reported to earn in excess of £50 million a year!

Quick Quiz The emphasis on road safety throughout Europe has increased over the last 25 years. Not only are cars packed with safety technology and devices but roads are also designed to be safer with the use of safety barriers and better road surfaces, for example. Is there a case, therefore, for believing that if people feel that they are safer in their cars there is an incentive to drive faster because the marginal cost is now outweighed by the marginal benefit?

Consider how a seat belt law alters a motorist's cost-benefit calculation. Seat belts make accidents less costly because they reduce the likelihood of injury or death. In other words, seat belts reduce the benefits to slow and careful driving. People respond to seat belts as they would to an improvement in road conditions – by faster and less careful driving. The end result of a seat belt law, therefore, is a larger number of accidents and so it will affect both motorists and pedestrians. The decline in safe driving has a clear, adverse impact on pedestrians, who are more likely to find themselves in an accident but (unlike the motorists) don't have the benefit of added protection.

At first, this discussion of incentives and seat belts might seem like idle speculation. Yet a 1981 study of seat belt laws in eight European countries commissioned by the UK Department of Transport showed that the laws did appear to have had many of these effects. Similar evidence was also presented in a 1975 study of US seat belt laws by the American economist Sam Peltzman. It does indeed seem that seat belt laws produce both fewer deaths per accident and more accidents. The net result is little change in the number of motorist deaths and an increase in the number of pedestrian deaths.

This is an example of the general principle that people respond to incentives. Many incentives that economists study are more straightforward than those of the car-safety laws. No one is surprised that people drive smaller cars in Europe, where petrol taxes

are relatively high, than in the United States, where petrol taxes are lower. Yet, as the seat belt example shows, policies can have effects that are not obvious in advance. When analysing any policy, we must consider not only the direct effects but also the indirect effects that work through incentives. If the policy changes incentives, it will cause people to alter their behaviour.

Quick Quiz List and briefly explain the four principles of individual decision making.

1. trade offs
2. opportunity cost
3. marginal benefits and cost
4. incentives

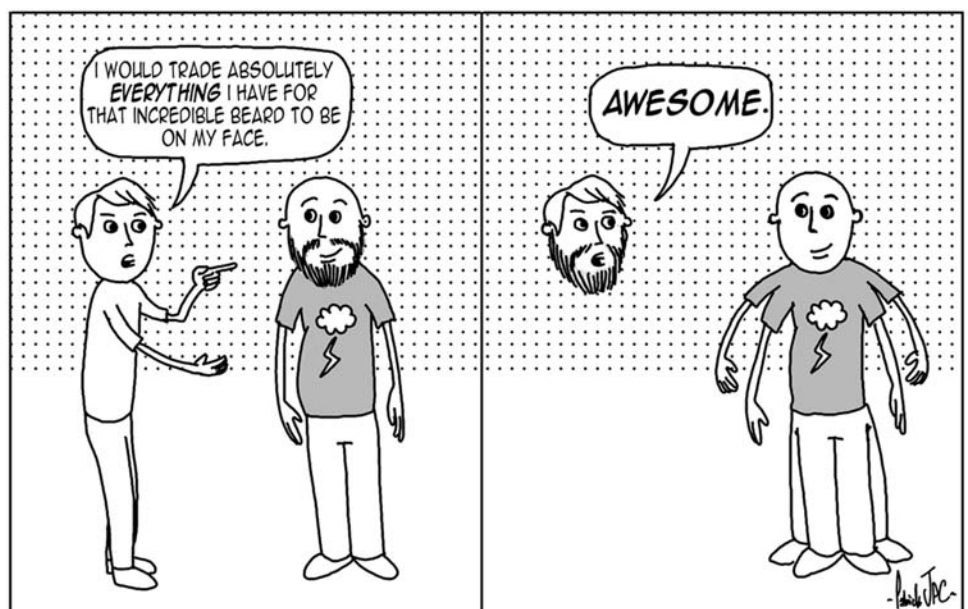
HOW PEOPLE INTERACT

The first four principles discussed how individuals and businesses make decisions. As we go about our lives, many of our decisions and the decisions of businesses affect not only ourselves but other people and businesses as well. The next three principles concern how people and businesses interact with one another.

Principle 5: Trade Can Make Everyone Better Off

The Americans, South Africans and the Japanese are often mentioned in the news as being competitors to Europeans in the world economy. In some ways this is true, because American, South African and Japanese firms do produce many of the same goods as European firms. Airbus and Boeing compete for the same customers in the market for aircraft. Toyota and Citroën compete for the same customers in the market for cars. South African and American fruit growers compete in the same market as European fruit growers and South African and American wine producers compete in the same market as French, Spanish and Italian wine makers.

Yet it is easy to be misled when thinking about competition among countries. Trade between Europe and South Africa or the United States, or between Europe and Japan, is not like a sports contest where one side wins and the other side loses (a zero-sum game). In fact, the opposite is true: trade between two economies can make each economy better off.



WHAT IS BUSINESS – THE TRANSFORMATION PROCESS

One of the recurring themes of this book is that of decision making. Successful business is all about making decisions which help the business meet its aims and objectives and, as we have seen, economics can be seen as being a science of decision making. If business is about decision making, what is business? In its simplest terms **business activity involves taking a series of inputs and producing an output**. The output could be a physical good or a service. The business might provide these goods and services to someone who actually consumes the good or service (the final consumer) or to another business who may either act as an intermediary in getting goods and services to the final consumer or who will do something to those goods and services before selling them on to a final consumer.

Business activity where the business sells goods and services to a final consumer is referred to as **B2C business**. Where a business sells a good or service to another business, this is referred to as **B2B business**. In recent years there are also other forms of activity that could be classed as business activity where consumers interact with other consumers via social networking sites or specialist websites such as eBay, Amazon Marketplace, eBid, OZtion, uBid and Overstock, which facilitate this type of trade. This is referred to as **C2C business**.

B2C business business activity where the business sells goods and services to a final consumer

B2B business business activity where the business sells goods and services to another business

C2C business business activity where consumers exchange goods and services often facilitated by a third party such as an online auction site

transformation process the process in which businesses take factor inputs and process them to produce outputs which are then sold

factors of production a classification of inputs used in business activity which includes land, labour, capital and enterprise

Pitfall Prevention The term 'business' in a question is generic – when considering answers to questions be sure to specify what type of business you are referring to so that you contextualize your answers and show some awareness that different businesses may be affected in different ways.

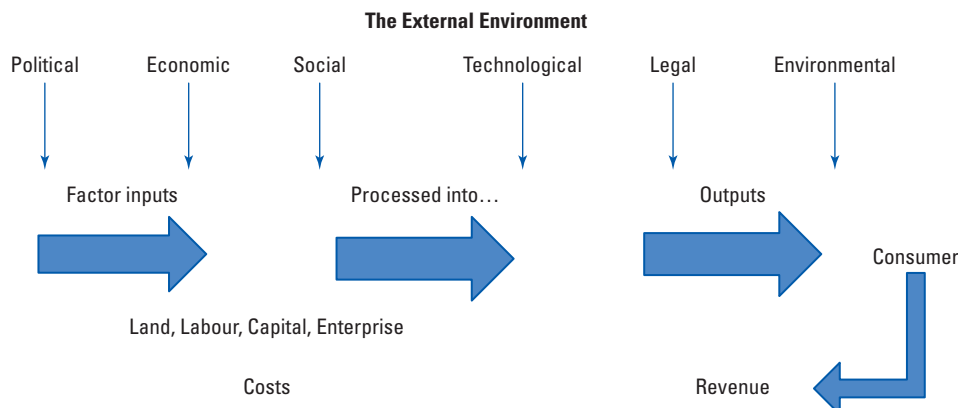
Factors of Production

There is a common feature which characterizes business activity. This feature is a **transformation process** summarized in Figure 3.1. Any business has to utilize inputs, referred to as **factors of production**, and does something with them to produce an output – a

FIGURE 3.1

The Transformation Process and the External Environment

This diagram is a representation of the business transformation process. Businesses take in inputs which incur costs and combine those inputs in different ways to produce an output which is then sold and generates revenue. Over time the revenue has to exceed the costs in order for the business to survive. The size of the difference between costs and revenues will vary according to different businesses and the aims and objectives of these businesses. Businesses do not carry out this process in isolation, they are affected by various influences classed as political, economic, social, technological, legal and environmental.



semi-finished product or commodity (raw materials such as rubber, cocoa, coffee, wheat, tin, ores, etc.), which is then sold on to another business or a finished product which is sold to a consumer.

Economists classify these factors of production in four main ways, although there are some that argue for only three factors of production. The four are **land, labour, capital and enterprise**. Some class enterprise as a specialist form of labour but we will assume it is a separate factor of production.

Land is a term that includes all the natural resources of the Earth and so might not only include pieces of land on which factories or offices are built, or which is farmed, but also things such as fish in the sea, minerals and ores from the ground and so on.

Labour is all the physical and mental effort that is provided by humans in production. This then includes human activity ranging from the work of a chief executive officer (CEO) at the head of a large public company right through to the person who has to clean the toilets. Look around you at any time and you can see many examples of this human activity.

In everyday language we use the term capital to refer to money. Economists use the term capital in a different way although the two are linked. **Capital** refers to anything that is not used for its own sake but which makes a contribution to production. This might include equipment and machinery, buildings, offices, shops, computers, main-frames, desks, chairs, etc. Of course, in order to get capital, businesses need money but economists view money as a medium of exchange and so it is often more informative to look at what money has been used to purchase rather than how much money has been spent. This is because in making a decision to buy anything we have to make a sacrifice, the opportunity cost involved, and this can tell us a great deal about the relative value that businesses put on decision making.

Quick Quiz Choose one product and write down some specific examples of the three factors of production covered so far which are used in the production of this product.

Factors of production like land, labour and capital need organizing before they combine to produce outputs. Iron ore in the ground is useless until someone organizes to bring the labour and capital to bear to extract it and process it ready for another business to use in many different ways. A chemical company will not discover new processes unless humans combine with land and capital to work out what these processes might be and design them so that they are cost effective and viable. As individuals we would have problems cutting and styling our own hair unless someone with the necessary skill brings together the factors of production to enable us to sit and watch as our hair is transformed. That requires land for the salon building, equipment such as sinks, taps, chairs, scissors, dryers, colourings, chemicals, etc., not to mention someone or a group of people taking the risk of setting up the business in the first place.

This is the factor of **enterprise**. Entrepreneurs take the risk of organizing factors of production to generate business activity and in return hope to get a number of rewards which might include profit but might also be less obvious things such as self-satisfaction, personal challenge and the desire to take more control over one's life.

We might often think of entrepreneurs as being exceptional individuals who have seemingly become incredibly successful and very wealthy. The names usually quoted in the same breath as entrepreneurs include Sergey Brin and Larry Page, Richard Branson, Mark Zuckerberg, Debbie Fields, Azim Premji and Lakshmi Mittal, amongst others, but these tend to be extreme examples. The reality is that the world is full of millions of people being entrepreneurial. They might include an individual who has set himself up in business as a painter and decorator, an electrician, builder, plumber, florist, carpet fitter, a child in a

land all the natural resources of the Earth which can be used in production

labour all the human effort, mental and physical, which is used in production

capital any item used in production which is not used for its own sake but for what it contributes to production



Look around you any time and business activity is everywhere. A busy city centre is full of people engaging in some sort of business activity. Every non-human thing in this image has been part of the transformation process.

enterprise the act of taking risks in the organization of factors of production to generate business activity

poverty stricken area of India making some money out of recycling rubbish in some way, a farmer running a dairy herd, a financial advisor, tyre fitter and many other examples.

An enterprise is an organized activity that:

Produces or provides goods/services
Has some level of planning, structure, and risk-taking
Aims to earn income or profit

hence, no!



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Can this be classed as an entrepreneurial activity?

The skills necessary to be an entrepreneur are well documented but the extent individuals possess and utilize these skills varies in each case. The reasons why some entrepreneurs go on to make millions whilst others struggle to barely make a living is not simply to do with the degree of determination, initiative, planning, access to finance, asking the right questions, acting on hunches, taking risks, being willing to work hard and make things happen, thinking ahead and thinking creatively, it is also to do with being in the right place at the right time and having a large degree of luck.

Entrepreneurs take risks and many of them fail: the rate of business failure in the three to five years after start up is estimated to be around 30 per cent, although getting precise data is difficult because we have to be careful how we define 'business failure'. It is fairly clear that a business fails if it has to file for insolvency or, in the case of a sole trader, bankruptcy, but if the owner sells on the business after a few years because they do not feel they are getting enough of a return is that also an example of failure?



what if... the number of business failures rises above the 50 per cent level, would this mean that it is not worth taking the risks to start up a new business?

It is clear that setting up a new business is challenging. Potentially high failure rates do not put off millions of people around the world from starting businesses and many will try again after (sometimes many) failures in the hope that lessons have been learned and the next time will see things work.

The skills and qualities of entrepreneurs are many but perhaps the most important is the willingness to take risks. This is one reason why some economists prefer to class enterprise as a factor of production in its own right rather than seeing it as just another form of labour.

Business activity is about bringing these factors of production together to generate a product which is then sold. It is essentially a transformation process, therefore, with some types of business activity being very much more complex and risky than others. This is one reason why prices might be higher for some products compared to others and why some types of labour generate more income than do others.

The Transformation Process

How these factors are brought together, in what proportions and how they work together in the transformation process, could be very different, even in the same type of business operating in the same industry. Rarely are two firms producing cars or chemicals the same although they may have many similarities. One of the key elements of the transformation process is adding value and this could be at any stage in this process.

Added value is what a business does to inputs to convert them to outputs which customers (businesses or final consumers) are prepared to pay for. Adding value could be in the form of a piece or technology that makes a consumer's life much easier in some way or does the job the product is designed for more effectively or more stylishly than other rival products on the market. It might even be that a business creates a product or service that no one has thought of before and which people are prepared to pay enough money for, and over a long enough period (often repeatedly), to enable the business to cover the costs of producing that product and to provide a sufficient return to those who own the business to persuade them to keep producing. There is a great deal of complexity which arises out of this relatively simple statement but it is at the heart of what business activity is about and how a business can survive. If it cannot add value then the business will ultimately fail.

Some products will fail because they do not meet market needs, by which we mean that there are not enough people willing to pay the price being asked which is sufficient to cover the costs of production and provide the return.

added value the difference between the cost of factor inputs into production and the amount consumers are prepared to pay (the value placed on the product by consumers)

Added Value = Selling Price – Cost of Inputs
the increment of cost at each step

Pitfall Prevention Added value is the difference between the cost of inputs and the price consumers are 'prepared to pay' which can be affected by non-tangible things like perceived brand value.

The entrepreneur may have thought there was a market and whilst there invariably are some people who will buy the product, the key is whether there are *enough* people or businesses willing to pay for the product.

In other cases, a perfectly good product which has a market will fail because some other business comes along and offers a product which does something more and better. In other cases the product may fail because times have changed and there is simply no need for that product any more.

Business activity, therefore, is dynamic.

Quick Quiz Think of two products which are now not produced in the volumes they were (if at all). Write down a brief list of why these products have ceased to have any commercial value.

The transitions in mobile phone technology is an excellent example of this dynamic process. The very idea of having a phone which could be used anywhere is relatively new, perhaps only 30 years old, but the changes over that period of time in what these products look like, their size and what they can do have been significant. Initially, merely being able to contact and speak to another person away from the house was a major step forward.

Then being able to send short messages was seen as a revolutionary step forward. After that combining a mobile phone with a device that could access the Internet was a goal of businesses in this Industry, but very quickly it became not only accessing the Internet but being able to send emails, record and transmit video, watch TV, play music, record voices, download and read books, act as a calculator, a personal messaging system, diary, satellite navigation system and so on, which have all become part of our mobile phone to the extent that in many cases people rarely use them for actually making a phone call any more!

It is difficult to imagine the many hours of development and technological change that have led us to the situation we are currently in with mobile phones, and it is probably even harder for us to imagine what these devices might be like in ten years' time. However, the point is that somebody has sat and thought about these things; they have asked what else could the technology allow us to do, what new technologies do we need to enable us to provide some even more wonderful things in the future? What sort of things do people want from these devices and perhaps equally important, what do they not want?

There were probably many different cases of products, phones, technologies that never actually made it past design or market research stage. Many ideas that never worked, but businesses in the industry took those risks, marshalled the factors of production and made the transformations necessary to create that dynamic process whereby competition led us to seeing new products and new technologies which we have been prepared to pay for and which we presumably think improve our lives (although there are always going to be disadvantages). If the advantages or costs outweigh the disadvantages then we tend to buy them and businesses will produce products as a result.

THE PESTLE FRAMEWORK

We have already mentioned that business activity is dynamic. Change will be happening all the time in any business and how they adapt to this change will be an important part of the extent to which the business is a success. In responding to change, there are some factors which the business will have some control over and others which the business does not have control over.

The business can have some control over the inputs it buys and how it combines those inputs to produce its outputs. However, there are a number of external factors over which it has very little control but which it has to respond and react to.

CASE STUDY

A Complex Transformation Process

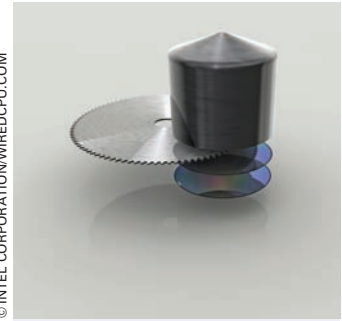
It is entirely possible that you might be reading this case study on a computer device of some kind, perhaps a laptop, a smart phone, tablet etc. One of the key elements of these technologies is the silicon chip. Intel is one of the largest chip makers in the world and the following case gives some indication as to the complexity of the transformation process.

One of the key inputs is sand which includes large amounts of silicon. The silicon is purified by being melted and then cooled into an ingot – a cylindrical shape around 300 mm in diameter weighing 100 kg. This process is undertaken in Japan by companies like Toshiba Ceramics. Other companies then take these ingots and slice them into thin wafers around 1 mm thick. Each wafer is then polished and

Intel buys them in this state ready for manufacturing at its plants in Arizona and Oregon in the United States.

At these plants (called ‘fabs’) – which, incidentally are identical in design and building orientation – the wafers are etched with integrated circuits which build layers and which are the result of hundreds of individual processes. Once these processes are complete the wafers are then shipped to Intel’s assembly and test plants in Malaysia. Here the wafers are tested and then sliced into pieces called dies and tested again to ensure they work. Those that do not pass this test are discarded and the ones that do are packaged and sent back to warehouses in Arizona. At this stage the packaging is anonymous so that it is not clear that they are from Intel to help reduce the risk of theft in transportation. From the warehouses in Arizona the chips are then shipped to computer manufacturing plants across the world and to different manufacturers. The plants can be in Brazil, Taiwan, China, Malaysia, Ireland and to other parts of the US such as Texas and Tennessee, depending on the manufacturer. Once the chips are put into the device this may then be shipped either to a retail outlet or direct to the customer. The humble chip in your device is likely to have travelled thousands of miles during its production process as it is transformed from sand into an extremely sophisticated electronic component capable of helping process millions of operations in a short amount of time.

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A silicon ingot being sliced into a wafer.

In order to help understand and analyse these external factors, a framework is used which summarizes a number of broad, sometimes highly related and interacting areas, which business has to work within. This framework is referred to as the PESTLE framework with the acronym standing for:

- Political
- Economic
- Social
- Technological
- Legal
- Environment

We will take a brief look at each one in turn.

Political

Politics refers to power – who has power, who makes decisions and how it affects individuals and business. Power can be wielded by local governments, national governments and supranational governments (where decision making or laws are made by groups or states outside national boundaries) such as the European Union (EU). In some countries power lies in the hands of a relatively small number of people, possibly linked to a royal family, tribal or religious group. In others the military may be an important element in the way in which political authority is framed.

In most countries in the EU, government is democratic with political parties submitting themselves for election periodically. Different systems determine who forms governments but one particular party or group of parties in coalition may have been given the power by the people to make decision, policies and laws which affect individuals and businesses in various ways. In other countries it is a ruling elite or the military who make decisions and establish laws and the people in the country may have a very limited,

or no say, in the political process. In still other countries, the rule of law may barely exist and in this case business activity may be very difficult to carry out.

Changes to laws, directions in policy or regulations can all affect businesses in different ways. Laws on employment, employee rights and responsibilities, health and safety, taxation, planning, trade, advertising and business governance amongst many other things, all affect business and invariably raise the cost of doing business by either involving the business in additional time, form filling or procedures. In some cases laws or regulations may be passed with the aim of helping a business by giving grants or special dispensation to operate. In this case there may also be a cost in terms of certain conditions to which the business has to adhere to get the benefit. For example, there may be a relaxation of planning regulations which mean a business can establish new premises more quickly but the *quid pro quo* (something given in return for something) is that the business has to remain at the premises for a certain period of time or employ a certain number of people.

Economic

Businesses have to operate within an economic environment. This relates to the extent of economic activity in different 'economies' which could include a very local economy, a regional economy, a national or supranational or global economy. There are also bodies which have supranational powers – the authority to act across different nations – such as the International Monetary Fund (IMF).

Economic activity, as we have seen, refers to the amount of buying and selling that takes place. This activity can be looked at within a local area, a region, nation and on a global scale. The rate at which buying and selling (or the number of transactions) takes place varies at different time periods for different reasons as we will see in more detail in later chapters.

Businesses are affected by these fluctuations in economic activity. For example, a restaurant will be affected by decisions of people to decide to go out for a meal which may, in turn, be affected by how confident these people feel that they will continue to have a job in the future or whether they have recently lost their job. If a restaurant finds that the number of people they serve in a week is falling then they will have to adjust the supplies that they purchase and this will then have an effect on other businesses. Those supplying wine, soft drinks, fresh fruit and vegetables and other ingredients will then be affected by falling sales.

Such an example might be characteristic of a decline in the local economy and can often happen when an area is highly dependent on a single employer who may either scale back operations or even close down. A similar case will occur over regions and whole countries. The south of Italy, for example, suffers from a lower level of economic activity than the north and as such standards of living in the south are lower and opportunities for employment and growth are lower in return. There are similar situations which occur in parts of the former eastern Germany and across different parts of the Middle East. In the Palestinian territories, for example, economic activity is significantly affected by the ongoing political situation and this means that this region has a lower level of economic activity.

Countries can be compared in terms of economic activity and we can see from such comparisons that there are very widely differing levels of economic activity. In some African countries recorded economic activity is tiny in comparison to others in the region such as South Africa and Nigeria.

In recent years much has been made of the global economy and at the time of writing, forecasts for global economic activity are gloomy to say the least and come after a period of slower global economic growth which followed the financial crisis of 2007–2009.

What we can see from the above is that businesses are affected by the swings in economic activity both locally, nationally and globally. In many cases these swings might be

triggered by some crisis such as the eurozone debt crisis, or a financial crisis in the banking system or by changes in interest rates, but the effects are magnified by the changes in confidence levels in individuals, businesses and governments.

It is also possible to classify these effects as microeconomic or macroeconomic. The **microeconomic environment** refers to factors and issues that affect an individual firm operating in a particular market or industry. Changes in economic activity can affect some firms in a positive way and others in a negative way. For example, regardless of the level of economic activity, funeral directors may experience relatively stable levels of trade although in times of economic slowdown families may choose to spend smaller amounts on funerals or choose cheaper options than they may do when the economy is performing more strongly.

In times of weak economic growth retail businesses such as supermarkets may also find that whilst there may be changes to the type of products people buy, the volume of trade does not decline that much, meaning they are relatively insulated from declines in economic activity. Retail businesses which sell high end products such as electronic goods or fashion items may find that they are very badly hit by economic slowdown and sales may fall dramatically. Other businesses such as second-hand shops, pawn shops or low price discount stores may actually find their business increases in times of economic slowdown. These are all examples of specific businesses in particular markets.

The **macroeconomic environment** refers to the national or global economy within which the business operates. The things which can affect businesses from macroeconomic changes include variations in exchange rates, interest rates (which may be linked), policies on taxation, planning, competition and so on. Changes in these macroeconomic factors can affect the level of economic activity in the economy as a whole and as a result impact on businesses.

If exchange rates change then businesses will be affected in different ways depending on the extent to which they buy and sell products from abroad and in what proportions. Costs could rise or fall (or a combination of the two) and demand could also rise or fall and the effects can be highly complex in businesses which trade extensively across different regions of the world.

Social

Businesses are affected by various trends, fashions, moods and changes in society. The move to improving the equality between men and women in the workplace, for example, has led to businesses having to adapt their processes and their attitudes to employment, the way their businesses operate and how they monitor the attitudes and behaviour of workers.

Social changes affect our attitudes to things like recycling and the publicity which has been given to the problem of climate change and/or global warming has meant that many businesses now report the extent to which they have taken steps to monitor energy use, recycling, the use of natural resources and where they source raw materials. Building and office construction is changing to try and find ways to improve efficiency and make them 'greener'.

There are broad social changes that are also having an impact on businesses, such as the changing structure of the population. Many European and Middle East economies, for example, are experiencing an ageing population with an increasing proportion of the population over the age of 65. This creates both opportunities and threats to businesses. Retirement ages may well rise in countries as governments struggle to afford state pensions and this changes employment dynamics. Businesses that offer pension support have already found that they have to adjust the type of pension they offer. Final salary schemes, where the pension the employee receives is based on a proportion of their salary at retirement, have been phased out and replaced by contributory pension schemes simply because some businesses were finding that they could not afford to sustain final salary schemes as people were living longer.

microeconomic environment
factors and issues that affect an individual firm operating in a particular market or industry

macroeconomic environment
the national or global economy within which the business operates



Every individual is part of the wider economy and their decisions affect not only the immediate businesses they interact with but the economy as a whole.



YURI ARCURI/SHUTTERSTOCK

An ageing population requires resources to be diverted to different uses in order to meet the needs of a different population structure – this provides opportunities for businesses as well as challenges.

For some businesses the ageing population provides opportunities to develop products and services which are targeted at the needs of the growing number of people who are over 65, who tend to be more affluent and who are still relatively healthy and active despite their age. The pattern of housing demand changes, with smaller homes required to take account not only of single pensioners, but also the rising number of single families, which has followed rising divorce rates in many countries.

Manufacturers are looking to develop vehicles which cater for the needs of older drivers through the provision of more intuitive technologies such as automatic parallel parking, sensors which mean the car can effectively 'see round corners', have active safety systems which warn the driver of potential hazards or danger ahead, voice operated functions and wi-fi capability in the car.

Other social changes such as the growth in the use of social networking sites, viral messaging and the Internet have opened up opportunities but also present threats. Facebook and Twitter provide the chance for businesses to showcase themselves and have their brand and message spread to large numbers of people very quickly and at low cost. The flip side of this is that the degree of control a business has over messaging and the reporting of the business is very weak.

Employees can, sometimes innocently, compromise the business and damage the brand or reputation simply through an injudicious use of 140 characters or an ill-judged piece of behaviour which is subsequently broadcast to millions on YouTube or Facebook. It is fair to say that most businesses are still trying to understand the social networking phenomenon and how best to utilize it. The problem is that social networking tends to change more quickly than the ability of a business to understand it and work out how to use it most effectively.

Technological

technology the application or use of knowledge in some way which enables individuals or businesses to have greater control over their environment

It is tempting to think of technology as some electronic gadget but the definition of technology is much wider. **Technology** is the use of knowledge in some way which enables individuals or businesses to have greater control over their environment. Businesses constantly think of ways in which they can employ knowledge in this way because it can help to reduce costs, improve technical and productive efficiency and in many cases social and allocative efficiency. Technology can also help give a firm competitive advantage where the advantage has the key characteristics of being distinctive and defensible.

The last 50 years have seen an explosion in technological developments that have provided both opportunities for businesses and also threats. How businesses respond to these opportunities and threats is a crucial element of business. Technological developments can also help to provide some answers to the most pressing problems that humans face, including the effect on the environment of business activity, how to feed a rapidly growing human population, how to provide access to the essentials of life such as water, how to treat killer diseases, save animals and plants from extinction and tackle global poverty.

Quick Quiz Outline two examples where you think technology has improved our lives and two examples where you think technology has not led to an improvement in human welfare.

Legal

Laws and regulation can be national or supranational. The legal framework covers all aspects of society and businesses have to abide by these laws. A strong legal system

which is respected is fundamental to the principle of good governance, which in turn helps provide confidence in the way in which a business operates and so promotes trade. As we have seen in Chapter 1, trade is beneficial most of the time but businesses will be reluctant to trade and customers put off buying if they do not have confidence in business activity.

For example, customers want to know that if they buy a litre of fuel from a petrol station they do actually get a litre of fuel dispensed from the pump; investors need to know that the information on which they base decisions is as accurate and truthful as possible, if a business comes up with a new idea, process or invention that they can protect the investment in time, money and intellectual capital that they have made.

Much of business success relies on confidence. Confidence between businesses, confidence between businesses and customers and the legal and regulatory framework, which is adhered to by most and which builds in adequate incentives to be adhered to, is an important part of this.

Laws and regulation govern the way in which financial accounts are reported, how labour markets work, what health and safety measures businesses need to put in place, how they can describe and advertise products, what information consumers must be given, what minimum standards must be met, how much pollution a business can create and many more.

Whilst a strong and respected legal and regulatory framework provides confidence, it also comes at a cost; businesses have to pay to implement legal and regulatory requirements and this not only means higher costs for them and possibly an effect on margins, but might also mean higher prices or more inconvenience for consumers. For example, data protection laws mean that husbands and wives often get frustrated at the fact that a business will not discuss issues relating to a spouse or partner.



what if... a business sees an opportunity to sell its goods into a new market in an emerging economy but a report tells them that governance and the rule of law is weak. Should the business enter that market?

Environment

It is now rare for any business to operate without some recognition of the impact of its operations on the environment. This awareness may be as a result of a conscious policy decision to manage its operations to take account of that impact, or through being forced by law or regulation to do so.

Economic growth across many countries around the world has meant that resource use is expanding and as we saw in Chapter 1, resources are scarce in relation to demand. There is also concern about how we use resources, not only in terms of productive and technical efficiency but also from social and allocative efficiency and how we manage the results of resource use in terms of the waste products generated and the impact on ecosystems and land use.

One of the major themes of the last 30 years has been a growing concern that the consequences of human activity could be having a negative effect on the wider ecosystem. There are plenty of studies to suggest that carbon emissions, largely produced by human activity, have been a direct cause of a gradual rise in average global temperatures which in turn could lead to greater volatility in global weather patterns and also bring about a thawing of polar ice caps, rising sea levels and subsequent effects on those living in low-lying areas of the world.

Major efforts have been made to get global agreement on reducing carbon emissions and finding more environmentally friendly ways of producing goods, services and



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The potential for business activity to impact on the environment always exists. The argument centres on the extent to which governments, laws and regulations can help reduce the risk.

energy. This has not been easy to achieve. One of the reasons is that the richer countries who have been accused of being responsible for these carbon emissions are now asking for everyone to take the pain and cost of adjustment and poorer countries are suggesting that such a move would jeopardize their efforts to grow and better provide for their people. They argue that the rich nations are the ones who caused the problems so the rich nations should be the ones who take most of the pain.

To counter this the rich nations say that there is little point in them taking action to reduce emissions if the emerging nations are going to more than replace any reductions they might make several times over in the coming years as they grow rapidly. It seems that whilst there is some consensus that the planet does face a problem, who is responsible and how it should be tackled is less in agreement.

One aspect of thinking like an economist is the necessity of thinking critically, of not accepting everything you hear or read without questioning its validity and reliability. In many countries in Europe, recycling is an obvious and significant feature of everyday life. Almost every business has recycling policies and facilities, universities vie with one another to be the 'greenest' institution, households, local government and businesses are required to recycle and to meet targets set by national and supranational government.

As business economists, we need to be thinking about the costs and benefits of recycling and as a result whether it makes economic sense. Not all recycling is 'good'; if, for example, the amount of resources necessary to recycle metal cans into other products was greater than the cost of producing the cans from scratch, would it be a sensible business decision to do it?

Equally, we need to remember to critically examine claims from 'scientists' about carbon emissions and climate change. If businesses are going to be required to make what are quite possibly very expensive and significant decisions on resource use and allocation, are the reasons for making those decisions based on sound information? How reliable are the studies carried out into the effects of rising carbon emissions? Does the idea of 'global average temperature' actually mean anything? Simply, businesses have to ask the right questions to get the right information in order to make more informed decisions.

Pitfall Prevention Whilst we classify external factors using the PESTLE framework, in analysing real business situations it is often not easy to simply classify factors affecting a business in a simple way – the factors tend to be interrelated and cause and effect are not readily identifiable.

JEOPARDY PROBLEM

A business producing top-of-the-range smartphones with the very latest technology goes into insolvency. What is most likely to have caused this?

SHAREHOLDER VALUE AND STAKEHOLDERS

We have seen how business activity is a transformation process but we have to ask ourselves why businesses carry out this activity and for whom? We can use two concepts to provide at least part of the answer: shareholder value and stakeholders.

THE MARKET FORCES OF SUPPLY AND DEMAND

We saw in Chapter 2 how a market is made up of two parties – the buyer and the seller. The buyer represents demand and the seller supply. In this chapter we are going to look at a fundamental aspect of business economics, the operation of markets and the interaction between these two ‘forces’.

Poor weather conditions in parts of Europe can have an effect on the yield of wheat crops and as a result businesses using wheat in the production of food products face higher costs. High levels of economic growth in China cause the demand for steel to rise and this pushes up prices for businesses across the rest of the world, meaning the price of both semi-finished and finished products rises where steel is a component part in production.

If a report is published linking food products with health risks, firms producing these foods face falling prices and a possible collapse in their markets. A change in exchange rates for currencies can have different effects on different businesses depending on the extent to which they trade with other businesses and customers abroad. Airlines know that they can charge higher prices at certain times of the year to certain destinations than at other times where they may have to cut fares to fill aircraft. What do these events have in common? They all show the workings of supply and demand.

Supply and *demand* are the two words that economists use most often – and for good reason. Supply and demand are the forces that make market economies work. We refer to them as forces because they act in different ways and cause prices to change – the factor that links the two forces. Supply and demand determine the quantity of each good produced and the price at which it is sold. If you want to know how any event or policy will affect the economy and businesses, you must think first about how it will affect supply and demand.

This chapter introduces the theory of supply and demand. It considers how sellers and buyers behave and how they interact with one another. It shows how supply and demand determine prices in a market economy and how prices, in turn, allocate the economy’s scarce resources.

At this point it is important to note an important distinction. We use the terms ‘price’ and ‘cost’ regularly in everyday life – often interchangeably. In this book we will refer to the two terms in a distinct way. **Price** is the amount of money a buyer (a business or a consumer) has to give up in order to acquire something. **Cost** refers to the payment to factor inputs in production. When we discuss suppliers we will be referring to cost in this sense.

price the amount of money a buyer (a business or a consumer) has to give up in order to acquire something

cost refers to the payment to factor inputs in production

market a group of buyers and sellers of a particular good or service

MARKETS AND COMPETITION

The terms *supply* and *demand* refer to the behaviour of businesses and people as they interact with one another in markets. A **market** is a group of sellers and buyers of a particular good or service. The sellers as a group determine the supply of the product and the buyers as a group determine the demand for the product. Before discussing how sellers and buyers behave, let’s first consider more fully what we mean by a ‘market’ and the various types of markets we observe in the economy.

Competitive Markets

Markets take many forms. Sometimes markets are highly organized, such as the markets for many agricultural commodities and for metals. In these markets, buyers and sellers meet at a specific time and place, where an auctioneer helps set prices and arrange sales.

Many businesses rely on these highly organized markets and are affected by them because they have little control over the prices they have to pay for these products which can affect their costs and margins considerably both in a positive and a negative way.

More often, markets are less organized. For example, consider the market for perfume. Businesses manufacturing and selling perfume are varied and seek to offer different products for sale which they hope will be distinctive and popular. The buyers of perfume do not all meet together at any one time. These buyers are individuals, all of whom have different tastes. One person's ideal fragrance is another person's obnoxious smell and they do not all gather together in a room to shout out the prices they are willing to pay. There is no auctioneer calling out the price of perfume. Each seller of perfume posts a price for a bottle of perfume in their shop, and each buyer either walks past the shop because they do not want perfume or if they do will go to different shops and try different perfumes before making their decision.

Even though it is not organized, the group of perfume sellers and buyers forms a market. Each seller is aware that her product is similar but different to that offered by other sellers. Each buyer knows that there are several sellers from which to choose. The price of perfume and the quantity of perfume sold are not determined by any single buyer or seller. Rather, price and quantity are determined by all sellers and buyers as they interact in the marketplace.

The market for perfume, like most markets in the economy, is competitive. **Competition** exists when two or more firms are rivals for customers. Each firm strives to gain the attention and custom of buyers in the market. Economists use the term *competitive market* in a different way to mean something very specific. A **competitive market** is a market in which there are many buyers and many sellers so that each has a negligible impact on the market price. In the perfume market, sellers may have some limited control over price because of the way they can differentiate their product from competitors. In describing a competitive market in this chapter, we are going to look at a more specific definition of a competitive market and in subsequent chapters look at firm behaviour when the assumptions of perfect competition are relaxed. We are going to look at how sellers and buyers interact in competitive markets and see how the forces of supply and demand determine both the quantity of the good sold and its price.

competition a situation when two or more firms are rivals for customers. Each firm strives to gain the attention and custom of buyers in the market

competitive market a market in which there are many buyers and many sellers so that each has a negligible impact on the market price

Competition: Perfect and Otherwise

The assumptions outlined above refer to markets that are referred to as *perfectly competitive*. Perfectly competitive markets are defined by two main characteristics: (1) the goods being offered for sale are all the same (homogenous) and as a result buyers have no preference between one seller or another; and (2) the buyers and sellers are so numerous that no single buyer or seller can influence the market price. Because buyers and sellers in perfectly competitive markets must accept the price the market determines, they are said to be *price takers*.

There are some markets in which the assumption of perfect competition applies to a very large degree. In the wheat market, for example, there are tens of thousands of farmers who sell wheat and millions of consumers who use wheat and wheat products. Because no single buyer or seller can influence the price of wheat, each takes the price as given.

The reason for making this assumption is so that we can look at how markets operate under these 'ideal' conditions and what the expected outcomes are. If we then observe in reality that these outcomes do not occur as we expect, then we can analyse what imperfections exist which help to explain this behaviour.

Not all goods and services, therefore, are sold in perfectly competitive markets. Some markets have only one seller, and this seller sets the price. Such a seller is called a *monopoly*. Your local water company, for instance, may be a monopoly. Residents in your area probably have only one water company from which to buy this service.

quantity supplied the amount of a good that sellers are willing and able to sell

law of supply the claim that, other things being equal, the quantity supplied of a good rises when the price of the good rises

Some markets fall between the extremes of perfect competition and monopoly. One such market, called an *oligopoly*, has a few sellers that do not always compete aggressively. Airline routes are an example. If a route between two cities is serviced by only two or three carriers, the carriers may avoid rigorous competition so they can keep prices high. Another type of market is *monopolistically or imperfectly competitive*; it contains many sellers but each offers a slightly different product. Because the products are not exactly the same, each seller has some ability to set the price for its own product. An example is the market for magazines. Magazines compete with one another for readers and anyone can enter the market by starting a new one, but each magazine offers different articles and can set its own price.

Despite the diversity of market types we find in the world, we begin by studying perfect competition. Perfectly competitive markets are the easiest to analyse. Moreover, because some degree of competition is present in most markets, many of the lessons that we learn by studying supply and demand under perfect competition apply in more complicated markets as well.

Quick Quiz What is the market for processing chips in a computer or mobile phone made up of? • What are the characteristics of a competitive market for milk?

SUPPLY

We are going to begin our look at markets by considering producers – businesses – and examine the behaviour of sellers. To focus our thinking and provide a context for our analysis, let's consider producers of rape seed which is used to make cooking oil.

The Supply Curve: The Relationship Between Price and Quantity Supplied

The **quantity supplied** of any good or service is the amount that sellers are willing and able to sell. There are many determinants of quantity supplied, but price plays a special role in our analysis. When the price of rape seed is high, selling rape seed is profitable, and so sellers are willing to supply more. Sellers of rape seed work longer hours, devote more planting to rape, invest in research and development on improvements to rape seed growing and hire extra workers in order to ensure supplies to the market rise. By contrast, when the price of rape seed is low, the business is less profitable, and so growers are willing to plant less rape. At a low price, some growers may even choose to shut down, and their quantity supplied falls to zero. Because the quantity supplied rises as the price rises and falls as the price falls, we say that the quantity supplied is *positively related* to the price of the good. This relationship between price and quantity supplied is called the **law of supply**: other things being equal, when the price of a good rises, the quantity producers are willing to supply also rises, and when the price falls, the quantity supplied falls as well.

The table in Figure 4.1 shows the quantity Tramontana, a rape seed grower, is willing to supply, at various prices of rape seed. By convention, the price is on the vertical axis and the quantity supplied on the horizontal axis. At a price below €0.50 per tonne, Tramontana does not supply any rape seed at all. As the price rises, it is willing to supply a greater and greater quantity. This is the **supply schedule**, a table that shows the relationship between the price of a good and the quantity supplied, holding constant everything else that influences how much producers of the good want to sell.

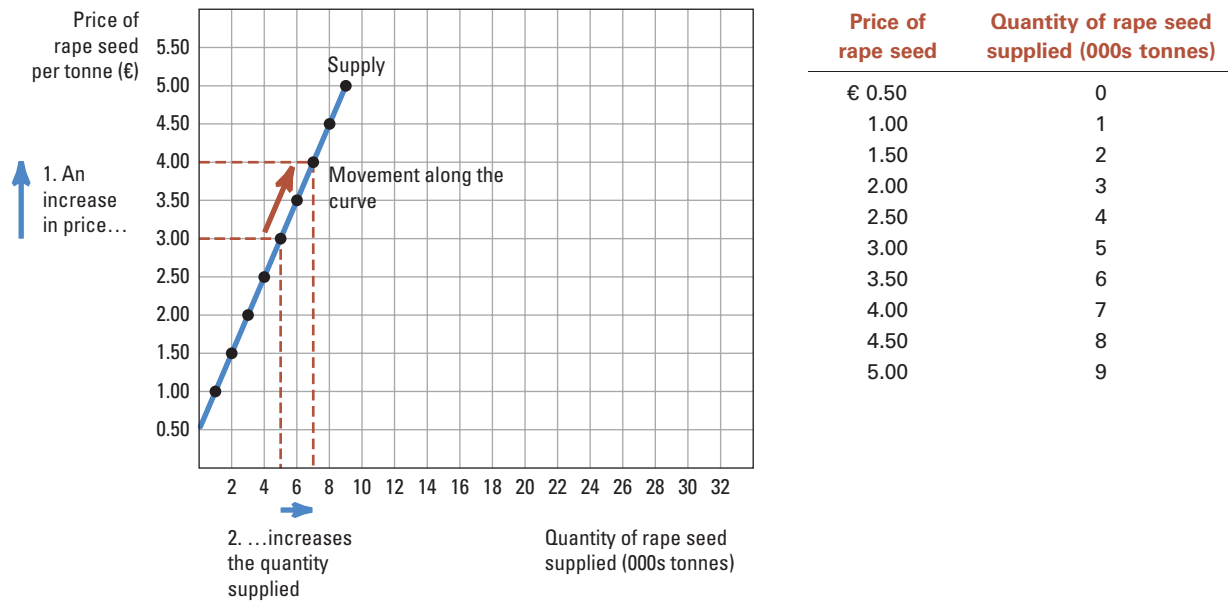


The distinctive yellow flowers of rape plants make an attractive sight in many countries at certain times of the year. The seed from the plants provides a valuable source of cooking oil.

supply schedule a table that shows the relationship between the price of a good and the quantity supplied

FIGURE 4.1**Tramontana's Supply Schedule and Supply Curve**

The supply schedule shows the quantity supplied at each price. This supply curve, which graphs the supply schedule, shows how the quantity supplied of the good changes as its price varies. Because a higher price increases the quantity supplied, the supply curve slopes upward.



The graph in Figure 4.1 uses the numbers from the table to illustrate the law of supply. The curve relating price and quantity supplied is called the supply curve. The **supply curve** slopes upward because, other things being equal, a higher price means a greater quantity supplied.

supply curve a graph of the relationship between the price of a good and the quantity supplied

Market Supply versus Individual Supply

Market supply is the sum of the supplies of all sellers. The table in Figure 4.2 shows the supply schedules for two rape seed producers – Tramontana and Sedona. At any price, Tramontana's supply schedule tells us the quantity of rape seed Tramontana is willing to supply, and Sedona's supply schedule tells us the quantity of rape seed Sedona is willing to supply. The market supply is the sum of the two individual supplies.

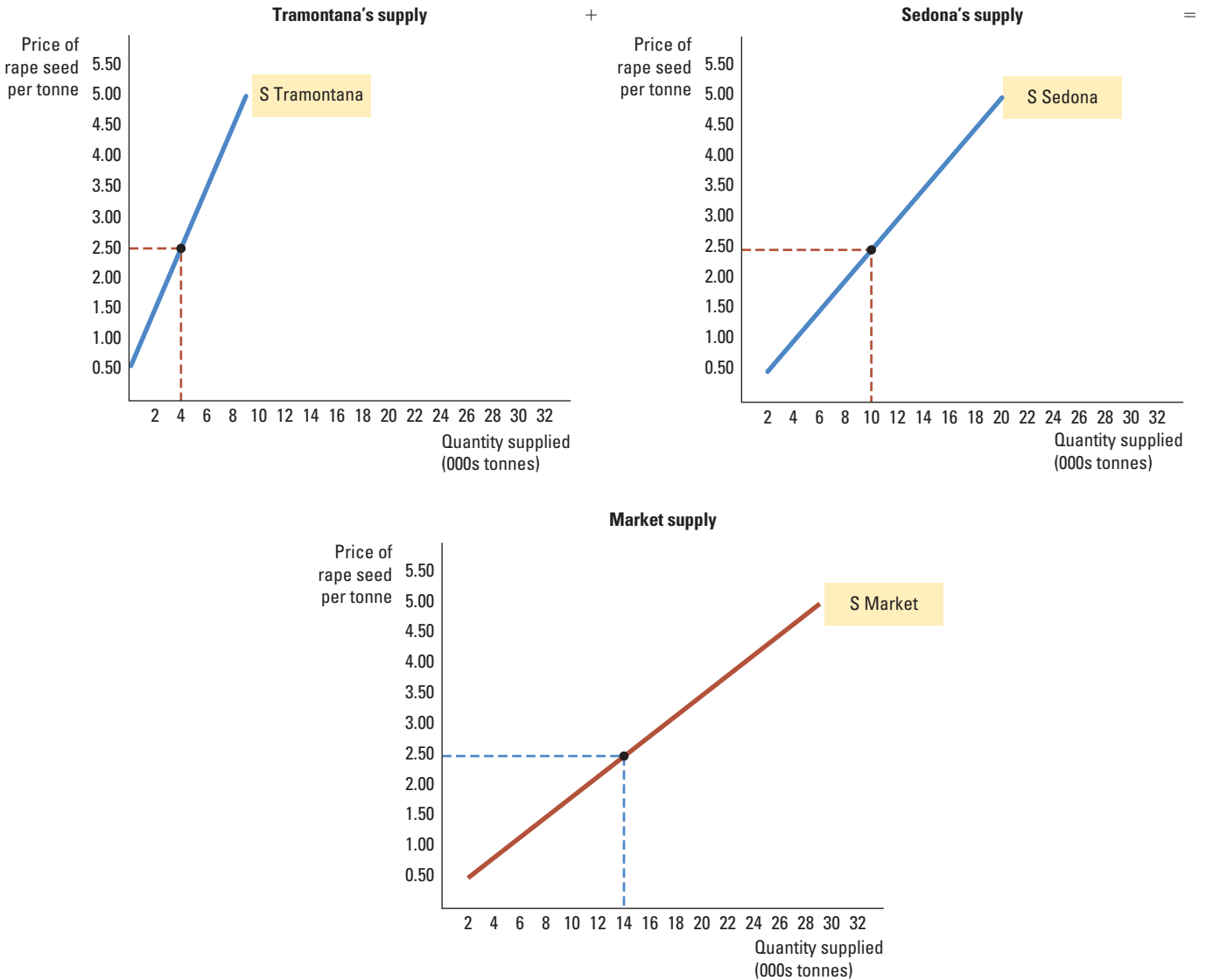
The graph in Figure 4.2 shows the supply curves that correspond to the supply schedules. We sum the individual supply curves *horizontally* to obtain the market supply curve. That is, to find the total quantity supplied at any price, we add the individual quantities found on the horizontal axis of the individual supply curves. The market supply curve shows how the total quantity supplied varies as the price of the good varies. In reality, the market supply will be the amount all producers in the market (rape seed manufacturers and sellers in this example) are willing to offer for sale at each price.

Pitfall Prevention Be careful to ensure that you distinguish between individual and market supply in your analysis – the behaviour of one individual business may be different from the whole industry.

FIGURE 4.2

Market Supply as the Sum of Individual Supplies

The quantity supplied in a market is the sum of the quantities supplied by all the sellers at each price. Thus, the market supply curve is found by adding horizontally the individual supply curves. At a price of €2.50, Tramontana is willing to supply 4000 tonnes of rape seed and Sedona is willing to supply 10 000 tonnes. The quantity supplied in the market at this price is 14 000 tonnes of rape seed.



Price of rape seed (per tonne)	Tramontana (000s)		Sedona (000s)		Market (000s)
€0.50	0	+	2	=	2
1.00	1		4		5
1.50	2		6		8
2.00	3		8		11
2.50	4		10		14
3.00	5		12		17
3.50	6		14		20
4.00	7		16		23
4.50	8		18		26
5.00	9		20		29

Shifts versus Movements along the Supply Curve

A distinction must be made between a shift in the supply curve and a movement along the supply curve. A shift in the supply curve is caused by a factor affecting supply other than a change in price. The factors affecting supply are outlined below. If any of these factors change, then the amount sellers are willing to offer for sale changes, whatever the price. The shift in the supply curve is referred to as an *increase or decrease in supply*. A movement along the supply curve occurs when there is a change in price. This may occur because of a change in demand conditions. The factors affecting supply are assumed to be held constant. A change in price leads to a movement along the supply curve and is referred to as a *change in quantity supplied*.

Shifts in the Supply Curve

The supply curve for rape seed shows how much rape seed producers are willing to offer for sale at any given price, holding constant all the other factors beyond price that influence producers' decisions about how much to sell. This relationship can change over time, which is represented by a shift in the supply curve. For example, suppose the price of fertilizer falls. Because fertilizer is an input into producing rape seed, the fall in the price means producing rape seed is now cheaper – the same quantity of rape seed can be made at lower cost which makes selling rape seed more profitable. This raises the supply of rape seed: at any given price, sellers are now willing to offer for sale a larger quantity. Thus, the supply curve for rape seed shifts to the right.

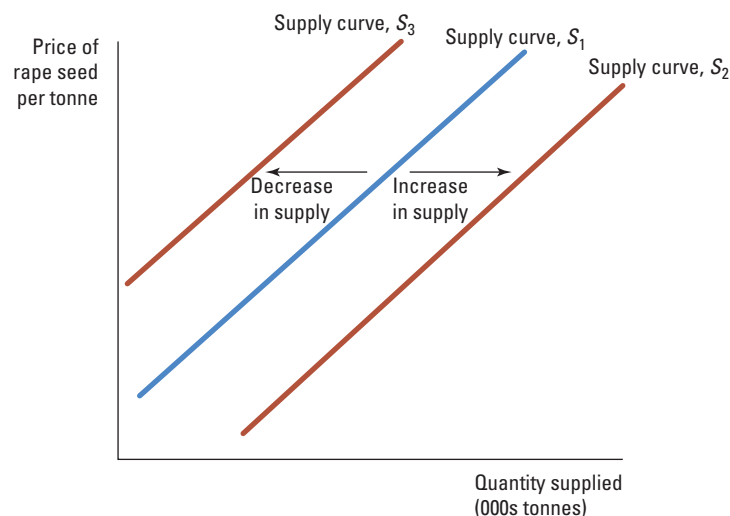
Figure 4.3 illustrates shifts in supply. Any change that raises quantity supplied at every price, such as a fall in the price of fertilizer, shifts the supply curve to the right and is called an *increase in supply*. Similarly, any change that reduces the quantity supplied at every price shifts the supply curve to the left and is called a *decrease in supply*.

There are many variables that can shift the supply curve. Here are some of the most important.

FIGURE 4.3

Shifts in the Supply Curve

Any change that raises the quantity that sellers are willing to produce and offer for sale at a given price shifts the supply curve to the right. Any change that lowers the quantity that sellers are willing to produce and offer for sale at a given price shifts the supply curve to the left.



Input Prices To produce their output of rape seed, sellers use various inputs: fertilizer, fuel for tractors, weed killer, harvesting machines and different types of packaging for distribution at various stages in the production process. Growers will also have to pay for computers, machinery, farm buildings and the labour of workers which is used throughout the planting, growing, harvesting and distribution process. When the price of one or more of these inputs rises, producing rape seed is less profitable and firms supply fewer tonnes of rape seed. If input prices rise substantially, some firms might shut down and supply no rape seed at all. If input prices fall for some reason, then production may be more profitable and there is an incentive to supply more at each price. Thus, the supply of a good is negatively related to the price of the inputs used to make the good.

Technology The technology for turning the inputs into rape seed is yet another determinant of supply. This technology might be related to the quality and viability of seed, improvements in plant breeding to create more uniform and productive plants, or in the use of fertilizers to improve soil quality and growth. Advances in technology increase productivity allowing more to be produced using fewer factor inputs. As a result costs, both total and unit, may fall and supply increases. The invention of harvesting machines, for example, reduces the amount of labour necessary to gather and process rape seed. By reducing firms' costs, the advance in technology raises the supply of rape seed.

Expectations The amount of rape seed firms supply today may depend on their expectations of the future. For example, if growers expect the price of rape seed to rise in the future, they may put some of their current stock into storage and supply less to the market today. If government reports from health departments suggest that using cooking oil made from rape seed reduces the chance of heart disease then producers might reasonably expect an increase in sales and so plant more fields to rape in anticipation.

The Number of Sellers Market supply will be affected by the number of firms in the industry. In the EU there are around 20 million tonnes of rape seed produced a year and at a price of approximately €370 per tonne this means a market value around €7.4 billion. If there were more farmers switching to rape production, then the amount of rape seed produced would be likely to rise. Equally, if some of the growers that currently plant rape closed down their operations the amount of rape seed produced would be likely to fall each year.

Natural/Social Factors There are often many natural or social factors that affect supply. These include such things as the weather affecting crops, natural disasters, pestilence and disease, changing attitudes and social expectations (for example, over the production of organic food, the disposal of waste, reducing carbon emissions, ethical supply sourcing and so on) can all have an influence on production decisions. Some or all of these may have an influence on the cost of inputs into production.

Summary The supply curve shows what happens to the quantity supplied of a good when its price varies, holding constant all the other variables that influence sellers. When one of these other variables changes, the supply curve shifts. Table 4.1 lists all the variables that influence how much producers choose to sell of a good.

Quick Quiz Make up an example of a supply schedule for apples, and graph the implied supply curve • Give an example of something that would shift this supply curve • Would a change in the price of apples shift this supply curve?

TABLE 4.1**Variables That Influence Sellers**

This table lists the variables that affect how much producers choose to sell of any good. Notice the special role that the price of the good plays: a change in the good's price represents a movement along the supply curve, whereas a change in one of the other variables shifts the supply curve.

Variable	A change in this variable ...
Price	Is represented as a movement along the supply curve
Input prices	Shifts the supply curve
Technology	Shifts the supply curve
Expectations	Shifts the supply curve
Number of sellers	Shifts the supply curve

DEMAND

We now turn to the other side of the market and examine the behaviour of buyers. Once again, to focus our thinking, let's consider the market for rape seed.

The Demand Curve: The Relationship Between Price and Quantity Demanded

The **quantity demanded** of any good is the amount of the good that buyers are willing and able to purchase. As we shall see, many things determine the quantity demanded of any good, but when analysing how markets work, one determinant plays a central role – the price of the good. If the price of rape seed rose, people would buy fewer tonnes of rape seed. Food manufacturers and retailers might switch to another form of cooking oil such as sunflower oil. If the price of rape seed fell to €1 per tonne, people would buy more. Because the quantity demanded falls as the price rises and rises as the price falls, we say that the quantity demanded is *negatively related* to the price. This relationship between price and quantity demanded is true for most goods in the economy and, in fact, is so pervasive that economists call it the **law of demand**: other things being equal, when the price of a good rises, the quantity demanded of the good falls, and when the price falls, the quantity demanded rises.

The table in Figure 4.4 shows how many tonnes of rape seed Hanse, a food manufacturer, is willing and able to buy each year at different prices of rape seed. If rape seed were free, Hanse would be willing to take 10000 tonnes of rape seed. At €2 per tonne, Hanse would be willing to buy 6000 tonnes of rape seed. As the price rises further, he is willing to buy fewer and fewer tonnes of rape seed. When the price reaches €5.00 per tonne, Hanse would not be prepared to buy any rape seed at all. This table is a **demand schedule**, a table that shows the relationship between the price of a good and the quantity demanded, holding constant everything else that influences how much consumers of the good want to buy.

The graph in Figure 4.4 uses the numbers from the table to illustrate the law of demand. The price of rape seed is on the vertical axis, and the quantity of rape seed demanded is on the horizontal axis. The downward sloping line relating price and quantity demanded is called the **demand curve**.

quantity demanded the amount of a good that buyers are willing and able to purchase at different prices

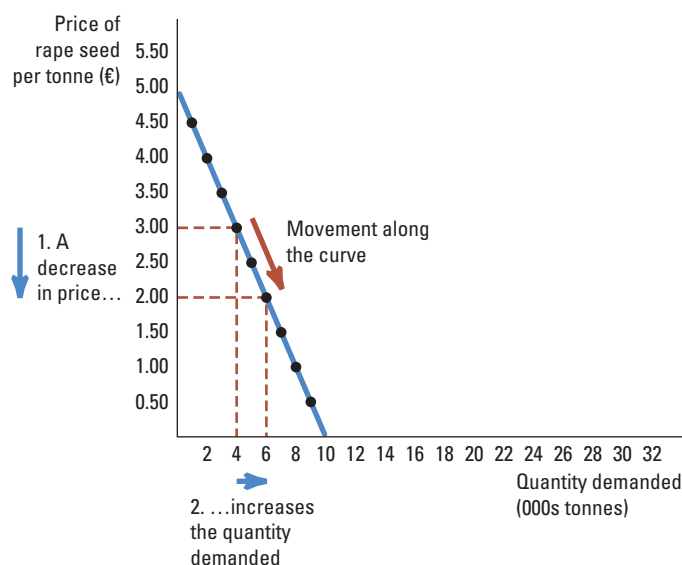
law of demand the claim that, other things being equal, the quantity demanded of a good falls when the price of the good rises

demand schedule a table that shows the relationship between the price of a good and the quantity demanded

demand curve a graph of the relationship between the price of a good and the quantity demanded

FIGURE 4.4**Hanse's Demand Schedule and Demand Curve**

The demand schedule shows the quantity demanded at each price. The demand curve, which graphs the demand schedule, shows how the quantity demanded of the good changes as its price varies. Because a lower price increases the quantity demanded, the demand curve slopes downward.



Price of rape seed Per tonne	Quantity of rape seed demanded (000s tonnes)
€0.00	10
0.50	9
1.00	8
1.50	7
2.00	6
2.50	5
3.00	4
3.50	3
4.00	2
4.50	1
5.00	0

Market Demand versus Individual Demand

The demand curve in Figure 4.4 shows an individual's demand for a product. To analyse how markets work, we need to determine the *market demand*, which is the sum of all the individual demands for a particular good or service.

The table in Figure 4.5 shows the demand schedules for rape seed of two food manufacturers – Hanse and Michelle. At any price, Hanse's demand schedule tells us how many tonnes of rape seed he would be willing and able to buy at different prices, and Michelle's demand schedule tells us how many tonnes of rape seed she is willing and able to buy. The market demand at each price is the sum of the two individual demands.

The graph in Figure 4.5 shows the demand curves that correspond to these demand schedules. As we did with the market supply we sum the individual demand curves *horizontally* to obtain the market demand curve.

Because we are interested in analysing how markets work, we shall work most often with the market demand curve. The market demand curve shows how the total quantity demanded of a good varies as the price of the good varies, while all the other factors that affect how much consumers want to buy, such as incomes and taste, amongst other things, are held constant.

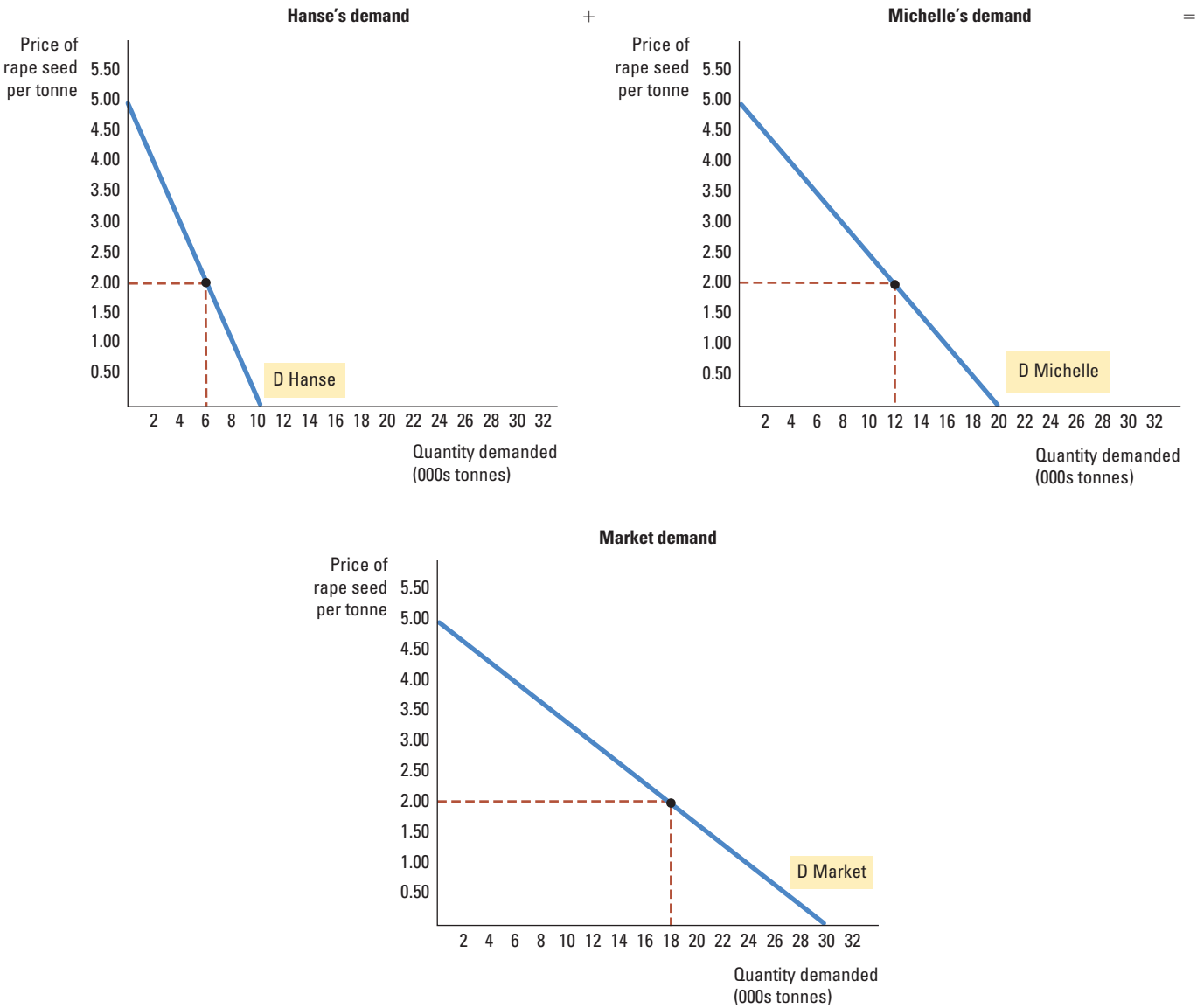
Shifts versus Movements along the Demand Curve

As with supply, we must ensure that we distinguish between a shift in the demand curve and a movement along the demand curve. A shift in the demand curve is caused by a factor affecting demand other than a change in price. The factors affecting demand are

FIGURE 4.5

Market Demand as the Sum of Individual Demands

The quantity demanded in a market is the sum of the quantities demanded by all the buyers at each price. Thus, the market demand curve is found by adding horizontally the individual demand curves. At a price of €2, Hanse would like to buy 6000 tonnes of rape seed but Michelle would be prepared to buy 12 000 tonnes of rape seed. The quantity demanded in the market at this price, therefore, is 18 000 tonnes of rape seed.



Price of rape seed per tonne	Hanse		Michelle		Market (000s tonnes)
€0.00	10	+	20	=	30
0.50	9		18		27
1.00	8		16		24
1.50	7		14		21
2.00	6		12		18
2.50	5		10		15
3.00	4		8		12
3.50	3		6		9
4.00	2		4		6
4.50	1		2		3
5.00	0		0		0

outlined below. If any of these factors change, then the amount consumers wish to purchase changes, whatever the price. The shift in the demand curve is referred to as an *increase or decrease in demand*. A movement along the demand curve occurs when there is a change in price. This may occur because of a change in supply conditions. The factors affecting demand are assumed to be held constant. A change in price leads to a movement along the demand curve and is referred to as a *change in quantity demanded*.

Movement Along the Demand Curve

We are going to briefly look at the economics behind a movement along the demand curve. Let us assume that the price of a particular variety of rape seed (Excalibur) falls, whilst all other rape seed varieties (Vision, Dimension, Vistive and Fashion) prices remain constant. We know that the fall in price will lead to an increase in quantity demanded. There are two reasons for this increase:

1. **The income effect.** If we assume that incomes remain constant then a fall in the price of Excalibur means that growers who buy this variety can now afford to buy more with their income. In other words, their *real income*, what a given amount of money can buy at any point in time, has increased and part of the increase in quantity demanded can be put down to this effect.
2. **The substitution effect.** Now that Excalibur is lower in price compared to other rape seed varieties, some growers will choose to substitute the more expensive varieties with the now cheaper Excalibur. This switch accounts for the remaining part of the increase in quantity demanded.

Shifts in the Demand Curve

The demand curve for rape seed shows how many tonnes of rape seed people are willing to buy at any given price, holding constant the many other factors beyond price that influence consumers' buying decisions. As a result, this demand curve need not be stable over time. If something happens to alter the demand at any given price, the demand curve shifts. For example, suppose European health authorities discovered that people who regularly use rape seed oil live longer, healthier lives. The discovery would raise the demand for rape seed. At any given price, buyers would now want to purchase a larger quantity of rape seed at all prices and the demand curve for rape seed would shift.

Figure 4.6 illustrates shifts in demand. Any change that increases the quantity demanded at every price, such as our imaginary discovery by the European health authorities, shifts the demand curve to the right and is called *an increase in demand*. Any change that reduces the demand at every price shifts the demand curve to the left and is called *a decrease in demand*.

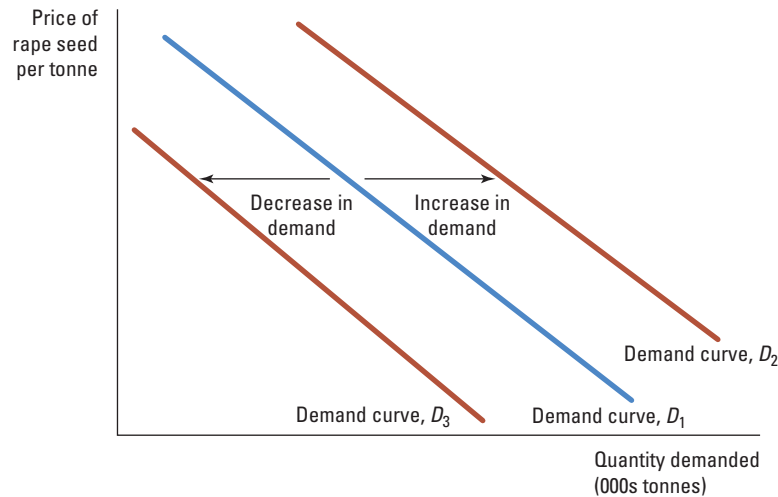
There are many variables that can shift the demand curve. Here are the most important.

normal good a good for which, other things being equal, an increase in income leads to an increase in demand (and vice versa)

Income What would happen to the demand for rape seed if unemployment increases? Most likely, it would fall (how much it would fall is another question and will be dealt with in the next chapter) because of lower incomes. Lower incomes mean that people have less to spend in total, so they are likely to spend less on some – and probably most – goods. If the demand for a good falls when income falls, the good is called a **normal good**.

FIGURE 4.6**Shifts in the Demand Curve**

Any change that raises the quantity that buyers wish to purchase at a given price shifts the demand curve to the right. Any change that lowers the quantity that buyers wish to purchase at a given price shifts the demand curve to the left.



Not all goods are normal goods. If the demand for a good rises when income falls, the good is called an **inferior good**. An example of an inferior good might be bus rides. As income falls, people are less likely to buy a car or take a taxi and more likely to take the bus. As income falls, therefore, demand for bus rides tends to increase. There is a wide range of rape seed oil on the market and we might see consumer behaviour changing when incomes change. More expensive, high-quality luxury oil might be a normal good, whereas oil that could be regarded as being 'standard' might see demand rise when incomes fall as consumers switch from the more expensive, luxury cooking oil to the cheaper, 'standard' oil.

inferior good a good for which, other things being equal, an increase in income leads to a decrease in demand (and vice versa)

Prices of Related Goods Suppose that the price of sunflower seed oil falls. The law of demand says that people will buy more sunflower seed oil. At the same time, people will probably buy less rape seed oil. Because sunflower seed oil and rape seed oil can both be used for cooking, they satisfy similar desires. When a fall in the price of one good reduces the demand for another good, the two goods are called **substitutes**. Substitutes are often pairs of goods that are used in place of each other, such as beef steak and Wiener schnitzel, pullovers and sweatshirts, and cinema tickets and DVD rentals. The more closely related substitute products are, the more effect we might see on demand if the price of one of the substitutes changes.

substitutes two goods for which an increase in the price of one leads to an increase in the demand for the other

Suppose that the price of woks fall. According to the law of demand, people will buy more woks. Yet, in this case, people will probably buy more rape seed oil as well, because rape seed oil and woks tend to be used together. When a fall in the price of one good raises the demand for another good, the two goods are called **complements**. Complements are often pairs of goods that are used together, such as petrol and cars, computers and software, bread and cheese, strawberries and cream, and bacon and eggs. As with substitutes not only do we need to identify what goods can be classed as complementary, we also need to be aware of the strength of the relationship between the two goods.

complements two goods for which an increase in the price of one leads to a decrease in the demand for the other (and vice versa)

Tastes The most obvious determinant of demand are tastes and fashions. If people like rape seed oil, they buy more of it. Economists are increasingly interested in understanding and explaining people's tastes. The developments in neuroscience mean that we now have an increasing understanding of why people make decisions and this has come into the realm of economics. This helps economists examine what happens, and why, when tastes change. This knowledge is also very important to businesses seeking to get a better understanding of their market, how consumer's behave and why they behave in the ways they do.

Expectations Buyer's expectations about the future may affect their demand for a good or service today. For example, if food manufacturers expect to earn higher revenues next month, they may be more willing to spend some of their current cash reserves buying rape seed oil. As another example, if buyers expect the price of rape seed to fall tomorrow, they may be less willing to buy rape seed at today's price.

The Size and Structure of the Population A larger population, other things being equal, will mean a higher demand for all goods and services. Changes in the way the population is structured also influences demand. Many European countries have an ageing population and this leads to a change in the demand. Goods and services required by the elderly increase in demand as a result. The demand for retirement homes, insurance policies suitable for elderly drivers and smaller cars may increase as a result.

Pitfall Prevention Many students confuse movements along and shifts in demand and supply curves. Using the correct phrasing (change in supply/demand refers to a shift in the curve, change in quantity supplied/demanded refers to a movement along the curve) is one way to help prevent this confusion.

Summary The demand curve shows what happens to the *quantity demanded* of a good when its price varies, holding constant all the other variables that influence buyers. When one or more of these other variables changes, the demand curve shifts leading to an *increase or decrease in demand*. Table 4.2 lists all the variables that influence how much consumers choose to buy of a good.

TABLE 4.2

Variables That Influence Buyers

This table lists the variables that affect how much consumers choose to buy of any good. Notice the special role that the price of the good plays: a change in the good's price represents a movement along the demand curve, whereas a change in one of the other variables shifts the demand curve.

Variable	A change in this variable ...
Price	Is represented as a movement along the demand curve
Income	Shifts the demand curve
Prices of related goods	Shifts the demand curve
Tastes	Shifts the demand curve
Expectations	Shifts the demand curve
Number of buyers	Shifts the demand curve

CASE STUDY

Harvests and Forward Markets

Farmers operate in a highly competitive market – there are many thousands of relatively small firms producing a product that is largely homogenous and farmers are price takers – they have little control over the price they get for their output.

Futures markets exist to enable farmers to buy and sell a specified quantity at an agreed price at some point in the future. Using the futures market can be a useful device for businesses looking to hedge themselves against volatile prices. Farming is a particularly notorious industry for such volatile prices, dependent as it is on climate to determine the size and quality of the harvest. In recent years high temperatures in Russia, disease in Asia, dry weather in Europe and wet weather in Canada have combined to put pressure on the supply of grain, with prices for UK wheat changing dramatically over different periods.

In January 2010, for example, the price of British wheat was around £105 per tonne but by August 2010 it had risen to nearly £160 per tonne, a rise of just over 52 per cent. The price of barley also rose sharply from around £75 per tonne in mid-June 2010 to about £175 per tonne in August of the same year. In early 2012, wheat prices stood at around £146 per tonne. By comparison, in 2009 wheat was trading at around £90 per tonne. It can be seen from this limited example just how volatile prices can be over periods of time and how, as a result, many farmers find it difficult to be able to forecast their revenues with any certainty. High prices in 2012, for example, might benefit some arable farmers who are able to harvest high quality products; for livestock farmers the situation is reversed. High prices of grain mean that their input costs rise because of the role grain plays in livestock feed.

It is precisely this volatility which leads some farmers to ‘sell’ their output on the forward markets. There is a risk involved in doing this, however. If wheat prices remain relatively high in the period after contracts are agreed, then UK farmers harvesting at that time may well benefit from selling their output at much higher prices, but some will not do so because they entered into contracts to sell their output at agreed prices some time ago. If farmers make a decision to ‘sell’ their crops at prices around £90–£100 per tonne it may, at the time, seem like a good price, but if actual prices at the time of sale are nearer the £146 per tonne mark then these farmers might be wishing they had hung on and sold their harvest later.

On the flip side, farmers who do lose out this way may decide to enter into forward agreements for next year’s crop and if the prices they can get are based around current prices they may benefit in a year’s time. Who knows – next year may be a bumper harvest and world grain prices fall sharply as a result. This just highlights the difficult decisions that farmers have to make when they are price takers and operate in volatile markets in which they have limited control over supply, even if they have specific and highly organized markets to help reduce risk.



FOTOKOSTICSHUTTERSTOCK

Stocks can be important in determining how prices on markets for agricultural products change over a period of time.

Quick Quiz Make up an example of a demand schedule for pizza, and graph the implied demand curve • Give an example of something that would shift this demand curve • Would a change in the price of pizza shift this demand curve?

DEMAND AND SUPPLY TOGETHER

Having analysed supply and demand separately, we now combine them to see how they determine the quantity of a good sold in a market and its price.

Equilibrium

equilibrium a situation in which the price has reached the level where quantity supplied equals quantity demanded

equilibrium price the price that balances quantity supplied and quantity demanded

equilibrium quantity the quantity supplied and the quantity demanded at the equilibrium price

surplus a situation in which quantity supplied is greater than quantity demanded

Figure 4.7 shows the market supply curve and market demand curve together. Equilibrium is defined as a state of rest, a point where there is no force acting for change. Economists refer to supply and demand as being *market forces*. In any market the relationship between supply and demand exerts force on price. If supply is greater than demand or vice versa, then there is pressure on price to change. Notice, however, that there is one point at which the supply and demand curves intersect. This point is called the market's **equilibrium**. The price at this intersection is called the **equilibrium price**, and the quantity is called the **equilibrium quantity**. Here the equilibrium price is €2.00 per tonne, and the equilibrium quantity is 7000 tonnes bought and sold.

At the equilibrium price, the quantity of the good that buyers are willing and able to buy exactly balances the quantity that sellers are willing and able to sell. The equilibrium price is sometimes called the *market-clearing price* because, at this price, everyone in the market has been satisfied: buyers have bought all they want to buy, and sellers have sold all they want to sell – there is neither a shortage nor a surplus.

The actions of buyers and sellers naturally move markets towards the equilibrium of supply and demand. To see why, consider what happens when the market price is not equal to the equilibrium price.

Suppose first that the market price is above the equilibrium price, as in panel (a) of Figure 4.8. At a price of €2.50 per tonne, the quantity suppliers would like to sell at this price (10000 tonnes) exceeds the quantity which buyers are willing to purchase (4000 tonnes). There is a **surplus** of the good: suppliers are unable to sell all they want at the going price. A surplus is sometimes called a situation of *excess supply*. When there is a surplus in the rape seed market, sellers of rape seed find they cannot sell all the supplies

FIGURE 4.7

The Equilibrium of Supply and Demand

The equilibrium is found where the supply and demand curves intersect. At the equilibrium price, the quantity supplied equals the quantity demanded. Here the equilibrium price is €2: at this price, 7000 tonnes of rape seed are supplied and 7000 tonnes are demanded.

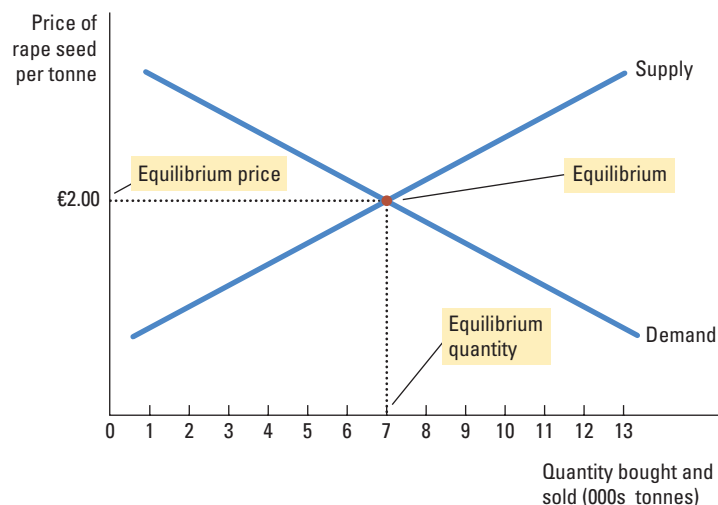
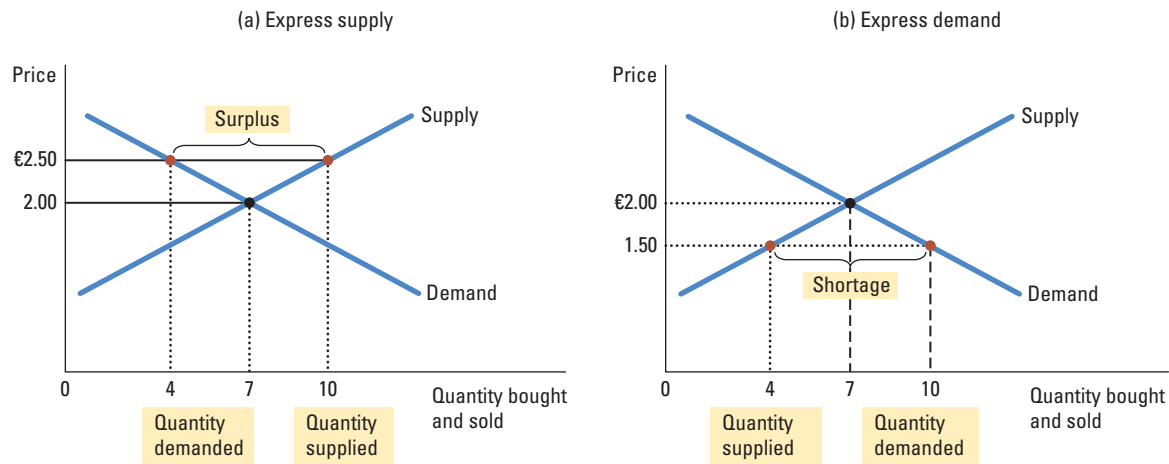


FIGURE 4.8**Markets Not in Equilibrium**

In panel (a), there is a surplus. Because the market price of €2.50 is above the equilibrium price, the quantity supplied (10000 tonnes) exceeds the quantity demanded (4000 tonnes). Suppliers try to increase sales by cutting the price of rape seed, and this moves the price toward its equilibrium level. In panel (b), there is a shortage. Because the market price of €1.50 is below the equilibrium price, the quantity demanded (10000 tonnes) exceeds the quantity supplied (4000 tonnes). With too many buyers chasing too few goods, suppliers can take advantage of the shortage by raising the price. Hence, in both cases, the price adjustment moves the market towards the equilibrium of supply and demand.



they have and so the market responds to the surplus by cutting prices. Falling prices, in turn, increase the quantity demanded and decrease the quantity supplied. Prices continue to fall until the market reaches the equilibrium.

Suppose now that the market price is below the equilibrium price, as in panel (b) of Figure 4.8. In this case, the price is €1.50 per tonne, and the quantity of the good demanded exceeds the quantity supplied. There is a **shortage** of the good: demanders are unable to buy all they want at the going price. A shortage is sometimes called a situation of *excess demand*. When a shortage occurs in the rape seed market, buyers may find they cannot acquire all the supplies they need. With too many buyers chasing too few goods, the suppliers respond to the shortage by raising prices without losing sales. As the price rises, quantity demanded falls, quantity supplied rises and the market once again moves toward the equilibrium.

Thus, the activities of the many buyers and sellers automatically push the market price towards the equilibrium price. Once the market reaches its equilibrium, all buyers and sellers are satisfied, and there is no upward or downward pressure on the price. How quickly equilibrium is reached varies from market to market, depending on how quickly prices adjust. In most free markets under an assumption of high levels of information available to buyers and sellers, surpluses and shortages are only temporary because prices eventually move towards their equilibrium levels (we will see the significance of the word 'free' later in the book). Indeed, this phenomenon is so pervasive that it is called the **law of supply and demand**: the price of any good adjusts to bring the quantity supplied and quantity demanded for that good into balance.

shortage a situation in which quantity demanded is greater than quantity supplied

law of supply and demand the claim that the price of any good adjusts to bring the quantity supplied and the quantity demanded for that good into balance

JEOPARDY PROBLEM

The market for bicycles has seen falling prices but not a change in the amount of bicycles bought and sold. Explain how this situation might have come about. Use diagrams to illustrate.

FYI



This FYI will be helpful if you have to use maths in your course. If you do not have to use maths then you can safely move on to the next section without affecting your overall understanding of this chapter.

Functions

In economics a lot of use is made of functions. Demand and supply equations are two examples of functions. Typically, functions are expressed as:

$$Y = f(x)$$

or simply $f(x)$

This means that the value of Y is dependent on the value of the terms in the bracket – in our example above there is only one value, x , so the value of Y is dependent on the value of x .

We know from this chapter that there are a number of factors affecting demand and supply. The general form of the function in such a case would look like:

$$Y = f(x_1, \dots, x_n)$$

where $x_1, \dots, x_n$ represents a range of variables.

Given the determinants of demand and supply we could write the demand and supply functions as:

$$D = f(P_n, P_{n-1}, \dots, P_{n-1}, Y, T, P, A, E)$$

Where:

- P_n = Price
- $P_{n-1}, \dots, P_{n-1}$ = Prices of other goods – substitutes and complements
- Y = Incomes – the level and distribution of income
- T = Tastes and fashions
- P = The level and structure of the population
- A = Advertising
- E = Expectations of consumers

and:

$$S = f(P_n, P_{n-1}, \dots, P_{n-1}, H, N, F_1, \dots, F_m, E, S_f)$$

Where:

- P_n = Price
- $P_{n-1}, \dots, P_{n-1}$ = Profitability of other goods in production and prices of goods in joint supply
- H = Technology
- N = Natural shocks
- $F_1, \dots, F_m$ = Costs of production
- E = Expectations of producers
- S_f = Social factors

Linear Equations

Both demand and supply can be represented as linear equations and be drawn as straight line graphs.

A linear equation normally looks like:

$$y = a + bx$$

In this equation, y = the value plotted on the vertical axis (the dependent variable)

x is the value on the horizontal axis (the independent variable)

a is a constant and b is the slope of the line or its gradient.

Remember that demand looks at the relationship between price and the quantity demanded and supply is the relationship between price and the quantity supplied.

In both cases, the quantity demanded and supplied are dependent on the price. So, price is the independent variable and the quantity the dependent variable.

At this point we hit a bit of a snag.

Students of pure maths will notice that in economics, supply and demand graphs are the wrong way round – normally, the vertical Y axis represents the dependent variable and the X axis the independent variable. In supply and demand graphs, price, the independent variable, is drawn on the Y axis and quantity demanded and supplied, the dependent variable, on the X axis.

The switch is attributed to Alfred Marshall (1842–1924) who developed supply and demand analysis in the latter part of the 19th century. It is important, therefore, to remember which is the dependent variable and which the independent variable as we progress through the analysis.

Applying the relationship between price and quantity demanded and supplied we get typical equations such as:

$$Q_d = 2100 - 2.5p$$

$$Q_s = -10 + 6p$$

In the case of the demand curve the minus sign in front of the price variable tells us that there is a negative relationship between price and quantity demanded whereas the plus sign in front of the price in the supply equation tells us that there is a positive relationship between price and quantity supplied.

You may also see demand and supply equations which look like:

$$P = 840 - 0.4Q_d \text{ or}$$

$$P = -120 + 0.8Q_s$$

The equation $P = 840 - 0.4Q_d$ is just the inverse of the demand equation $Q_d = 2100 - 2.5p$. We found this by adopting the following method:

$$Q_d = 2100 - 2.5p$$

$$Q_d + 2.5p = 2100$$

$$2.5p = 2100 - Q_d$$

$$\frac{2.5p}{2.5} = \frac{2100 - Q_d}{2.5}$$

$$P = 840 - 0.4 Q_d$$

Remember, however, that because of the switching of supply and demand curves as noted above, you can get some odd looking equations which do not fit with the graphical analysis.

The important thing to remember when manipulating linear equations of

this sort is that whatever you do to one side of the equation (multiply, add, divide or subtract a number or element) you must do the same thing to the other side.

Finding Price and Quantity

If we take the original two equations:

$$Q_d = 2100 - 2.5p$$

$$Q_s = -10 + 6p$$

We can dissect them in a bit more detail in relation to the standard $y = a + bx$ linear equation we first introduced.

In our equations, the quantity demanded and supplied are one of the variables in the equation. In this case they are the dependent variable. Their value depends upon the price – the independent variable.

In the case of the demand curve, the quantity demanded will be 2100 minus 2.5 times whatever the price is. If price is €6 then quantity demanded will be $2100 - 2.5(6) = 2085$. If price is €16 then quantity demanded will be $2100 - 2.5(16) = 2060$.

Looking at supply, if the price were €8 then the quantity supplied would be $-10 + 6(8) = 38$ and if price were €16 then quantity supplied would be $-10 + 6(16) = 86$.

If we used the other two equations we looked at:

$$P = 840 - 0.4Q_d$$

Or:

$$P = -120 + 0.8Q_s$$

Then we can arrive at values for P or Q assuming we have at least one of these two variables.

For demand, if $p = €6$ then the quantity demanded would be:

$$P = 840 - 0.4Q_d$$

$$0.4Q_d = 840 - 6$$

$$\frac{0.4Q_d}{0.4} = \frac{834}{0.4}$$

$$Q_d = 2085$$

In the case of supply, if price = €8:

$$P = -120 + 0.8Q_s$$

$$8 = -120 + 0.8Q_s$$

$$\frac{8}{0.8} = \frac{-120 + 0.8Q_s}{0.8}$$

$$10 = -150 + Q_s$$

$$10 + 150 = Q_s$$

$$Q_s = 160$$

Finding Market Equilibrium The substitution method

We know that in equilibrium, demand equals supply ($D = S$).

To find the market equilibrium, therefore, we set the demand and supply equations equal to each other and solve for P and Q.

Take the following demand and supply equations:

$$Q_d = 32 - 3p$$

$$Q_s = 20 + 4p$$

We know that in equilibrium:

$$Q_d = Q_s$$

So, equilibrium in this market will be where:

$$32 - 3p = 20 + 4p$$

This now allows us to solve for P and so find the equilibrium price:

$$32 - 3p = 20 + 4p$$

Subtract 20 from both sides and add 3p to both sides to get:

$$32 - 20 = 4p + 3p$$

$$12 = 7p$$

$$P = €1.71 \text{ (rounded to the nearest whole cent)}$$

We can now substitute the equilibrium price into our two equations to find the equilibrium quantity rounded to the nearest whole number:

$$Q_d = 32 - 3p$$

$$Q_d = 32 - 3(1.71)$$

$$Q_d = 32 - 5.13$$

$$Q_d = 26.87$$

$$Q_d = 27$$

$$Q_s = 20 + 4p$$

$$Q_s = 20 + 4(1.71)$$

$$Q_s = 20 + 6.84$$

$$Q_s = 26.84$$

$$Q_s = 27$$

Note the figures for Q_d and Q_s before rounding differ slightly because we had to round the price. Now look at this example:

$$P = 3 + 0.25Q_s$$

$$P = 15 - 0.75Q_d$$

In this case the equations are defined in terms of price but the principle of working out equilibrium is the same as we have used above.

First, set the two equations equal to each other:

$$3 + 0.25Q_s = 15 - 0.75Q_d$$

Then solve for Q:

Add $0.75Q_d$ to both sides and then subtract 3 from both sides to get:

$$0.75Q_d + 0.25Q_s = 15 - 3$$

$$Q = 12$$

Substitute $Q = 12$ into one of the equations to find P.

$$P = 3 + 0.25Q_s$$

$$P = 3 + 0.25(12)$$

$$P = 6$$

To check, also substitute into the demand equation:

$$P = 15 - 0.75Q_d$$

$$P = 15 - 0.75(12)$$

$$P = 15 - 9$$

$$P = 6.$$

There is another way to find both the quantity and the price and that is through adopting the approach of solving simultaneous equations. Simultaneous equations require us to find two or more unknowns. In our case it is two unknowns, price and quantity.

The elimination method

Look at the following two equations:

$$Q_d = 20 - 2p$$

$$Q_s = 2 + 2p$$

In this case, the terms are all neatly aligned above each other so it is a

relatively simple task to add the two together. Note that we are trying to find equilibrium so $Q_d = Q_s$ so the value of Q is the same. Adding the two together we get:

$$Q_d = 20 - 2p$$

$$Q_s = 2 + 2p$$

$$2Q = 22$$

$$Q = 11$$

Notice that in the above equations we have a very convenient fact that the coefficient of p in each case is the same but with opposite signs. This makes this example very easy to eliminate p to isolate the Q value. This is not always the case, however, but it is important to remember that having two equal values with opposite signs allows us to get rid of them! We will come back to this later.

We can now use the fact that we know Q to find the equilibrium price by substituting Q into one of the equations thus:

$$Q_d = 20 - 2p$$

$$11 = 20 - 2p$$

$$2p = 20 - 11$$

$$2p = 9$$

$$P = 4.5$$

It is always worth checking your answer to make sure you have made no mistakes along the way so in this case we will substitute our known value of Q into the second equation to check we get the same answer ($p = 4.5$). So:

$$Q_s = 2 + 2p$$

$$11 = 2 + 2p$$

$$11 - 2 = 2p$$

$$9 = 2p$$

$$P = 4.5$$

Sometimes we may have equations where the p and Q values are both on the same side of the equation. In this case we have to use a different technique – the elimination method.

Take the following two equations:

$$-3p + 4Q = 5 \quad (1)$$

$$2p - 5Q = -15 \quad (2)$$

We have labelled these two equations (1) and (2) to allow us to keep track of what we are doing and reduce the risk of making an error.

Remember above when we noted the fact that having a nice convenient equation where the coefficient was equal but the signs opposite enabling us to be able to eliminate one of the values to help solve the equation for the other unknown? That is what we need to do with these two equations. We have to choose to manipulate the two equations to make either the ' p ' terms or the ' Q ' terms have the same coefficient but opposite signs. A knowledge of factors and lowest common denominators is useful here!

In this example we are going to manipulate the equations to get rid of the ' p ' terms. This allows us to isolate the ' Q ' terms and thus solve for Q and then find p .

This is how we do this:

$$-3p + 4Q = 5 \quad (1)$$

$$2p - 5Q = -15 \quad (2)$$

To eliminate p , multiply (1) by 2 and (2) by 3

$$-6p + 8Q = 10 \quad (3)$$

$$6p - 15Q = -45 \quad (4)$$

Add together (3) and (4)

$$-6p + 8Q = 10 \quad (3)$$

$$6p - 15Q = -45 \quad (4)$$

$$-7Q = -35$$

Divide both sides by -7

$$Q = 5$$

We can now substitute Q into equations (1) and (2) to find (and check) p

If $Q = 5$ then:

$$-3p + 4(5) = 5$$

$$-3p + 20 = 5$$

$$20 - 5 = 3p$$

$$15 = 3p$$

$$P = 5$$

$$2p - 5(5) = -15$$

$$2p - 25 = -15$$

$$2p = -15 + 25$$

$$2p = 10$$

$$p = 5$$

In this case the equilibrium price is £5 and the equilibrium quantity is 5.

Three Steps to Analysing Changes in Equilibrium

So far we have seen how supply and demand together determine a market's equilibrium, which in turn determines the price of the good and the amount of the good that buyers purchase and sellers produce. Markets are dynamic – demand and supply change all the time and in some markets these changes may be almost every second of every day – in foreign exchange markets, for example. The equilibrium price and quantity depend on the position of the supply and demand curves. When some event shifts one (or both) of these curves, the equilibrium in the market changes. The analysis of such a change is called *comparative statics* because it involves comparing two unchanging situations – an initial and a new equilibrium.

FYI



Prices as Signals

Our analysis so far has only brushed the surface of the way markets operate. Economists have conducted extensive research into the nature and determinants of both demand and supply. It is beyond the scope of this book to go into too much detail on these issues but it is useful to have a little bit of background knowledge on this to help understand markets more effectively.

At the heart of research into demand and supply is why buyers and sellers behave as they do. The development of magnetic resonance imaging (MRI) techniques has allowed researchers to investigate how the brain responds to different stimuli when making purchasing decisions (referred to as *neuroeconomics*). As time goes by our understanding of buyer and seller behaviour will improve and theories will have to be adapted to accommodate this new understanding.

However, much of the theory behind how markets work relies on the assumption of rational behaviour, defined in terms of humans preferring more to less and taking into account information prior to making a decision. The main function of price in a free market is to act as a signal to both buyers and sellers to help in decision making.

For buyers, price tells them something about what they have to give up to acquire the benefits that having the good will confer on them. These benefits are referred to as the *utility* (satisfaction) derived from consumption. If I am willing to pay €10 to go and watch a movie then economists will assume that the value of the benefits I gain from watching the movie is greater than the next best alternative – what else I could have spent my €10 on. Principles 1 and 2 of the *Ten Principles of Economics* state that people face trade-offs and that the cost of something is what you have to give up to acquire it. This is fundamental to the law of demand. At higher prices, the sacrifice being made in terms of the value of the benefits gained from alternatives is greater and so we may be less willing to do so as a result. If the price of a ticket for the movie was €20 (other things being equal) then it might have to be a very good movie to persuade us that giving up what else €20 could buy is worth it.

For sellers price acts as a signal in relation to the profitability of production. For most sellers, increasing the amount of a good produced will incur some additional input costs. A higher price is required in order to compensate for the additional cost and to also enable the

producer to gain some reward from the risk they are taking in production. That reward is termed profit.

If prices are rising in a free market then this acts as a different but related signal to buyers and sellers. Rising prices to a seller means that there is a shortage and thus there is an incentive to expand production because the seller knows that she will be able to sell what she produces. For buyers, a rising price changes the nature of the trade-off they have to face. They will now have to give up more in order to acquire the good and they will have to decide whether the value of the benefits they will gain from acquiring the good is worth the extra price they have to pay.

What we do know is that for both buyers and sellers, there are many complex processes that occur in decision making. Whilst we do not fully understand all these processes yet, economists are constantly searching for new insights that might help them understand the workings of markets more fully. All of us go through these complex processes every time we make a purchasing decision – although we may not realize it! Having some appreciation of these processes is fundamental to thinking like an economist.

When analysing how some event affects a market, we proceed in three steps. First, we decide whether the event shifts the supply curve, the demand curve or, in some cases, both curves. Secondly, we decide whether the curve shifts to the right or to the left. Thirdly, we use the supply and demand diagram to compare the initial and the new equilibrium, which shows how the shift affects the equilibrium price and quantity. It is important in the analysis that the process by which equilibrium changes is understood and that the changes involved are not instantaneous (although some schools of thought do refer to instantaneous changes in markets) – some markets will take longer to adjust to changes than others. Table 4.3 summarizes the three steps. To see how this recipe is used, let's consider various events that might affect the market for rape seed.

Example: A Change in Demand Suppose that a government-sponsored research project finds that using rape seed oil helps reduce the risk of heart disease and

TABLE 4.3**A Three-Step Programme for Analysing Changes in Equilibrium**

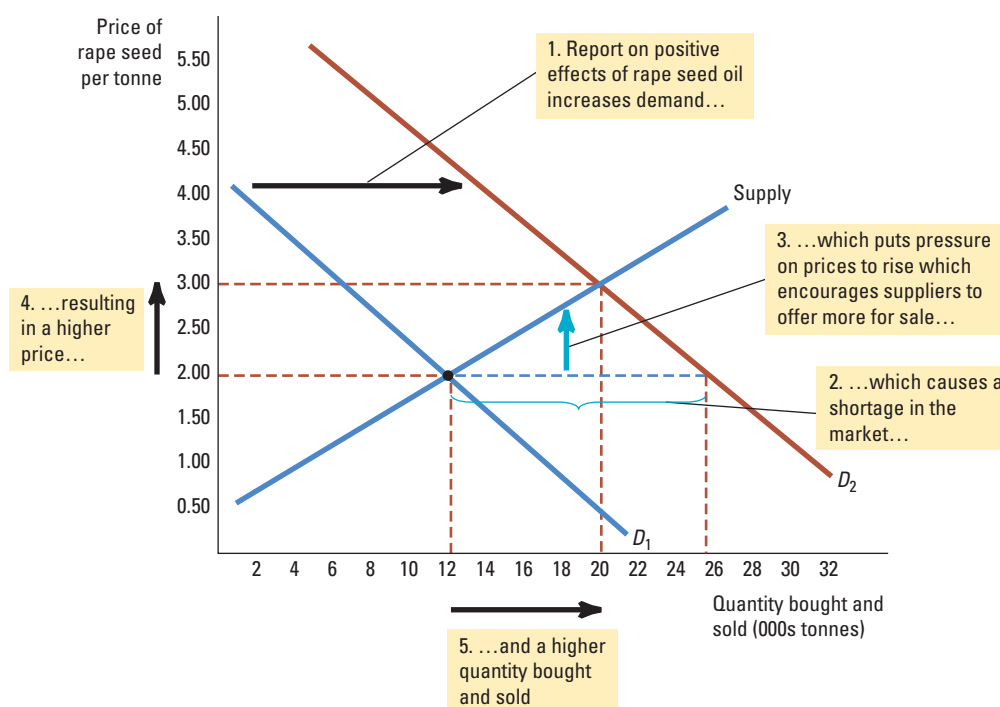
1. Decide whether the event shifts the supply or demand curve (or perhaps both).
2. Decide in which direction the curve shifts.
3. Use the supply and demand diagram to see how the shift changes the equilibrium price and quantity.

strokes. How does this event affect the market for rape seed? To answer this question, let's follow our three steps.

1. The news has a direct effect on the demand curve by changing people's taste for rape seed oil. That is, the report changes the amount of rape seed oil that people want to buy at any given price.
2. Because the report incentivizes people to want to use more rape seed oil, the demand curve shifts to the right. Figure 4.9 shows this increase in demand as the shift in the demand curve from D_1 to D_2 . This shift indicates that the quantity of rape seed oil demanded is higher at every price. The shift in demand has led to a shortage of rape seed oil in the market. At a price of €2.00 buyers now want to buy 26 000 tonnes of rape seed but sellers are only offering 12 000 tonnes for sale at this price. There is a shortage of 14 000 tonnes.

FIGURE 4.9**How an Increase in Demand Affects the Equilibrium**

An event that raises quantity demanded at any given price shifts the demand curve to the right. The equilibrium price and the equilibrium quantity both rise. Here, the report linking using rape seed oil to reductions in risk of ill health causes buyers to demand more rape seed. The demand curve shifts from D_1 to D_2 , which causes the equilibrium price to rise from €2.00 to €3.00 and the equilibrium quantity to rise from 12 000 to 20 000 tonnes.



3. As Figure 4.9 shows, the shortage starts to force up prices and encourages growers to plant more rape. The additional production incurs extra costs and so a higher price is required to compensate sellers. This raises the equilibrium price from €2.00 to €3.00 and the equilibrium quantity from 12 000 to 20 000 tonnes. In other words, the report increases the price of rape seed and the quantity of rape seed bought and sold.

Shifts in Curves versus Movements along Curves – a Reminder

When the report drives up the price of rape seed, the quantity of rape seed that growers supply rises, but the supply curve remains in the same position. Economists say there has been an increase in ‘quantity supplied’ but no change in ‘supply’.

‘Supply’ refers to the position of the supply curve, whereas the ‘quantity supplied’ refers to the amount suppliers wish to sell at different prices. In this example, we assumed supply does not change. Instead, the report alters consumers’ desire to buy at any given price and thereby shifts the demand curve. The increase in demand creates a shortage. The shortage means there are more buyers looking to purchase rape seed than there are sellers willing to sell. As a result, price starts to creep up as buyers are prepared to pay higher prices to get products. When the price rises growers are willing to offer more rape seed for sale and so the quantity supplied rises. This increase in quantity supplied is represented by the movement along the supply curve. The shortage will continue to be competed away by price rising until supply and demand are once again brought into equilibrium. The final result will be a rise in equilibrium price and in the equilibrium amount bought and sold.

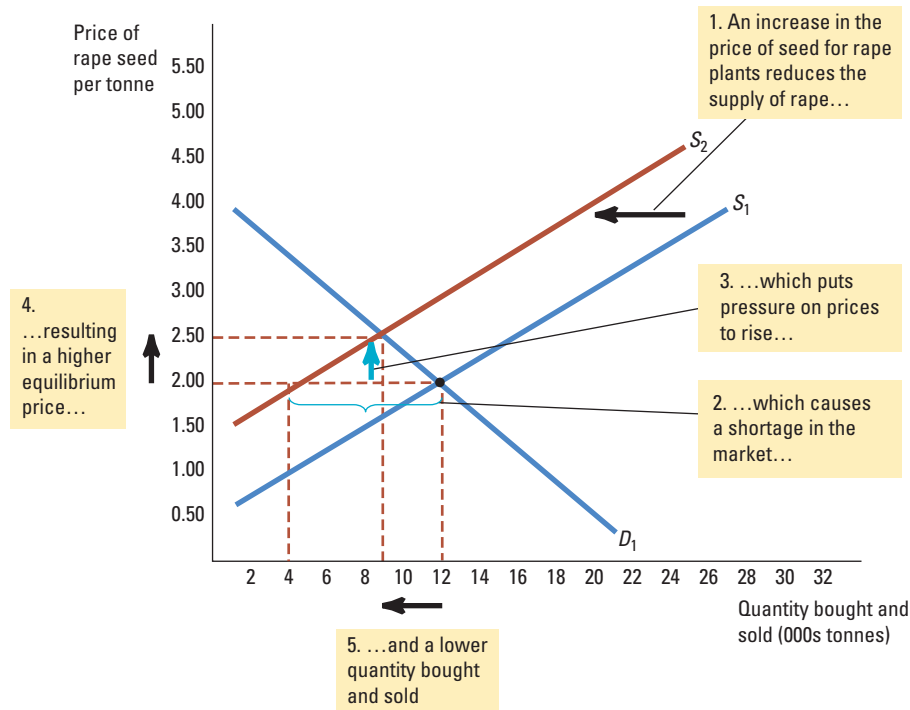
To summarize, a shift *in* the supply curve is called a ‘change in supply’, and a shift *in* the demand curve is called a ‘change in demand’. A movement *along* a fixed supply curve is called a ‘change in the quantity supplied’, and a movement *along* a fixed demand curve is called a ‘change in the quantity demanded’.

Example: A Change in Supply Suppose that, during another summer, bad weather destroys part of the seed crop for rape and drives up the world price of rape seed for planting. How does this event affect the market for rape seed? Once again, to answer this question, we follow our three steps.

1. The change in the price of seed, an input needed to grow rape, affects the supply curve. By raising the costs of production, it reduces the amount of rape seed that firms produce and sell at any given price. The demand curve does not change because the higher cost of inputs does not directly affect the amount of rape seed buyers wish to buy.
2. The supply curve shifts to the left because, at every price, the total amount that firms are willing and able to sell is reduced. Figure 4.10 illustrates this decrease in supply as a shift in the supply curve from S_1 to S_2 . At a price of €2.00 sellers are now only able to offer 4000 tonnes of rape seed for sale but demand is still 12 000 tonnes of rape seed. The shift in supply to the left has created a shortage in the market. Once again, the shortage will create pressure on price to rise as buyers look to purchase rape seed.
3. As Figure 4.10 shows, the shortage raises the equilibrium price from €2.00 to €2.50 and lowers the equilibrium quantity from 12 000 to 9000 tonnes. As a result of the increase in seed for growing, the price of rape seed rises, and the quantity of rape seed bought and sold falls.

FIGURE 4.10**How a Decrease in Supply Affects the Equilibrium**

An event that reduces supply at any given price shifts the supply curve to the left. The equilibrium price rises, and the equilibrium quantity falls. Here, an increase in the price of seed for growing rape plants (an input) causes sellers to supply less rape seed. The supply curve shifts from S_1 to S_2 , which causes the equilibrium price of rape seed to rise from €2.00 to €2.50 and the equilibrium quantity to fall from 12000 to 9000 tonnes.

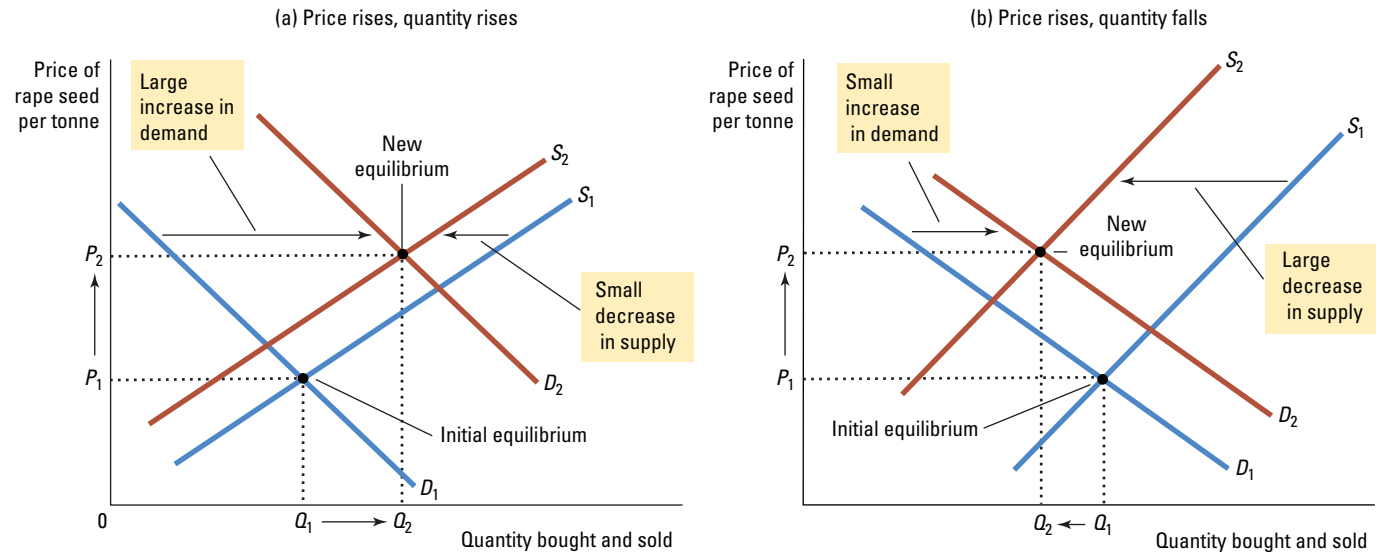


Example: A Change in Both Supply and Demand (i) Now suppose that the report and the bad weather occur during the same summer. To analyse this combination of events, we again follow our three steps.

1. We determine that both curves must shift. The report affects the demand curve because it alters the amount of rape seed that buyers want to buy at any given price. At the same time, when the bad weather drives up the price of seed for growing rape, it alters the supply curve for rape seed because it changes the amount of rape seed that firms want to sell at any given price.
2. The curves shift in the same directions as they did in our previous analysis: the demand curve shifts to the right, and the supply curve shifts to the left. Figure 4.11 illustrates these shifts.
3. As Figure 4.11 shows, there are two possible outcomes that might result, depending on the relative size of the demand and supply shifts. In both cases, the equilibrium price rises. In panel (a), where demand increases substantially while supply falls just a little, the equilibrium quantity also rises. By contrast, in panel (b), where supply falls substantially while demand rises just a little, the equilibrium quantity falls. Thus, these events certainly raise the price of rape seed, but their impact on the amount of rape seed bought and sold is ambiguous (that is, it could go either way).

FIGURE 4.11**A Shift in Both Supply and Demand (i)**

Here we observe a simultaneous increase in demand and decrease in supply. Two outcomes are possible. In panel (a), the equilibrium price rises from P_1 to P_2 , and the equilibrium quantity rises from Q_1 to Q_2 . In panel (b), the equilibrium price again rises from P_1 to P_2 , but the equilibrium quantity falls from Q_1 to Q_2 .

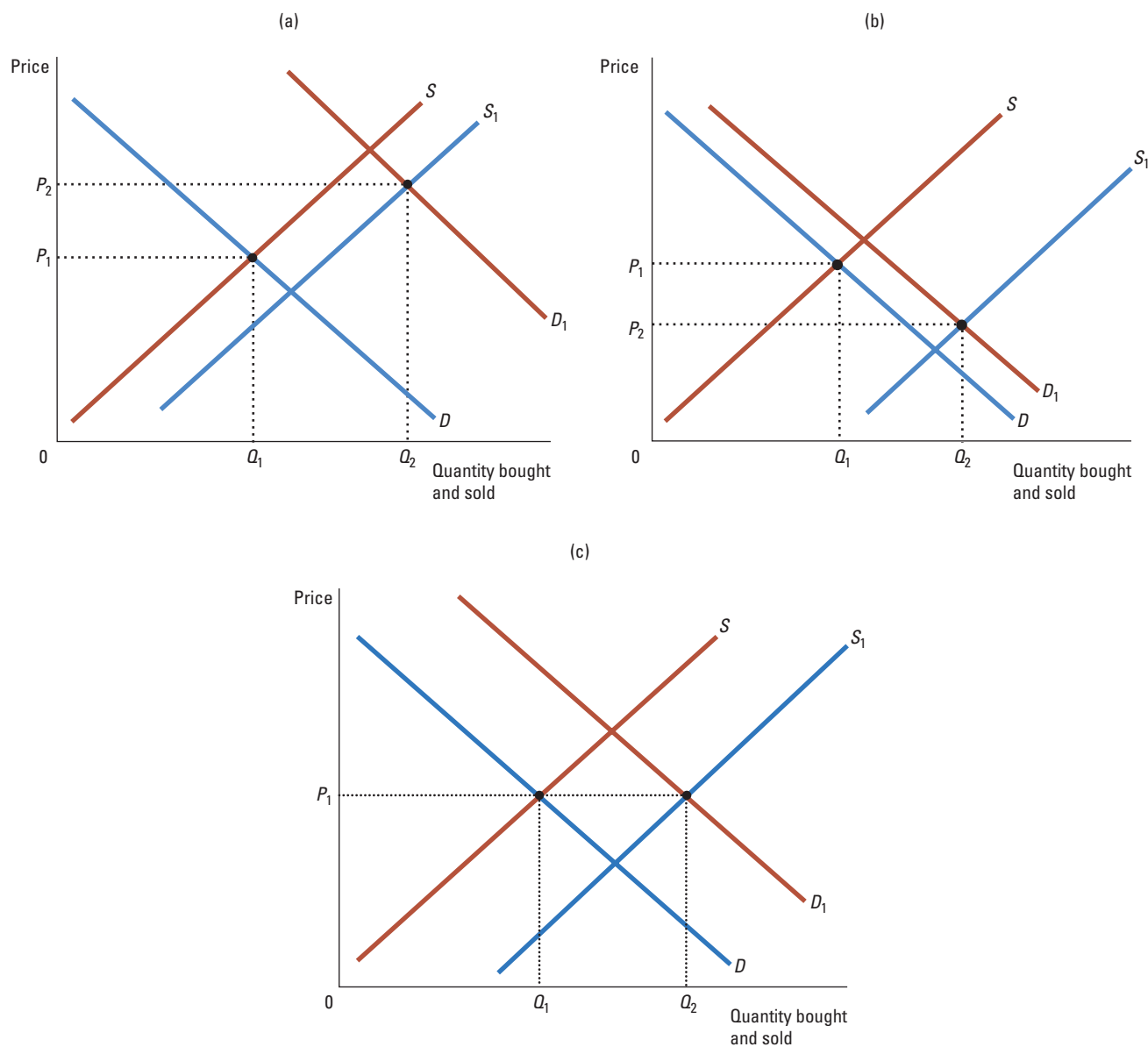


Example: A Change in Both Supply and Demand (ii) We are now going to look at a slightly different scenario but with both supply and demand increasing together. Assume that the details of the report have been leaked prior to official publication and the findings publicized on TV news stations. We know that the report is likely to increase demand for rape seed and so the demand curve will shift to the right. However, sellers' expectations that sales of rape seed will increase as a result of the forecasts mean that they take steps to expand production of rape seed. This would lead to a shift of the supply curve to the right – more rape seed is now offered for sale at every price. To analyse this particular combination of events, we again follow our three steps.

1. We determine that both curves must shift. The report affects the demand curve because it alters the amount of rape seed that buyers want to buy at any given price. At the same time, the expectations of producers alter the supply curve for rape seed because they change the amount of rape seed that firms want to sell at any given price.
2. Both demand and supply curves shift to the right: Figure 4.12 illustrates these shifts.
3. As Figure 4.12 shows, there are three possible outcomes that might result, depending on the relative size of the demand and supply shifts. In panel (a), where demand increases substantially while supply rises just a little, the equilibrium price and quantity rises. By contrast, in panel (b), where supply rises substantially while demand rises just a little, the equilibrium price falls but the equilibrium quantity rises. In panel (c) the increase in demand and supply are identical and so equilibrium price does not change. Equilibrium quantity will increase, however. Thus, these events have different effects on the price of rape seed although the amount of rape seed bought and sold in each case is higher. In this instance the effect on price is ambiguous.

FIGURE 4.12**A Shift in Both Supply and Demand (ii)**

Here, again, we observe a simultaneous increase in demand and supply. Three outcomes are possible. In panel (a) the equilibrium price rises from P_1 to P_2 and the equilibrium quantity rises from Q_1 to Q_2 . In panel (b), the equilibrium price falls from P_1 to P_2 but the equilibrium quantity rises from Q_1 to Q_2 . In panel (c), there is no change to the equilibrium price but the equilibrium quantity rises from Q_1 to Q_2 .



what if... Malmö FF, a Swedish football team, is taken over by a rich Middle East backer. The team is strengthened and produces a series of excellent results which propel them to the top of the Swedish Allsvenskan. Demand for tickets to the 24 000 capacity stadium increases dramatically. Draw a supply and demand diagram to illustrate this market situation. Would the club increase ticket prices? If so, what might fans' reaction be? If the club does not increase prices why might it make this decision? What do you think would be the practical effect on fans and

the prices they have to pay for tickets if prices were held at the original level? Assuming the improvement in the club's fortunes continues for several seasons, what might the long-term market situation look like?



The Swedbank Stadion, home of Malmö football club.

Summary

We have just seen four examples of how to use supply and demand curves to analyse a change in equilibrium. Whenever an event shifts the supply curve, the demand curve, or perhaps both curves, you can use these tools to predict how the event will alter the amount bought and sold in equilibrium and the price at which the good is bought and sold. Table 4.4 shows the predicted outcome for any combination of shifts in the two curves. To make sure you understand how to use the tools of supply and demand, pick a few entries in this table and make sure you can explain to yourself why the table contains the prediction it does.

TABLE 4.4

What Happens to Price and Quantity When Supply or Demand Shifts?

As a quick quiz, make sure you can explain each of the entries in this table using a supply and demand diagram.

	No change in supply	An increase in supply	A decrease in supply
No change in demand	P same Q same	P down Q up	P up Q down
An increase in demand	P up Q up	P ambiguous Q up	P up Q ambiguous
A decrease in demand	P down Q down	P down Q ambiguous	P ambiguous Q down

ELASTICITY AND ITS APPLICATION

For businesses, the price they charge for the products they produce is a vital part of their product positioning – what the product offering is in relation to competitors. We have seen in Chapter 4 how markets are dynamic and that price acts as a signal to both sellers and buyers; when prices change, the signal is altered and producer and consumer behaviour changes.

Imagine yourself as a producer of silicon chips for use in personal computers, laptops and a variety of other electronic devices. Because you earn all your income from selling silicon chips, you devote much effort to making your factory as productive as it can be. You monitor how production is organized, staff recruitment and motivation levels, check suppliers for cost effectiveness and quality, and study the latest advances in technology. You know that the more chips you manufacture, the more you will have available to sell, and the higher will be your income (assuming you sell them) and your standard of living.

One day a local university announces a major discovery. Scientists have devised a new material to produce chips which would help to increase computing power by 50 per cent. How should you react to this news? Should you use the new material? Does this discovery make you better off or worse off than you were before? In this chapter we will see that these questions can have surprising answers. The surprise will come from applying the most basic tools of economics – supply and demand – to the market for computer chips.

In any competitive market, such as the market for computer chips, the upward sloping supply curve represents the behaviour of sellers, and the downward sloping demand curve represents the behaviour of buyers. The price of the good adjusts to bring the quantity supplied and quantity demanded of the good into balance. To apply this basic analysis to understand the impact of the scientists' discovery, we must first develop one more tool: the concept of *elasticity* also referred to as *price sensitivity*. We know from Chapter 4 that when price rises, demand falls and supply rises. What we did not discuss in the chapter was *how far* demand and supply change in response to changes in price – in other words, how sensitive supply and demand is to a change in prices. When studying how some event or policy affects a market, we can discuss not only the direction of the effects but their magnitude as well. Elasticity, a measure of how much buyers and sellers respond to changes in market conditions, allows us to analyse supply and demand with greater precision.

PRICE ELASTICITY OF SUPPLY

elasticity a measure of the responsiveness of quantity demanded or quantity supplied to one of its determinants

When we introduced supply in Chapter 4, we noted that producers of a good offer to sell more of it when the price of the good rises, when their input prices fall or when their technology improves. To turn from qualitative to quantitative statements about quantity supplied, we use the concept of **elasticity**.

The Price Elasticity of Supply and its Determinants

price elasticity of supply a measure of how much the quantity supplied of a good responds to a change in the price of that good, computed as the percentage change in quantity supplied divided by the percentage change in price

The law of supply states that higher prices raise the quantity supplied. The **price elasticity of supply** measures how much the quantity supplied responds to changes in the price. Supply of a good is said to be *elastic* (or price sensitive) if the quantity supplied responds substantially to changes in the price. Supply is said to be *inelastic* (or price insensitive) if the quantity supplied responds only slightly to changes in the price.

The price elasticity of supply depends on the flexibility of sellers to change the amount of the good they produce. For example, seafront property has an inelastic supply

because it is almost impossible to produce more of it quickly – supply is not very sensitive to changes in price. By contrast, manufactured goods, such as books, cars and television sets, have relatively elastic supplies because the firms that produce them can run their factories longer in response to a higher price – supply is sensitive to changes in price.

Elasticity can take any value greater than or equal to zero. The closer to zero the more inelastic, and the closer to infinity the more elastic. We will look at the determinants of supply first and then look in more detail at how elasticity is computed.

The Determinants of Price Elasticity of Supply

The Time Period In most markets, a key determinant of the price elasticity of supply is the time period being considered. Supply is usually more elastic in the long run than in the short run. We can further distinguish between the short run and the very short run. Over very short periods of time, firms may find it impossible to respond to a change in price by changing output. In the short run firms cannot easily change the size of their factories or productive capacity to make more or less of a good but may have some flexibility. For example, it might take a month to employ new labour but after that time some increase in output can be accommodated. Overall, in the short run, the quantity supplied is not very responsive to the price. By contrast, over longer periods, firms can build new factories or close old ones, hire new staff and buy in more capital and equipment. In addition, new firms can enter a market and old firms can shut down. Thus, in the long run, the quantity supplied can respond substantially to price changes.

Productive Capacity Most businesses, in the short run, will have a finite capacity – an upper limit to the amount that they can produce at any one time determined by the amount of factor inputs they possess. How far they are using this capacity depends, in turn, on the state of the economy. In periods of strong economic growth, firms may be operating at or near full capacity. If demand is rising for the product they produce and prices are rising, it may be difficult for the firm to expand output to meet this new demand and so supply may be inelastic.

When the economy is growing slowly or is contracting, some firms may find they have to cut back output and may only be operating at 60 per cent of full capacity. In this situation, if demand later increased and prices started to rise, it may be much easier for the firm to expand output relatively quickly and so supply would be more elastic.

The Size of the Firm/Industry It is possible that as a general rule, supply may be more elastic in smaller firms or industries than in larger ones. For example, consider a small independent furniture manufacturer. Demand for its products may rise and in response the firm may be able to buy in raw materials (wood, for example), to meet this increase in demand. Whilst the firm will incur a cost in buying in this timber, it is unlikely that the unit cost for the material will increase. Compare this to a situation where a steel manufacturer increases its purchase of raw materials (iron ore, for example). Buying large quantities of iron ore on global commodity markets can drive up unit price and, by association, unit costs.

A study of the coffee industry in Papua New Guinea¹ found that the price elasticity of supply of smallholders was around 0.23 whereas that for large estates was much more inelastic at 0.04. The study suggested that these figures compared well to estimates derived in previous studies.

The response of supply to changes in price in large firms/industries, therefore, may be less elastic than in smaller firms/industries. This is also related to the number of firms in

¹<http://www.une.edu.au/bepp/working-papers/ag-res-econ/occasional-papers/agecop3.pdf>

the industry – the more firms there are in the industry the easier it is to increase supply, other things being equal.

The Mobility of Factors of Production Consider a farmer whose land is currently devoted to producing wheat. A sharp rise in the price of rape seed might encourage the farmer to switch use of land from wheat to rape seed relatively easily. The mobility of the factor of production land, in this case, is relatively high and so supply of rape seed may be relatively elastic.

A number of multinational firms that have plants in different parts of the world now build each plant to be identical. What this means is that if there is disruption to one plant the firm can more easily transfer operations to another plant elsewhere and continue production ‘seamlessly’. Car manufacturers provide another example of this interchangeability of parts and operations. The chassis, for example, may be identical across a range of branded car models. This is the case with some Audi, Volkswagen, Seat and Skoda models. This means that the supply may be more elastic as a result.



Different cars? Different brands maybe but there are more similarities to these two cars than there are differences.

Compare this to the supply of highly skilled oncology consultants. An increase in the wages of oncology consultants (suggesting a shortage exists) will not mean that a renal consultant or other doctors can suddenly switch to take advantage of the higher wages and increase the supply of oncology consultants. In this example, the mobility of labour to switch between different uses is limited and so the supply of these specialist consultants is likely to be relatively inelastic.

Ease of Storing Stock/Inventory In some firms, stocks can be built up to enable the firm to respond more flexibly to changes in prices. In industries where inventory build-up is relatively easy and cheap, the price elasticity of supply is more elastic than in industries where it is much harder to do this. Consider the fresh fruit industry, for example. Storing fresh fruit is not easy because it is perishable and so the price elasticity of supply in this industry may be more inelastic.

Computing the Price Elasticity of Supply

Now that we have discussed the price elasticity of supply in general terms, let's be more precise about how it is measured. Economists compute the price elasticity of supply as the percentage change in the quantity supplied divided by the percentage change in the price. That is:

$$\text{Price elasticity of supply} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

For example, suppose that an increase in the price of milk from €2.85 to €3.15 a litre raises the amount that dairy farmers produce from 90 000 to 110 000 litres per month. Using the midpoint method, we calculate the percentage change in price as:

$$\text{Percentage change in price} = (3.15 - 2.85)/3.00 \times 100 = 10\%$$

Similarly, we calculate the percentage change in quantity supplied as:

$$\text{Percentage change in quantity supplied} = (110\,000 - 90\,000)/100\,000 \times 100 = 20\%$$

In this case, the price elasticity of supply is:

$$\text{Price elasticity of supply} = \frac{20\%}{10\%} = 2$$

In this example, the elasticity of 2 reflects the fact that the quantity supplied moves proportionately twice as much as the price.

The Midpoint Method of Calculating Percentage Changes and Elasticities

If you try calculating the price elasticity of supply between two points on a supply curve, you will quickly notice an annoying problem: the elasticity for a movement from point A to point B seems different from the elasticity for a movement from point B to point A. For example, consider these numbers:

$$\begin{aligned}\text{Point A : Price} &= \text{€4 Quantity Supplied} = 80 \\ \text{Point B : Price} &= \text{€6 Quantity Supplied} = 125\end{aligned}$$

Going from point A to point B, the price rises by 50 per cent – the change in price divided by the original price $\times 100$ ($2/4 \times 100$) – and the quantity rises by 56.25 per cent – the change in quantity supplied divided by the original supply $\times 100$ ($45/80 \times 100$) – indicating that the price elasticity of supply is $56.25/50$, or 1.125. By contrast, going from point B to point A, the price falls by 33 per cent ($2/6 \times 100$), and the quantity falls by 36 per cent ($45/125 \times 100$), indicating that the price elasticity of supply is $36/33$, or 1.09.

Note, in the working above we have rounded the fall in price to the nearest whole number (33 per cent). If we had not used this rounding, the price elasticity of supply would be:

$$\frac{\left(\frac{85-125}{125}\right)}{\left(\frac{4-6}{6}\right)} = 1.08$$

One way to avoid this problem is to use the *midpoint method* for calculating elasticities. In the example above we used a standard way to compute a percentage change – divide the change by the initial level and multiply by 100. By contrast, the midpoint method computes a percentage change by dividing the change by the midpoint (or average) of the initial and final levels. For instance, €5 is the midpoint of €4 and €6. Therefore, according to the midpoint method, a change from €4 to €6 is considered a 40 per cent rise, because $((6 - 4)/5) \times 100 = 40$. Similarly, a change from €6 to €4 is considered a 40 per cent fall $((4 - 6)/5) \times 100 = -40$. Looking at the quantity, moving from Point A to Point B gives $(125 - 80)/102.5 \times 100 = 43.9$ per cent and for a price fall $(80 - 125)/102.5 \times 100 = -43.9$ per cent.

Because the midpoint method gives the same answer regardless of the direction of change (as indicated by the negative sign), it is often used when calculating price elasticities between two points. In our example, the midpoint between point A and point B is:

$$\text{Midpoint: Price} = \text{€5} \quad \text{Quantity} = 102.5$$

According to the midpoint method, when going from point A to point B, the price rises by 40 per cent, and the quantity rises by 43.9 per cent. Similarly, when going from point B to point A, the price falls by 40 per cent, and the quantity falls by 43.9 per cent. In both directions, the price elasticity of supply equals 1.1.

We can express the midpoint method with the following formula for the price elasticity of supply between two points, denoted (Q_1, P_1) and (Q_2, P_2) :

$$\text{Price elasticity of supply} = \frac{(Q_2 - Q_1)/[(Q_2 + Q_1)/2]}{(P_2 - P_1)/[(P_2 + P_1)/2]}$$

The numerator is the percentage change in quantity computed using the midpoint method, and the denominator is the percentage change in price computed using the midpoint method. If you ever need to calculate elasticities, you should use this formula.

The Variety of Supply Curves

Because the price elasticity of supply measures the responsiveness of quantity supplied to the price, it is reflected in the appearance of the supply curve (again, assuming we are using similar scales on the axes of diagrams being used). Figure 5.1 shows five cases. In the extreme case of a zero elasticity, as shown in panel (a), supply is *perfectly inelastic* and the supply curve is vertical. In this case, the quantity supplied is the same regardless of the price. As the elasticity rises, the supply curve gets flatter, which shows that the quantity supplied responds more to changes in the price. At the opposite extreme, shown in panel (e), supply is *perfectly elastic*. This occurs as the price elasticity of supply approaches infinity and the supply curve becomes horizontal, meaning that very small changes in the price lead to very large changes in the quantity supplied.

In some markets, the elasticity of supply is not constant but varies over the supply curve. Figure 5.2 shows a typical case for an industry in which firms have factories with a limited capacity for production. For low levels of quantity supplied, the elasticity of supply is high, indicating that firms respond substantially to changes in the price. In this region, firms have capacity for production that is not being used, such as buildings and machinery sitting idle for all or part of the day. Small increases in price make it profitable for firms to begin using this idle capacity. As the quantity supplied rises, firms begin to reach capacity. Once capacity is fully used, increasing production further

FIGURE 5.1**The Price Elasticity of Supply**

The price elasticity of supply determines whether the supply curve is steep or flat (assuming that the scale used for the axes is the same). Note that all percentage changes are calculated using the midpoint method.

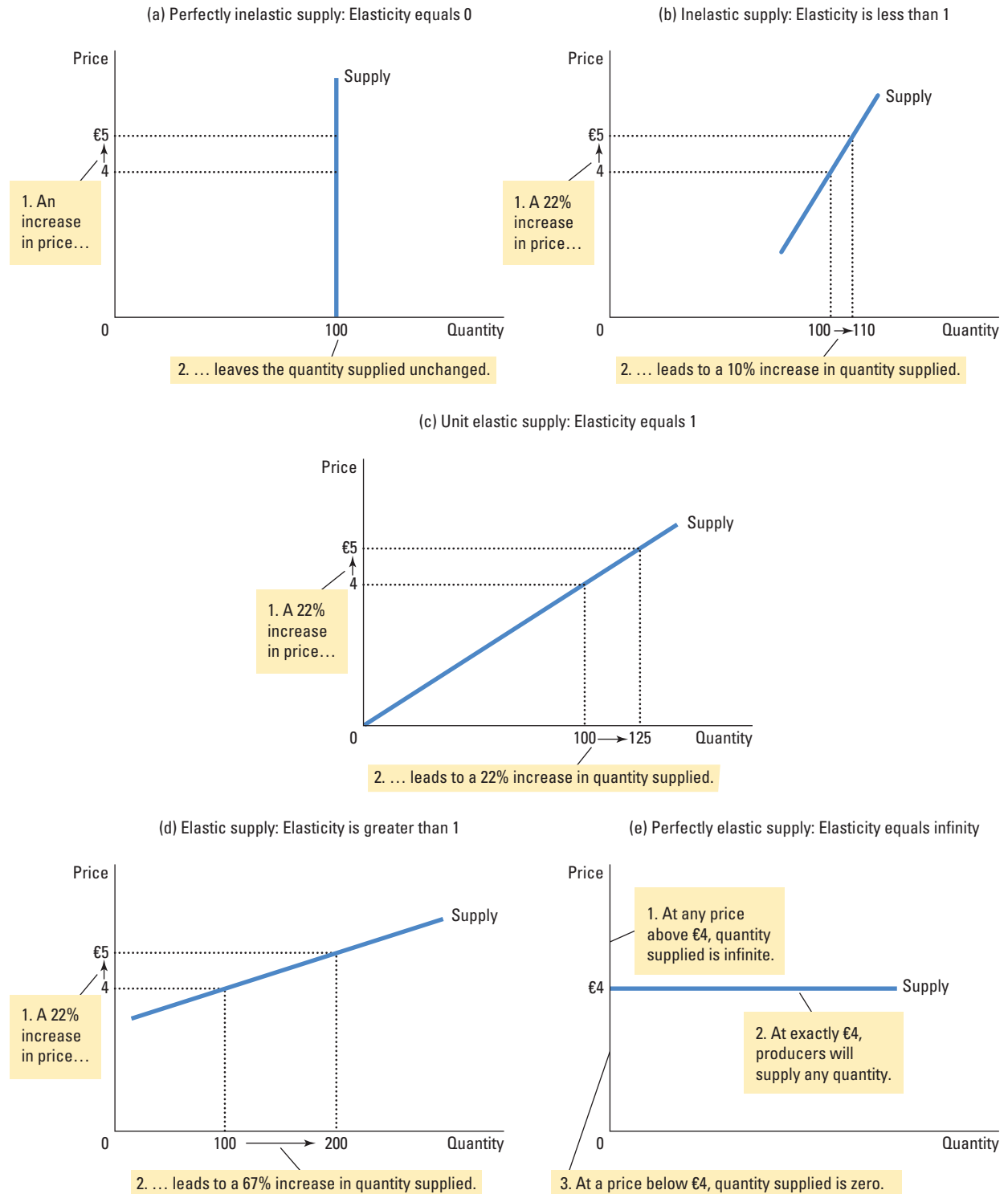
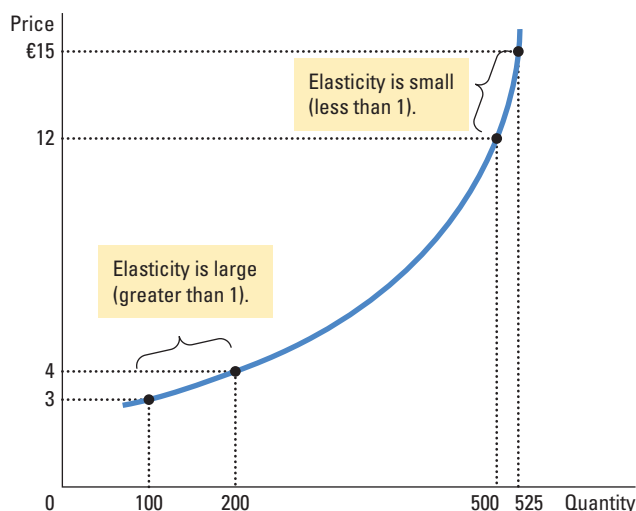


FIGURE 5.2**How the Price Elasticity of Supply Can Vary**

Because firms often have a maximum capacity for production, the elasticity of supply may be very high at low levels of quantity supplied and very low at high levels of quantity supplied. Here, an increase in price from €3 to €4 increases the quantity supplied from 100 to 200. Because the increase in quantity supplied of 67 per cent (computed using the midpoint method) is larger than the increase in price of 29 per cent, the supply curve is elastic in this range. By contrast, when the price rises from €12 to €15, the quantity supplied rises only from 500 to 525. Because the increase in quantity supplied of 5 per cent is smaller than the increase in price of 22 per cent, the supply curve is inelastic in this range.



requires the construction of new factories. To induce firms to incur this extra expense, the price must rise substantially, so supply becomes less elastic.

Figure 5.2 presents a numerical example of this phenomenon. In each case we have used the midpoint method and the numbers have been rounded for convenience. When the price rises from €3 to €4 (a 29 per cent increase, according to the midpoint method), the quantity supplied rises from 100 to 200 (a 67 per cent increase). Because quantity supplied moves proportionately more than the price, the supply curve has elasticity greater than 1. By contrast, when the price rises from €12 to €15 (a 22 per cent increase), the quantity supplied rises from 500 to 525 (a 5 per cent increase). In this case, quantity supplied moves proportionately less than the price, so the elasticity is less than 1.

Quick Quiz Define the price elasticity of supply • Explain why the price elasticity of supply might be different in the long run from in the short run.

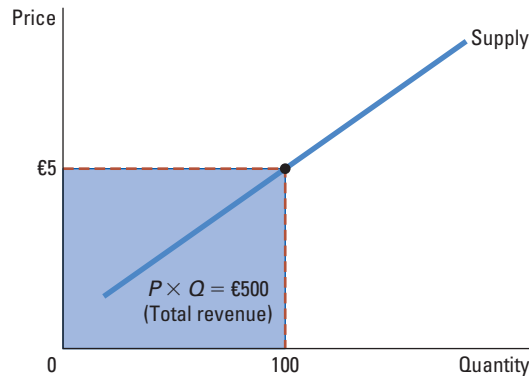
Total Revenue and the Price Elasticity of Supply

total revenue the amount received by sellers of a good, computed as the price of the good times the quantity sold

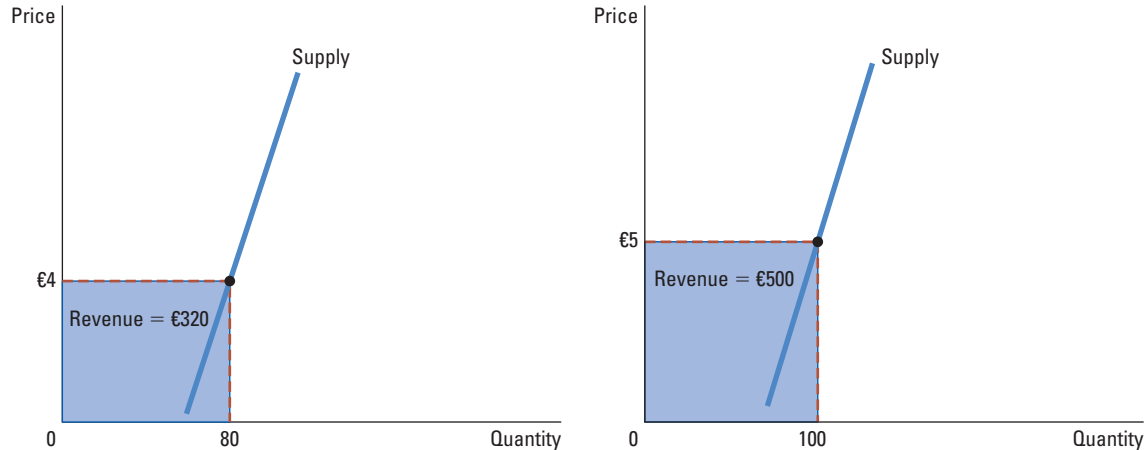
When studying changes in supply in a market we are often interested in the resulting changes in the **total revenue** received by producers. In any market, total revenue received by sellers is $P \times Q$, the price of the good times the quantity of the good sold. This is highlighted in Figure 5.3 which shows an upward sloping supply curve with an assumed price of €5 and a supply of 100 units. The height of the box under the supply curve is P and the width is Q . The area of this box, $P \times Q$, equals the total revenue received in this market. In Figure 5.3, where $P = €5$ and $Q = 100$, total revenue is $€5 \times 100$, or €500.

FIGURE 5.3**The Supply Curve and Total Revenue**

The total amount received by sellers equals the area of the box under the demand curve, $P \times Q$. Here, at a price of €5, the quantity supplied is 100 and the total revenue is €500.

**FIGURE 5.4****How Total Revenue Changes When Price Changes: Inelastic Supply**

With an inelastic supply curve, an increase in the price leads to an increase in quantity supplied that is proportionately smaller. Therefore, total revenue (the product of price and quantity) increases. Here, an increase in the price from €4 to €5 causes the quantity supplied to rise from 80 to 100, and total revenue rises from €320 to €500.

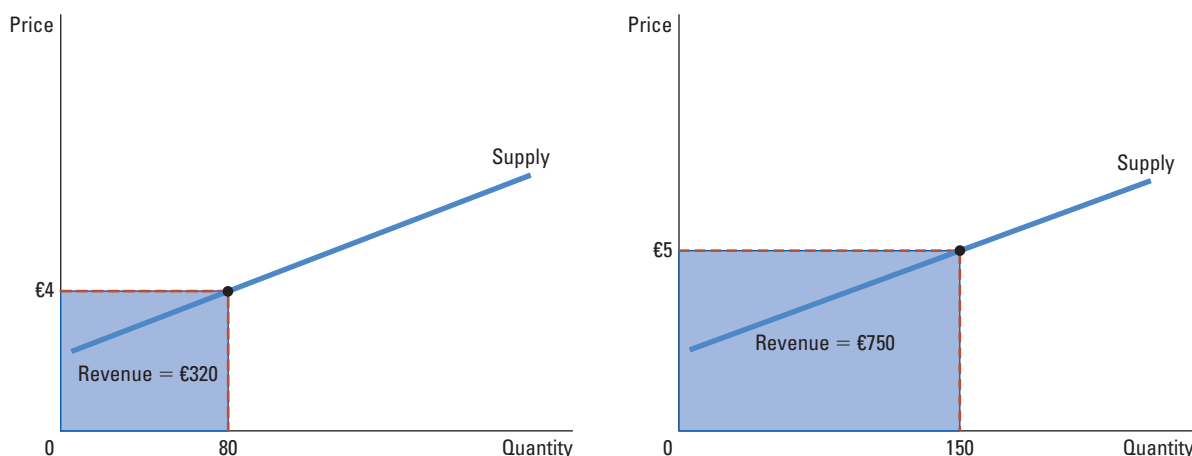


How does total revenue change as one moves along the supply curve? The answer depends on the price elasticity of supply. If supply is inelastic, as in Figure 5.4, then an increase in the price which is proportionately larger causes an increase in total revenue. Here an increase in price from €4 to €5 causes the quantity supplied to rise only from 80 to 100, and so total revenue rises from €320 to €500.

If supply is elastic then a similar increase in price brings about a much larger than proportionate increase in supply. In Figure 5.5, we assume a price of €4 and a supply of 80 with total revenue of €320. Now a price increase from €4 to €5 leads to a much greater than proportionate increase in supply from 80 to 150 with total revenue rising to €750.

FIGURE 5.5**How Total Revenue Changes When Price Changes: Elastic Supply**

With an elastic supply curve, an increase in the price leads to an increase in quantity supplied that is proportionately larger. Therefore, total revenue (the product of price and quantity) increases. Here, an increase in the price from €4 to €5 causes the quantity supplied to rise from 80 to 150, and total revenue rises from €320 to €750.



price elasticity of demand a measure of how much the quantity demanded of a good responds to a change in the price of that good, computed as the percentage change in quantity demanded divided by the percentage change in price

THE PRICE ELASTICITY OF DEMAND

Businesses cannot directly control demand. They can seek to influence demand (and do) by utilizing a variety of strategies and tactics but ultimately the consumer decides whether to buy a product or not. One important way in which consumer behaviour can be influenced is through a firm changing the prices of its goods (many firms do have some control over the price it can charge although as we have seen, in perfectly competitive markets this is not the case as the firm is a price taker). An understanding of the price elasticity of demand is important in anticipating the likely effects of changes in price on demand.

The Price Elasticity of Demand and its Determinants

The law of demand states that a fall in the price of a good raises the quantity demanded. The **price elasticity of demand** measures how much the quantity demanded responds to a change in price. Demand for a good is said to be *elastic* or price sensitive if the quantity demanded responds substantially to changes in the price. Demand is said to be *inelastic* or price insensitive if the quantity demanded responds only slightly to changes in the price.

The price elasticity of demand for any good measures how willing consumers are to move away from the good as its price rises. Thus, the elasticity reflects the many economic, social and psychological forces that influence consumer tastes. Based on experience, however, we can state some general rules about what determines the price elasticity of demand.

Availability of Close Substitutes Goods with close substitutes tend to have more elastic demand because it is easier for consumers to switch from that good to others. For example, butter and margarine are easily substitutable. A small increase in the price of butter, assuming the price of margarine is held fixed, causes the quantity of butter sold to fall by a relatively large amount. As a general rule, the closer the substitute the more elastic the good is because it is easier for consumers to switch from one to the other. By contrast, because eggs are a food without a close substitute, the demand for eggs is less elastic than the demand for butter.

Necessities versus Luxuries Necessities tend to have relatively inelastic demands, whereas luxuries have relatively elastic demands. People use gas and electricity to heat their homes and cook their food. If the price of gas and electricity rose together, people would not demand dramatically less of them. They might try and be more energy-efficient and reduce their demand a little, but they would still need hot food and warm homes. By contrast, when the price of sailing dinghies rises, the quantity of sailing dinghies demanded falls substantially. The reason is that most people view hot food and warm homes as necessities and a sailing dinghy as a luxury. Of course, whether a good is a necessity or a luxury depends not on the intrinsic properties of the good but on the preferences of the buyer. For an avid sailor with little concern over her health, sailing dinghies might be a necessity with inelastic demand and hot food and a warm place to sleep a luxury with elastic demand.

Definition of the Market The elasticity of demand in any market depends on how we draw the boundaries of the market. Narrowly defined markets tend to have more elastic demand than broadly defined markets, because it is easier to find close substitutes for narrowly defined goods. For example, food, a broad category, has a fairly inelastic demand because there are no good substitutes for food. Ice cream, a narrower category, has a more elastic demand because it is easy to substitute other desserts for ice cream. Vanilla ice cream, a very narrow category, has a very elastic demand because other flavours of ice cream are almost perfect substitutes for vanilla.

Proportion of Income Devoted to the Product Some products have a relatively high price and take a larger proportion of income than others. Buying a new suite of furniture for a lounge, for example, tends to take up a large amount of income whereas buying an ice cream might account for only a tiny proportion of income. If the price of a three-piece suite rises by 10 per cent, therefore, this is likely to have a greater effect on demand for this furniture than a similar 10 per cent increase in the price of an ice cream. The higher the proportion of income devoted to the product the greater the elasticity is likely to be.

Time Horizon Goods tend to have more elastic demand over longer time horizons. When the price of petrol rises, the quantity of petrol demanded falls only slightly in the first few months. Over time, however, people buy more fuel-efficient cars, switch to public transport and move closer to where they work. Within several years, the quantity of petrol demanded falls more substantially. Similarly, if the price of a unit of electricity rises much above an equivalent energy unit of gas, demand may fall only slightly in the short run because many people already have electric cookers or electric heating appliances installed in their homes and cannot easily switch. If the price difference persists over several years, however, people may find it worth their while to replace their old electric heating and cooking appliances with new gas appliances and the demand for electricity will fall.

Computing the Price Elasticity of Demand

The principles for computing price elasticity of demand are similar to that discussed when we looked at price elasticity of supply. The price elasticity of demand is computed as the percentage change in the quantity demanded divided by the percentage change in the price. That is:

$$\text{Price elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in price}}$$

For example, suppose that a 10 per cent increase in the price of a packet of breakfast cereal causes the amount you buy to fall by 20 per cent. Because the quantity demanded of a good is negatively related to its price, the percentage change in quantity will always have the opposite sign to the percentage change in price. In this example, the percentage change in price is a *positive* 10 per cent (reflecting an increase), and the percentage change in quantity demanded is a *negative* 20 per cent (reflecting a decrease). For this reason, price elasticities of demand are sometimes reported as negative numbers. In this book we follow the common practice of dropping the minus sign and reporting all price elasticities as positive numbers. (Mathematicians call this the *absolute value*.) With this convention, a larger price elasticity implies a greater responsiveness of quantity demanded to price.

Using this convention we calculate the elasticity of demand as:

$$\text{Price elasticity of demand} = \frac{20\%}{10\%} = 2$$

In this example, the elasticity is 2, reflecting that the change in the quantity demanded is proportionately twice as large as the change in the price.

Pitfall Prevention We have used the term ‘relatively’ elastic or inelastic at times throughout the analysis so far. It is important to remember that elasticity can be any value greater than or equal to 0. We can look at two goods, therefore, both of which are classed as ‘inelastic’ but where one is more inelastic than the other. If we are comparing good X, which has an elasticity of 0.2 and good Y, which has an elasticity of 0.5, then both are inelastic but good Y is relatively elastic by comparison. As with so much of economics, careful use of terminology is important in conveying a clear understanding.

Using the Midpoint Method

As with the price elasticity of supply, we use the midpoint method to calculate price elasticity of demand for the same reasons. We can express the midpoint method with the following formula for the price elasticity of demand between two points, denoted (Q_1, P_1) and (Q_2, P_2) :

$$\text{Price elasticity of demand} = \frac{(Q_2 - Q_1)/[(Q_2 + Q_1)/2]}{(P_2 - P_1)/[(P_2 + P_1)/2]}$$

The numerator is the proportionate change in quantity computed using the midpoint method, and the denominator is the proportionate change in price computed using the midpoint method.

The Variety of Demand Curves

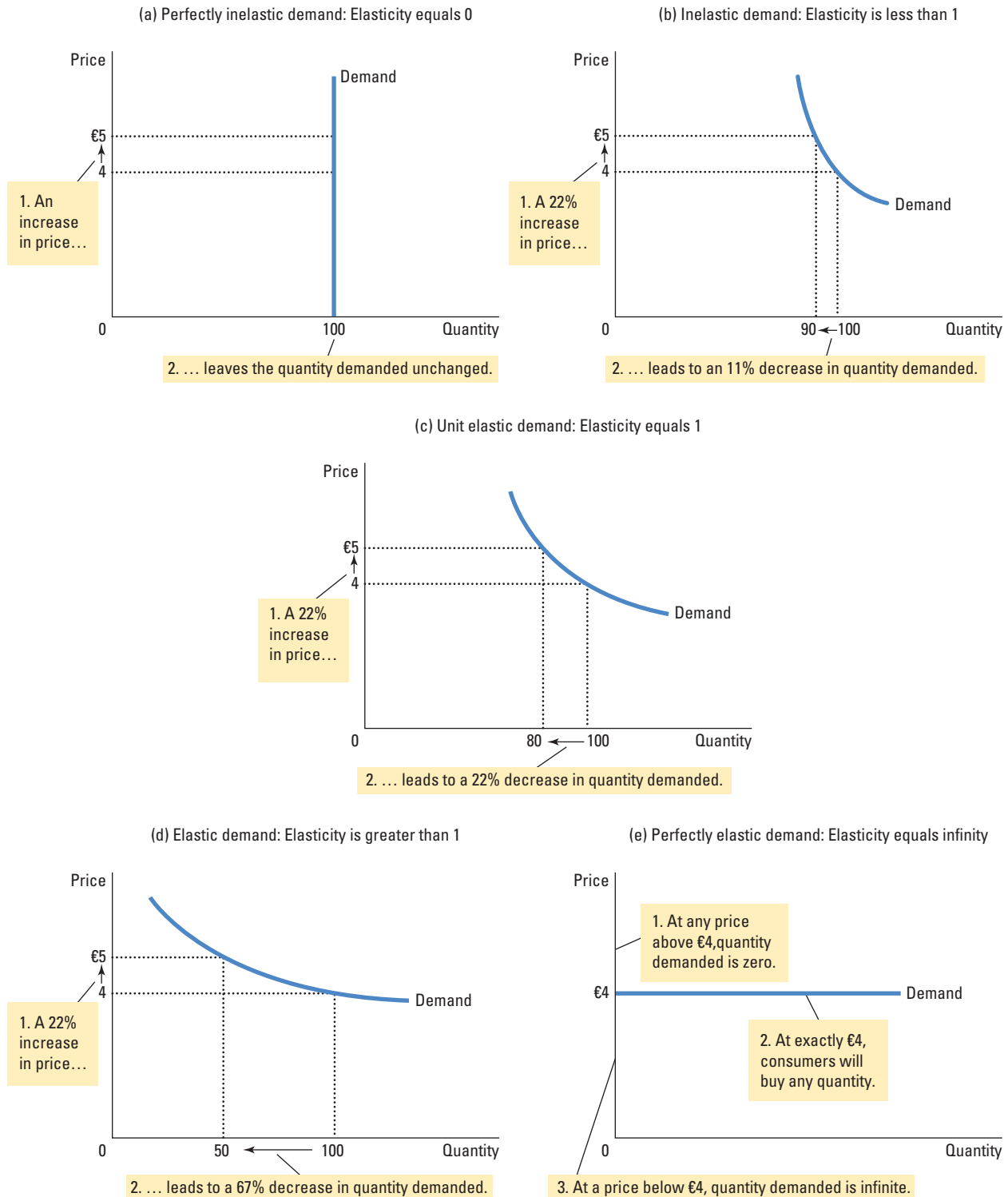
Economists classify demand curves according to their elasticity. Demand is *elastic* when the elasticity is greater than 1, so that quantity changes proportionately more than the price. Demand is *inelastic* when the elasticity is less than 1, so that quantity moves proportionately less than the price. If the elasticity is exactly 1, so that quantity moves the same amount proportionately as price, demand is said to have *unit elasticity*.

Because the price elasticity of demand measures how much quantity demanded responds to changes in the price, it is closely related to the slope of the demand curve. The following heuristic (rule of thumb) again, assuming we are using comparable scales on the axes, is a useful guide: the flatter the demand curve that passes through a given point, the greater the price elasticity of demand. The steeper the demand curve that passes through a given point, the smaller the price elasticity of demand.

Figure 5.6 shows five cases, each of which uses the same scale on each axis. This is an important thing to remember because simply looking at a graph and the shape of the

FIGURE 5.6**The Price Elasticity of Demand**

The steepness of the demand curve indicates the price elasticity of demand (assuming the scale used on the axes are the same). Note that all percentage changes are calculated using the midpoint method.



curve without recognizing the scale can result in incorrect conclusions about elasticity. In the extreme case of a zero elasticity shown in panel (a), demand is *perfectly inelastic*, and the demand curve is vertical. In this case, regardless of the price, the quantity demanded stays the same. As the elasticity rises, the demand curve gets flatter and flatter, as shown in panels (b), (c) and (d). At the opposite extreme shown in panel (e), demand is *perfectly elastic*. This occurs as the price elasticity of demand approaches infinity and the demand curve becomes horizontal, reflecting the fact that very small changes in the price lead to huge changes in the quantity demanded.

Total Expenditure, Total Revenue and the Price Elasticity of Demand

total expenditure the amount paid by buyers, computed as the price of the good times the quantity purchased

When studying changes in demand in a market, we are interested in the amount paid by buyers of the good which will in turn represent the total revenue that sellers receive. **Total expenditure** is given by the total amount bought multiplied by the price paid. We can show total expenditure graphically, as in Figure 5.7. The height of the box under the demand curve is P , and the width is Q . The area of this box, $P \times Q$, equals the total expenditure in this market. In Figure 5.7, where $P = €4$ and $Q = 100$, total expenditure is $€4 \times 100$, or €400.

For businesses, having some understanding of the price elasticity of demand is important in decision making. If a firm is thinking of changing price how will the demand for its product react? The firm knows that there is an inverse relationship between price and demand but the effect on its revenue will be dependent on the price elasticity of demand. It is entirely possible that a firm could reduce its price and increase total revenue. Equally, a firm could raise price and find its total revenue falling. At first glance this might sound counter-intuitive but it all depends on the price elasticity of demand for the product.

If demand is inelastic, as in Figure 5.8, then an increase in the price causes an increase in total expenditure. Here an increase in price from €1 to €3 causes the quantity demanded to fall only from 100 to 80, and so total expenditure rises from €100 to €240. An increase in price raises $P \times Q$ because the fall in Q is proportionately smaller than the rise in P .

We obtain the opposite result if demand is elastic: an increase in the price causes a decrease in total expenditure. In Figure 5.9, for instance, when the price rises from €4

FIGURE 5.7

Total Expenditure

The total amount paid by buyers, and received as expenditure by sellers, equals the area of the box under the demand curve, $P \times Q$. Here, at a price of €4, the quantity demanded is 100, and total expenditure is €400.

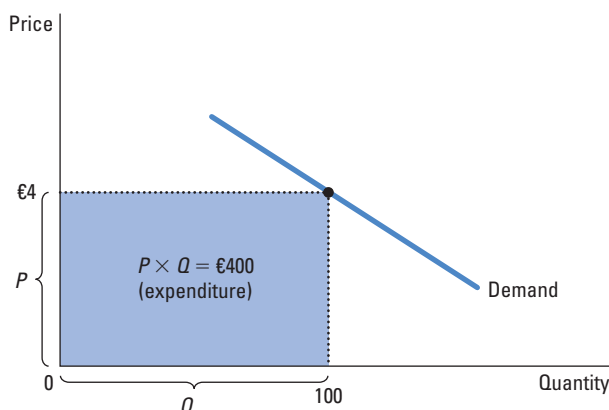
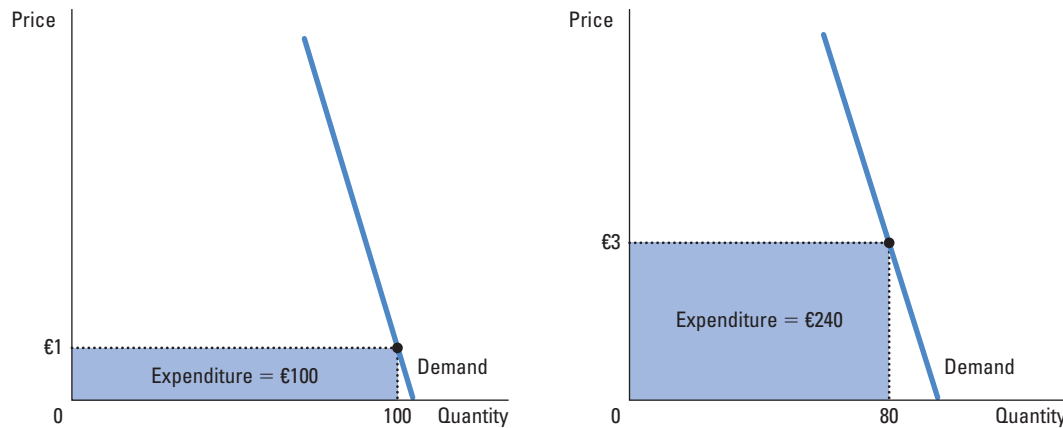
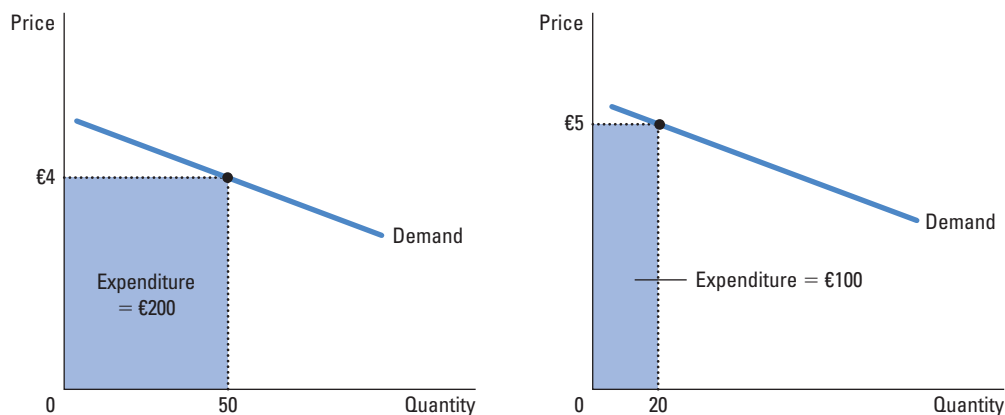


FIGURE 5.8**How Total Expenditure Changes When Price Changes: Inelastic Demand**

With an inelastic demand curve, an increase in the price leads to a decrease in quantity demanded that is proportionately smaller. Therefore, total expenditure (the product of price and quantity) increases. Here, an increase in the price from €1 to €3 causes the quantity demanded to fall from 100 to 80, and total expenditure rises from €100 to €240.

**FIGURE 5.9****How Total Expenditure Changes When Price Changes: Elastic Demand**

With an elastic demand curve, an increase in the price leads to a decrease in quantity demanded that is proportionately larger. Therefore, total expenditure (the product of price and quantity) decreases. Here, an increase in the price from €4 to €5 causes the quantity demanded to fall from 50 to 20, so total expenditure falls from €200 to €100.



to €5, the quantity demanded falls from 50 to 20, and so total expenditure falls from €200 to €100. Because demand is elastic, the reduction in the quantity demanded is so great that it more than offsets the increase in the price. That is, an increase in price reduces $P \times Q$ because the fall in Q is proportionately greater than the rise in P .

Although the examples in these two figures are extreme, they illustrate a general rule:

- When demand is inelastic (a price elasticity less than 1), price and total expenditure move in the same direction.
- When demand is elastic (a price elasticity greater than 1), price and total expenditure move in opposite directions.
- If demand is unit elastic (a price elasticity exactly equal to 1), total expenditure remains constant when the price changes.

revenue shown by the purple rectangle ($q \Delta p$). However, this is offset by the reduction in revenue caused by the fall in quantity demanded as a result of the change in price shown by the blue rectangle ($p \Delta q$). There is also an area indicated by the yellow rectangle which is equal to $\Delta p \Delta q$. This leaves us with a formula for the change in TR as:

$$\Delta TR = q \Delta p + p \Delta q + \Delta p \Delta q$$

Let us substitute some figures into our formula to see how this works in practice. Assume the original price of a product is 15 and the quantity demanded at this price is 750. When price rises to 20 the quantity demanded falls to 500.

Using the equation:

$$TR = pq + p \Delta q + \Delta pq + \Delta p \Delta q$$

TR is now:

$$TR = 15(750) + 15(-250) + 5(750) + 5(-250)$$

$$TR = 10\,000$$

The change in TR is:

$$\Delta TR = q \Delta p + p \Delta q + \Delta p \Delta q$$

$$\Delta TR = 750(5) + 15(-250) + 5(-250)$$

$$\Delta TR = 3750 - 3750 - 1250$$

$$\Delta TR = -1250$$

In this example the effect of the change in price has been negative on TR. We know from our analysis of price elasticity of demand that this means the percentage change in quantity demand was greater than

the percentage change in price – in other words, *ped* must be elastic at this point (>1). For the change in TR to be positive, therefore, the *ped* must be <1 .

We can express the relationship between the change in TR and *ped* as an inequality as follows:

$$ped = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} > 1$$

When price increases, revenue decreases if *ped* meets this inequality. Equally, for a price increase to result in a rise in revenue *ped* must meet the inequality below:

$$ped = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} < 1$$

JEOPARDY PROBLEM

A business selling plumbing equipment to the trade (i.e. professional plumbers only) increases the price of copper piping by 4 per cent and reduces the price of radiators by 5 per cent. A year later they analyse their sales figures and find that revenue for copper piping rose in the first three months after the price rise but then fell dramatically thereafter, while the revenue for sales of radiators also fell throughout the period.

Explain what might have happened to bring about this situation. Illustrate your answer with diagrams where appropriate.

APPLICATIONS OF SUPPLY AND DEMAND ELASTICITY

Can good news for the computing industry be bad news for computer chip manufacturers? Why do the prices of ski holidays in Europe rise dramatically over public and school holidays? At first, these questions might seem to have little in common. Yet both questions are about markets and all markets are subject to the forces of supply and demand. Here we apply the versatile tools of supply, demand and elasticity to answer these seemingly complex questions.

Can Good News for the Computer Industry Be Bad News for Chip Makers?

Let's now return to the question posed at the beginning of this chapter: what happens to chip manufacturers and the market for chips when scientists discover a new material for making chips that is more productive than silicon? Recall from Chapter 4 that we answer such questions in three steps. First, we examine whether the supply or demand curve

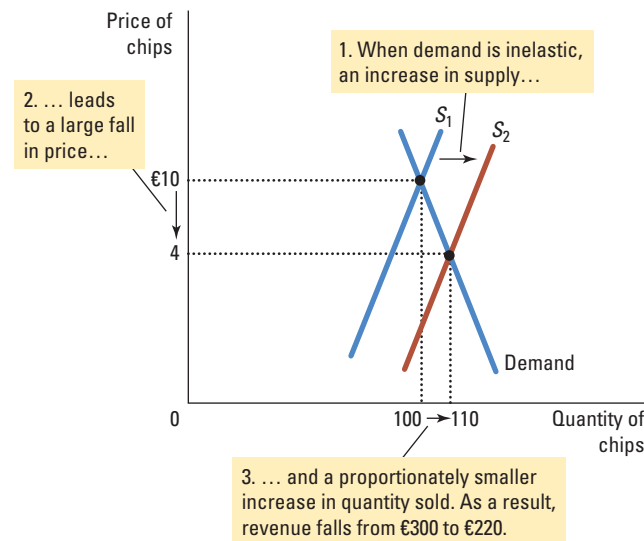
shifts. Secondly, we consider which direction the curve shifts. Thirdly, we use the supply and demand diagram to see how the market equilibrium changes.

This is a situation that is facing chip manufacturers. Scientists are investigating new materials to make computer chips. Such a material may allow the manufacturers to be able to work on building processing power at ever smaller concentrations and increase computing power considerably. In this case, the discovery of the new material affects the supply curve. Because the material increases the amount of computing power that can be produced on each chip, manufacturers are now willing to supply more chips at any given price. In other words, the supply curve for computing power shifts to the right. The demand curve remains the same because consumers' desire to buy chips at any given price is not affected by the introduction of the new material. Figure 5.15 shows an example of such a change. When the supply curve shifts from S_1 to S_2 , the quantity of chips sold increases from 100 to 110, and the price of chips falls from €10 per gigabyte to €4 per gigabyte.

FIGURE 5.15

An Increase in Supply in the Market for Computer Chips

When an advance in chip technology increases the supply of chips from S_1 to S_2 , the price of chips falls. Because the demand for chips is inelastic, the increase in the quantity sold from 100 to 110 is proportionately smaller than the decrease in the price from €10 to €4. As a result, manufacturers' total revenue falls from €1000 (€10 $\times$ 100) to €440 (€4 $\times$ 110).



But does this discovery make chip manufacturers better off? As a first stab at answering this question, consider what happens to the total revenue received by chip manufacturers; total revenue is $P \times Q$, the price of each chip times the quantity sold. The discovery affects manufacturers in two conflicting ways. The new material allows manufacturers to produce more chips with greater computing power (Q rises), but now each chip sells for less (P falls).

Whether total revenue rises or falls depends on the elasticity of demand. We can assume that the demand for chips is inelastic; in producing a computer, chips represent a relatively small proportion of the total cost but they also have few good substitutes. When the demand curve is inelastic, as it is in Figure 5.15, a decrease in price causes total revenue to fall. You can see this in the figure: the price of chips falls substantially, whereas the quantity of chips sold rises only slightly. Total revenue falls from €1000 to €440. Thus, the discovery of the new material lowers the total revenue that chip manufacturers receive for the sale of their products.

If manufacturers are made worse off by the discovery of this new material, why do they adopt it? The answer to this question goes to the heart of how competitive markets work. If each chip manufacturer is a small part of the market for chips, he or she takes

the price of chips as given. For any given price of chips, it is better to use the new material in order to produce and sell more chips. Yet when all manufacturers do this, the supply of chips rises, the price falls and manufacturers are worse off.

Although this example is only hypothetical, in fact a new material for making computer chips is being investigated. The material is called hafnium and is used in the nuclear industry. In recent years the manufacture of computer chips has changed dramatically. In the early 1990s, prices per megabyte of DRAM (dynamic random access memory) stood at around \$55 but fell to under \$1 by the early part of the new century. Manufacturers who were first involved in chip manufacture made high profits but as new firms joined the industry, supply increased and as the technology also spread, supply rose and prices fell. The fall in prices led to a number of firms struggling to stay in business. We assumed above that computer chip manufacturers were price takers but in reality the computer chip market is not perfectly competitive. However, the fact that so many smaller manufacturers struggled to survive as chip technology expanded at such a rapid rate from the early 1990s shows that even markets dominated now by a relatively small number of firms exhibit many of the features we have described so far.

When analysing the effects of technology, it is important to keep in mind that what is bad for manufacturers is not necessarily bad for society as a whole. Improvement in computing power technology can be bad for manufacturers who find it difficult to survive unless they are very large, but it represents good news for consumers of this computing power (ultimately the users of PCs, laptops, smartphones and so on) who pay less for computing.

Why Do Prices of Ski Holidays Differ so Much at Different Times of the Season?

Ski holidays in Europe are becoming ever more popular. There were over one million people from the UK who were part of the snowsports travel market in 2011². For an increasing number of people the pleasure of a holiday on the slopes is a part of the winter but people also face considerable changes in the prices that they have to pay for their holiday. For example, a quick check of a ski company website for the 2011–2012 season revealed the prices shown in Table 5.1 for seven-night ski trips per person to Austria leaving from London.

There is a considerable variation in the prices that holidaymakers have to pay – £490 being the greatest difference. Prices are particularly high leaving on 29 December and 12 February – why? The reason is that at this time of the season the demand for ski

TABLE 5.1

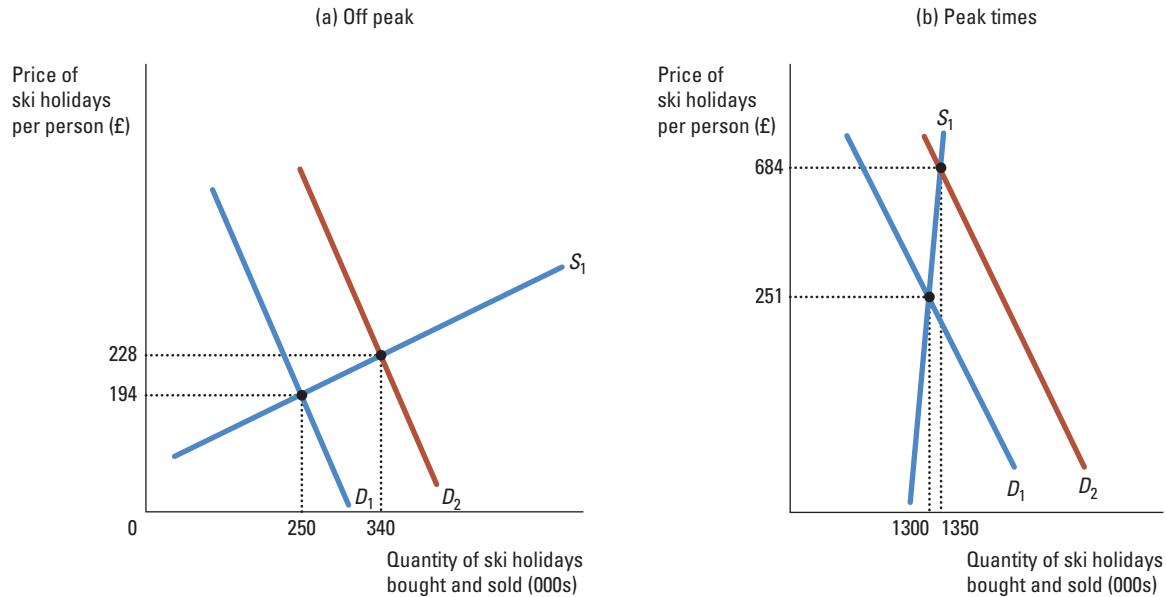
Prices for 7-night Ski Holidays in Austria, From London

Departure date	Price per person (£)
29 December 2011	262
8 January 2012	228
15 January 2012	194
22 January 2012	194
29 January 2012	194
5 February 2012	251
12 February 2012	684
19 February 2012	248
26 February 2012	392

²<http://www.skiclub.co.uk/assets/files/documents/snowsportsanalysis2011.pdf>

FIGURE 5.16**The Supply of Ski Holidays in Europe**

Panel (a) shows the market for ski holidays in off-peak times. The supply curve S_1 is relatively elastic in the short run. An increase in demand from D_1 to D_2 at this time leads to a relatively small increase in price because the increase can be accommodated by releasing some of the spare capacity that tour operators have. Panel (b) shows the market during peak times. The supply of holidays shown by the curve S_1 is relatively inelastic in the short run. If demand now increases from D_1 to D_2 the result will be a sharp rise in price.



holidays increases dramatically because they coincide with annual holiday periods; 29 December is part of the Christmas/New Year holidays and when schoolchildren are also on holiday; 12 February is also a major school holiday for many UK children.

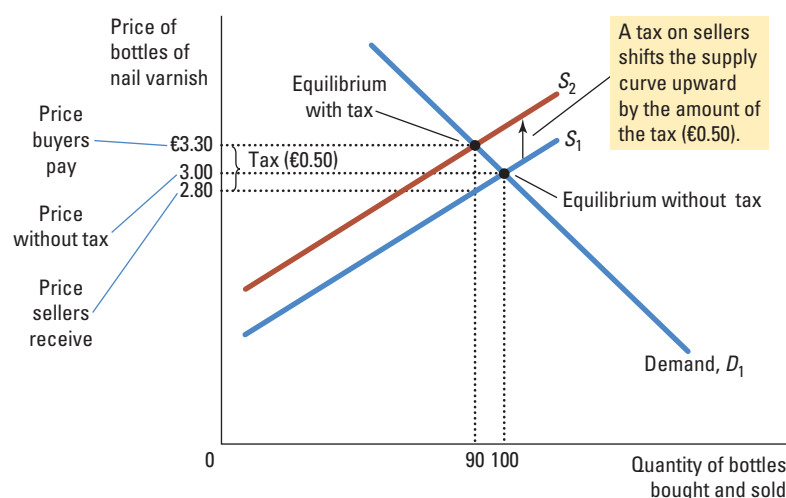
The supply of ski holidays does have a limit – there will be a finite number of accommodation places and passes for ski-lifts and so the elasticity of supply is relatively inelastic (see Panel b, Figure 5.16). It is difficult for tour operators to increase supply of accommodation or ski-passes easily in the short run in the face of rising demand at these times. The result is that the increase in demand for ski holidays at these peak times results in prices rising significantly to choke off the excess demand. If holiday-makers are able to be flexible about when they take their holidays then they will be able to benefit from lower prices for the same holiday. Away from these peak periods the demand for ski holidays is lower and so tour operators have spare capacity – the supply curve out of peak times is more elastic in the short run. If there was a sudden increase in demand in mid-January, for example, then tour operators would have the capacity to accommodate that demand so prices would not rise as much as when that capacity is strictly limited.

Cases for which supply is very inelastic in the short run but more elastic in the long run may see different prices exist in the market. Air and rail travel and the use of electricity may all be examples where prices differ markedly at peak times compared with off-peak times because of supply constraints and the ability of firms to be able to discriminate between customers at these times.

Quick Quiz How might a drought that destroys half of all farm crops be good for farmers? If such a drought is good for farmers, why don't farmers destroy their own crops in the absence of a drought?

FIGURE 6.8**A Tax on Sellers**

When a tax of €0.50 is levied on sellers, the supply curve shifts up by €0.50 from S_1 to S_2 . The equilibrium quantity falls from 100 to 90 bottles. The price that buyers pay rises from €3.00 to €3.30. The price that sellers receive (after paying the tax) falls from €3.00 to €2.80. Even though the tax is levied on sellers, buyers and sellers share the burden of the tax.



a quantity of nail varnish remover as if the price were €0.50 lower than it is. Put differently, to induce sellers to supply any given quantity, the market price must now be €0.50 higher to compensate for the effect of the tax. Thus, as shown in Figure 6.8, the supply curve shifts *upward* from S_1 to S_2 by exactly the size of the tax (€0.50).

Step Three Having determined how the supply curve shifts, we can now compare the initial and the new equilibrium. The figure shows that the equilibrium price of nail varnish remover rises from €3.00 to €3.30, and the equilibrium quantity falls from 100 to 90 bottles. The tax reduces the size of the nail varnish remover market and buyers and sellers share the burden of the tax. Because the market price rises, buyers pay €0.30 more for each bottle of nail varnish remover than they did before the tax was enacted. Sellers receive a higher price than they did without the tax, but the effective price (after paying the tax) falls from €3.00 to €2.80.

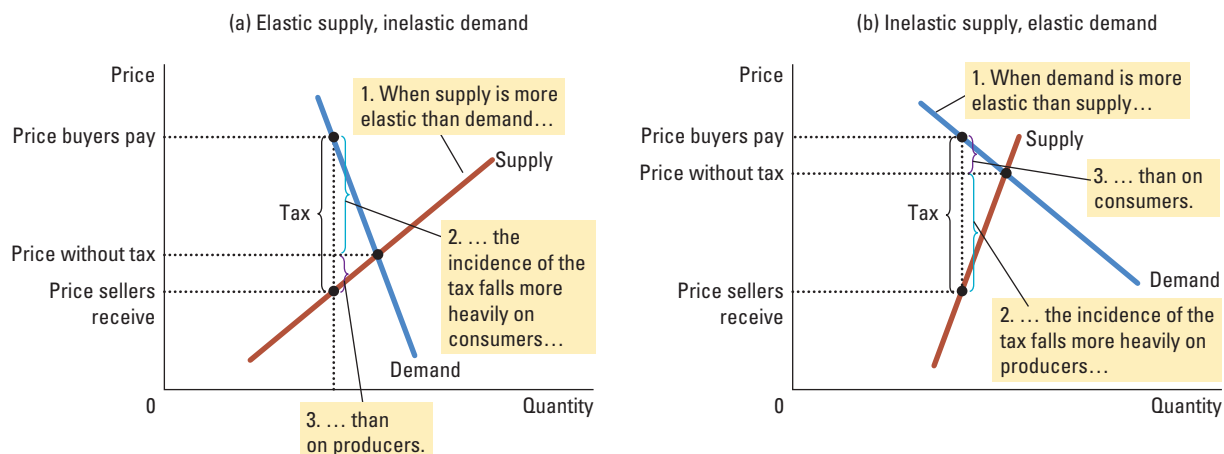
Implications A tax on sellers places a wedge between the price that buyers pay and the price that sellers receive. The wedge between the buyers' price and the sellers' price is the same, and would be the same regardless of whether the tax is levied on buyers or sellers. In reality most governments levy taxes on sellers rather than on buyers, however. The wedge shifts the relative position of the supply and demand curves. In the new equilibrium, buyers and sellers share the burden of the tax.

Elasticity and Tax Incidence

When a good is taxed, buyers and sellers of the good share the burden of the tax. But how exactly is the tax burden divided? Only rarely will it be shared equally. To see how the burden is divided, consider the impact of taxation in the two markets in Figure 6.9. In both cases, the figure shows the initial demand curve, the initial supply curve, and a tax that drives a wedge between the amount paid by buyers and the amount received by sellers. The difference in the two panels is the relative elasticity of supply and demand. Panel (a) of Figure 6.9 shows a tax in a market with very elastic supply and relatively

FIGURE 6.9**How the Burden of a Tax is Divided**

In panel (a), the supply curve is elastic and the demand curve is inelastic. In this case, the price received by sellers falls only slightly, while the price paid by buyers rises substantially. Thus, buyers bear most of the burden of the tax. In panel (b), the supply curve is inelastic and the demand curve is elastic. In this case, the price received by sellers falls substantially, while the price paid by buyers rises only slightly. Thus, sellers bear most of the burden of the tax.



inelastic demand. That is, sellers are very responsive to changes in the price of the good (so the supply curve is relatively flat), whereas buyers are not very responsive (so the demand curve is relatively steep). When a tax is imposed on a market with these elasticities, the price received by sellers does not fall much, so sellers bear only a small burden. By contrast, the price paid by buyers rises substantially, indicating that buyers bear most of the burden of the tax. If the price elasticity of demand is low (represented in Panel (a) by a steep curve) then demand will fall proportionately less in response to a rise in price – buyers are not very price sensitive. The seller can shift the burden of the tax onto the buyer, safe in the knowledge that demand will only fall by a relatively small amount.

Panel (b) of Figure 6.9 shows a tax in a market with relatively inelastic supply and very elastic demand (represented by a flatter curve). In this case, sellers are not very responsive to changes in the price (so the supply curve is steeper), while buyers are very responsive. The figure shows that when a tax is imposed, the price paid by buyers does not rise much, while the price received by sellers falls substantially. Thus, sellers bear most of the burden of the tax. In this case, sellers know that if they try to pass on the tax to buyers that demand will fall by a relatively large amount.

The two panels of Figure 6.9 show a general lesson about how the burden of a tax is divided: a tax burden falls more heavily on the side of the market that is less elastic. Why is this true? In essence, the elasticity measures the willingness of buyers or sellers to leave the market when conditions become unfavourable. A small elasticity of demand means that buyers do not have good alternatives to consuming this particular good. A small elasticity of supply means that sellers do not have good alternatives to producing this particular good. When the good is taxed, the side of the market with fewer good alternatives cannot easily leave the market and must, therefore, bear more of the burden of the tax.

Quick Quiz In a supply and demand diagram, show how a tax on car sellers of €1000 per car affects the quantity of cars sold and the price of cars. In your diagram, show the change in the price paid by car buyers and the change in price received by car sellers.

total revenue minus total cost, would increase. Hence, if marginal revenue is greater than marginal cost, as it is at Q_1 , the firm can increase profit by increasing production.

A similar argument applies when output is at Q_2 . In this case, marginal cost is greater than marginal revenue. If the firm reduced production by 1 unit, the costs saved (MC_2) would exceed the revenue lost (MR_2). Therefore, if marginal revenue is less than marginal cost, as it is at Q_2 , the firm can increase profit by reducing production. Where do these marginal adjustments to the level of production end? Regardless of whether the firm begins with production at a low level (such as Q_1) or at a high level (such as Q_2), the firm will eventually adjust production until the quantity produced reaches Q_{MAX} . This analysis confirms the general rule for profit maximization: at the profit-maximizing level of output, marginal revenue and marginal cost are exactly equal.

If we assume that the firm aims to maximize profit then it aims at an output level where $MC = MR$. In Figure 8.2, that quantity of output is Q_{MAX} .

FYI



The Mathematics of Profit Maximization

Assume that the equations for an inverse demand curve and total cost facing a firm are given as:

$$P = 230 - Q$$

and:

$$TC = 40 + Q^2$$

We know that we have to differentiate both equations to find the marginal revenue and marginal cost and that the power function rule is:

$$\frac{dy}{dx} = nx^{n-1}$$

$$\text{So, } MC = \frac{d(TC)}{dq} = -2q$$

$$TR = P \times Q$$

$$TR = (230 - Q)Q$$

$$TR = 230Q - Q^2$$

$$MR = \frac{d(TR)}{dq} = 230 - 2q$$

Setting the profit maximizing output at $MC = MR$ gives us:

$$-Q = 230 - 2Q$$

$$Q = 230$$

Profit would be maximized at an output level of 230. It should be noted that that the outcome figures are not always neat round numbers. The point, as has been made earlier, is that this is a model to help firms make decisions and as a result this firm might aim for an output level between two values and know that it is maximizing profit.

JEOPARDY PROBLEM

A firm finds itself in a position of seeing an increase in sales growth of 15 per cent over the previous year and productivity is up by 5 per cent. The firm's directors, however, have reported losses of €25 million compared to a profit of €15 million in the previous year. How could this have happened?

BREAK-EVEN ANALYSIS

break-even the level of output/sales at which total cost equals total revenue found by dividing the fixed costs by the contribution (selling price minus variable costs per unit)

When a new business starts up or if an existing business decides to develop a new product one goal may be to achieve break-even. **Break-even** refers to the output level at which the total costs of production are equal to the total revenue generated from selling that output. Many firms will look at break-even analysis as part of their planning tools.

They can look at the variables involved and how different figures plugged into these variables can affect decision making.

Figure 8.3 shows how we can represent break-even graphically. The vertical axis shows both costs and revenues and the horizontal axis shows output and sales. We know that a firm faces fixed costs which must be paid regardless of whether it produces any output. As a result the Fixed Cost curve FC is represented as a horizontal line at a cost of C_1 . Variable costs (VC) are zero when the firm produces no output but rise as output rises. The VC curve shows these costs rising in direct proportion to output. Total cost is the sum of variable costs and fixed cost ($TC = VC + FC$) and so the TC curve has a vertical intercept equal to the level of fixed costs (C_1) and then rises as output rises. The vertical distance between the TC and VC curve is constant because of the proportional relationship between output and VC assumed above.

The total revenue curve is dependent on the price that the firm chooses to charge for its product. The higher the price charged the steeper the TR curve. The break-even level of output in Figure 8.3 is the amount where $TC = TR$ which is depicted as Q_{BE} . At this output level the TC are C_1 and the TR is the same at R_1 shown by C_1/R_1 in Figure 8.3. At any output level below Q_{BE} the firm will find that its TC is greater than TR and so it will make a loss on that output. The loss made is represented by the area shown by the shaded triangle A.

At output levels above Q_{BE} , for example Q_1 , the firm's TR, shown by R_2 , is greater than its TC, shown by C_2 and as a result it makes profits on this output. The amount of profit made at output Q_1 is represented by the shaded triangle B.

If the firm were operating at Q_1 then it could experience a fall in sales and still be generating a profit provided sales did not fall below the break-even output Q_{BE} . The distance between the break-even output and current production where TR is greater than TC is called the **margin of safety**.

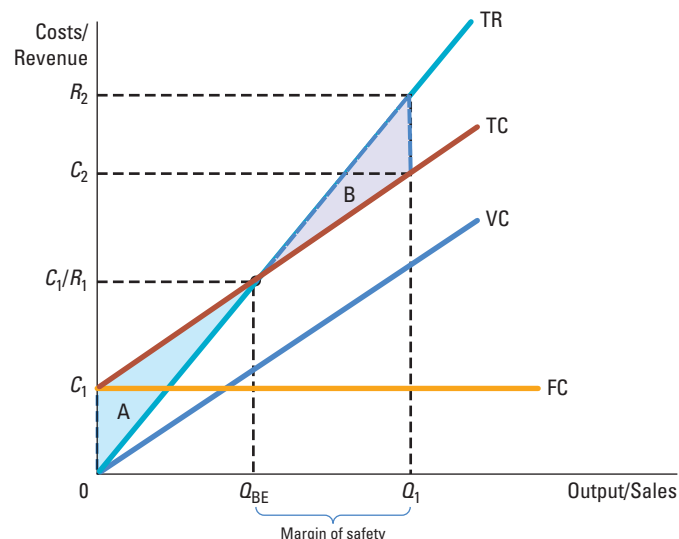
We can also look at the break-even point in a more mathematical way. A firm needs to aim at selling its output at a price greater than the variable cost of production.

margin of safety the distance between the break-even output and current production where total revenue is greater than total cost

FIGURE 8.3

Break-Even Analysis

A break-even chart shows a firm's total cost, made up of fixed and variable costs and its total revenue. The output level where $TR = TC$ is the break-even point shown by output Q_{BE} . Any output level below Q_{BE} would mean that the firm was producing at a loss because TC would be greater than TR but any output above Q_{BE} would mean the firm was operating at a profit. At output level Q_1 , the amount of profit is indicated by the shaded triangle B. The distance between Q_{BE} and Q_1 is called the margin of safety and denotes how far sales could fall before the firm starts to make losses.



contribution the difference between the selling price and the variable cost per unit

Imagine a firm producing chocolate bars. The ingredient costs are €0.30 per bar and the labour costs are estimated at €0.10 per bar. Total variable costs are €0.40 per bar. If the selling price of the chocolate bar is €0.60 then every bar sold covers the variable costs of production and leaves €0.20 which can be used to help pay off the fixed costs which also have to be paid. This sum is called the **contribution**. The contribution is the difference between the selling price and the variable cost per unit. Knowing the contribution per unit the break-even output can be given by:

$$\text{Break-even} = \frac{\text{Fixed costs}}{\text{Contribution per unit}}$$

The break-even point can be affected by the costs of production, therefore, any change in fixed costs, for example, will shift the FC line up or down and thus the TC curve. Equally, changes in raw materials costs can affect the variable costs and as a result the TC curve.

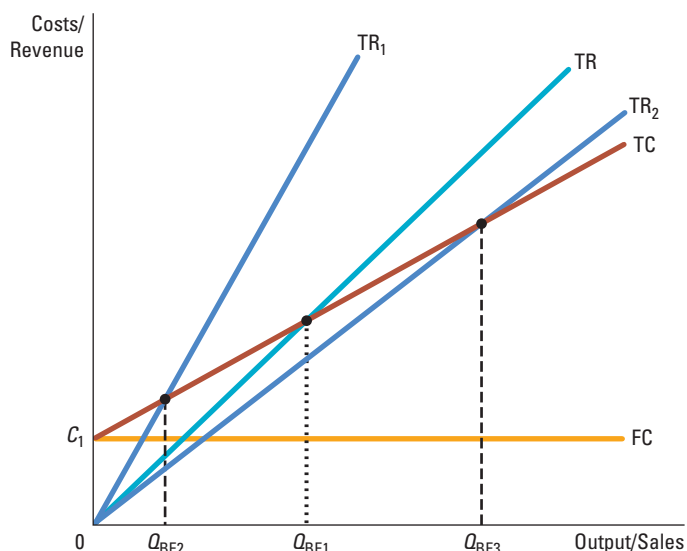
It may be that the firm has little control over changes to its costs but it can use break-even analysis to assess the impact of the changes and make decisions as a result. What many firms do have control over is the price they charge. We know that $TR = P \times Q$. If a firm changes its price the shape of the TR curve will alter. If a firm chooses to increase its price then the TR curve will pivot and become steeper as shown in Figure 8.4. With the TR curve now indicated as TR_1 , the firm will not have to sell as many products to cover its costs and so the break-even output will be lower at Q_{BE2} . A firm might choose to reduce its price in which case the TR curve will become flatter. In this case TR curve will now be represented as TR_2 and the firm will now have to sell more products in order to cover its costs and so the break-even output would rise to Q_{BE3} .

Limitations of break-even analysis As mentioned above, break-even is best seen as a planning tool and the outline given here is a simplistic one. There are a number of limitations with break-even analysis. Firstly there is an assumption that all output is sold. This is not the case for many firms where production takes place but products are not immediately sold and as such firms have stock (inventory).

FIGURE 8.4

The Effect on Break-Even Output of Changes in Price

Total revenue is found by multiplying price times the quantity sold. If the price is increased the TR curve will become steeper as shown by the curve TR_1 . As a result, the firm would now only need to produce output Q_{BE2} in order to break-even. If price was reduced the TR curve becomes flatter indicated by curve TR_2 and the firm would have to sell a higher output level, Q_{BE3} in order to break even.



Secondly, the shape of the TR and TC curves is unlikely to be smooth as increases in output can be associated with economies of scale and not every additional product a firm sells can be sold at the same price. Equally, given the dynamic nature of markets, any estimation of costs or revenues are not likely to be relevant for very long.

Thirdly, we know that if a firm changes price, demand will be affected. Break-even analysis says nothing about what the effect of a price change on actual sales will be, nor how long it will take a firm to sell the output level to break even. The effect on sales of a change in price will be dependent on the price elasticity of demand.

CASE STUDY

On 2 February 2011, Rupert Murdoch, the chairman and CEO of News Corporation, launched a new business product *The Daily*. *The Daily* is a newspaper designed exclusively for Apple's iPad. It has a mission to: 'provide the best news experience by combining world-class storytelling with the unique interactive capabilities of the iPad'. Customers can download the app and get two weeks free access before having to subscribe for US\$0.99 a week or \$39.99 a year.



NEWS INTERNATIONAL

Clearly there will have been a cost to setting the product up in the first place which has been estimated at \$30 million. The running costs have been reported to be \$500 000 a week or \$26 million a year. Given these figures we can get some idea of what the sales of subscriptions need to be for the product to break even. *The Daily* would need some 651 163 subscribers a year just to cover the running costs of the product, let alone the set up costs (which we can loosely refer to as the fixed costs) of \$30 million.

What are the chances that it will be able to recruit this number of subscribers? Well, the first thing to note is that the price being charged for subscriptions is relatively low, hence the high numbers needed to break even. To get some idea we might look at data on other subscriptions to online news services. French newspaper *Le Monde* is reported to have around 39 000 web-only subscribers, *The Times*, another of Murdoch's papers, reported subscribers to its digital version as rising to 110 000 in October 2011. A Long Island, New York daily newspaper, *Newsday*, was said to have received just 35 subscriptions after three months of launch. *The Wall Street Journal* is estimated to have around 400 000 online paid subscriptions and *Wired* magazine began its online life with 100 000 subscribers but this fell to 23 000 half a year later. Given these figures, some from established brands, the numbers for *The Daily* do not look promising. Even if subscribers can be encouraged to download the app and pay, Apple will take a cut from the subscription amount so the actual revenues would be lower than that estimated here.

However, a newspaper's revenue does not just come from subscriptions; advertising also brings in revenue. The next question is: is the amount of advertising spend sufficient to make up any gap between costs and subscriber revenue to help *The Daily* break even. Here again, things do not look that promising. Digital intelligence firm, eMarketer, estimates that the total spent on advertising in 2011 was around \$1 billion. *The Daily*, therefore, needs to get a share of that spending and given this spending covers all mobile advertising business, it has been

THE COSTS OF PRODUCTION

In Chapters 4 and 5 we looked at how markets operate. In this and subsequent chapters we are going to look in more detail at businesses and how they operate. We will refer to these businesses as ‘firms’.

The economy is made up of thousands of firms that produce the goods and services we enjoy every day: Mercedes Benz produces cars, Miele produce kitchen appliances and Nestlé produces food and drink. Some firms, such as these three, are large; they employ thousands of workers and have thousands of shareholders who share in the firms’ profits. Other firms, such as the local hairdresser’s shop or pizzeria, are small; they employ only a few workers and may be owned by a single person or family.

As we examine firm behaviour in more detail you will gain a better understanding of what decisions lie behind the supply curve in a market which we introduced in Chapter 4. In addition, it will introduce you to a part of economics called *industrial organization* – the study of how firms’ decisions regarding prices and quantities depend on the market conditions they face. The town in which you live, for instance, may have several restaurants but only one water supply company. How does this difference in the number of firms affect the prices in these markets and the efficiency of the market outcomes? The field of industrial organization addresses exactly this question.

Before we turn to these issues, however, we need to discuss the costs of production. All firms, from Air France to your local baker’s shop, incur costs as they make the goods and services that they sell. A firm’s costs are a key determinant of its production and pricing decisions. In this chapter, we define some of the variables that economists use to measure a firm’s costs, and we consider the relationships among them. We will also introduce a model to look at how a firm might maximize production given certain constraints such as its budgets and the factors of production it has available.

WHAT ARE COSTS?

We begin our discussion of costs at Primo’s Pizza Factory. Primo, the owner of the firm, buys flour, tomatoes, mozzarella cheese, salami and other pizza ingredients. He also buys



Pizza production, like most businesses, relies on some basic factor inputs – land, capital and labour. These are combined to produce products (pizzas in this example) which are then sold to customers and generate revenue. If, over time, the revenue generated by sales of pizzas is greater than the total cost of making them, the firm will make a profit.

the mixers and ovens and hires workers to run this equipment, all of which is housed inside the factory building. He then sells the resulting pizzas to consumers. By examining some of the issues that Primo faces in his business, we can learn some lessons about costs that apply to all firms in the economy.

Total Revenue, Total Cost and Profit

We have seen in Chapter 4 that the amount a firm sells multiplied by the price it sells the products at is the total revenue the firm receives. If Primo sells 500 000 pizzas a year and the average price of each pizza sold is €5 his total revenue will be $5 \times 500\,000 = \text{€}2.5$ million.

To produce those 500 000 pizzas, Primo will have had to employ labour, buy machinery and equipment, run and maintain that equipment, buy the raw materials to make the pizzas, pay rent or a mortgage on the factory, pay off loans he may have secured (both the sum borrowed and any interest charged), pay for market research, marketing costs such as advertising and promotion, administration costs (such as managing the firm's payroll, monitoring the finances, processing sales and purchase invoices), and many other everyday payments down to the cost of using the telephone, energy use, postage, maintaining the buildings and so on. These represent Primo's **total cost**.

If Primo subtracts all the costs over the year from the revenues received in the same year he will either have a surplus (i.e. his revenue will be greater than his costs) or possibly have spent more on costs than received in revenues. **Profit** is a firm's total revenue minus its total cost. That is:

$$\text{Profit} = \text{Total revenue} - \text{Total cost}$$

We can express this in the formula:

$$\pi = \text{TR} - \text{TC}$$

where the Greek letter pi (π) represents profit.

We will look at a firm's objectives in more detail in Chapter 8 but let us assume for the moment that Primo's objective is to make his firm's profit as large as possible – in other words, he wants to maximize profit. Primo needs to be able to measure his total revenue and his total costs. We know how to calculate total revenue, in many cases that is a relatively easy thing to do, but the measurement of a firm's total cost is more subtle and open to different interpretations.

Pitfall Prevention It is important to understand the distinction between profit and cash flow – the latter is the money flowing into and out of a business over a period of time, whereas profit takes into consideration the total revenue and total cost. A firm could be profitable but have cash flow problems which could force it out of business.

Costs as Opportunity Costs

When measuring costs at Primo's Pizza Factory, or any other firm, it is important to keep in mind one of the *Ten Principles of Economics* from Chapter 1: the cost of something is what you give up to get it. Recall that the *opportunity cost* of an item refers to all those things that must be forgone to acquire that item. When economists speak of a firm's cost of production, they include the opportunity costs of making its output of goods and services.

A firm's opportunity costs of production are sometimes obvious but sometimes less so. When Primo pays €1000 for a stock of flour, he can no longer use that €1000 to buy something else; he has to sacrifice what else that €1000 could have purchased. Similarly, when Primo hires workers to make the pizzas, the wages he pays are part of the firm's costs.

total cost the market value of the inputs a firm uses in production

profit total revenue minus total cost

Because these costs require the firm to pay out some money, they are called **explicit costs**. By contrast, some of a firm's opportunity costs, called **implicit costs**, do not require a cash outlay. Imagine that Primo is skilled with computers and could earn €100 per hour working as a programmer. For every hour that Primo works at his pizza factory, he gives up €100 in income, and this forgone income is also classed as part of his costs by an economist.

explicit costs input costs that require an outlay of money by the firm

This distinction between explicit and implicit costs highlights an important difference between how economists and accountants analyze a business. Economists are interested in studying how firms make production and pricing decisions. Because these decisions are based on both explicit and implicit costs, economists include both when measuring a firm's costs. By contrast, accountants have the job of keeping track of the money that flows into and out of firms. As a result, they measure the explicit costs but often ignore the implicit costs.

implicit costs input costs that do not require an outlay of money by the firm

The difference between economists and accountants is easy to see in the case of Primo's Pizza Factory. When Primo gives up the opportunity to earn money as a computer programmer, his accountant will not count this as a cost of his pizza business. Because no money flows out of the business to pay for this cost, it never shows up on the accountant's financial statements. An economist, however, will count the foregone income as a cost because it will affect the decisions that Primo makes in his pizza business. This is an important part of thinking like an economist. If the wage as a computer programmer rose from €100 to €500 per hour, the opportunity cost of running his pizza business might now change his decision making. The opportunity cost of running the business in terms of what Primo is sacrificing in foregone income has risen. Primo might decide he could earn more by closing the business and switching to computer programming.

The Cost of Capital as an Opportunity Cost An important implicit cost of almost every firm is the opportunity cost of the financial capital that has been invested in the business. Suppose, for instance, that Primo used €300 000 of his savings to buy his pizza factory from the previous owner. If Primo had instead left this money deposited in a savings account that pays an interest rate of 5 per cent, he would have earned €15 000 per year (assuming simple interest). To own his pizza factory, therefore, Primo has given up €15 000 a year in interest income. This forgone €15 000 is an implicit opportunity cost of Primo's business. An economist views the €15 000 in interest income that Primo gives up every year as a cost of his business, even though it is an implicit cost. Primo's accountant, however, will not show this €15 000 as a cost because no money flows out of the business to pay for it.

To explore further the difference between economists and accountants, let's change the example slightly. Suppose now that Primo did not have the entire €300 000 to buy the factory but, instead, used €100 000 of his own savings and borrowed €200 000 from a bank at an interest rate of 5 per cent. Primo's accountant, who only measures explicit costs, will now count the €100 000 interest paid on the bank loan every year as a cost because this amount of money now flows out of the firm. By contrast, according to an economist, the opportunity cost of owning the business is still €15 000. The opportunity cost equals the interest on the bank loan (an explicit cost of €10 000) plus the forgone interest on savings (an implicit cost of €5 000).

Economic Profit versus Accounting Profit

Now let's return to the firm's objective – profit. Because economists and accountants measure costs differently, they also measure profit differently. An economist measures a firm's **economic profit** as the firm's total revenue minus all the opportunity costs (explicit and implicit) of producing the goods and services sold. An accountant measures the firm's **accounting profit** as the firm's total revenue minus only the firm's explicit costs.

economic profit total revenue minus total cost, including both explicit and implicit costs

accounting profit total revenue minus total explicit cost

FIGURE 9.1**Economists versus Accountants**

Economists include all opportunity costs when analyzing a firm, whereas accountants measure only explicit costs. Therefore, economic profit is smaller than accounting profit.

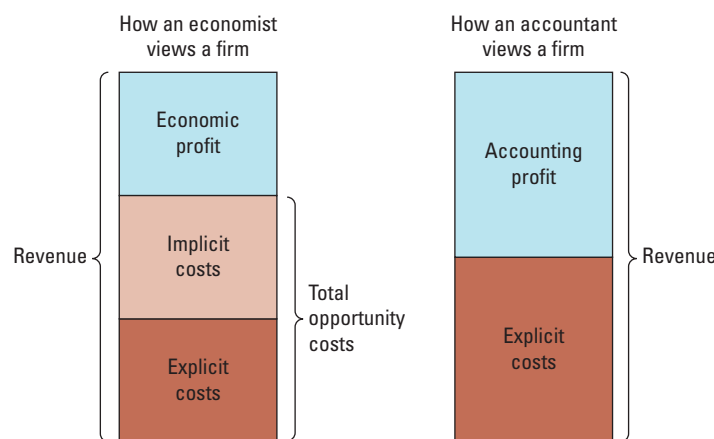


Figure 9.1 summarizes this difference. Notice that because the accountant ignores the implicit costs, accounting profit is usually larger than economic profit. For a business to be profitable from an economist's standpoint, total revenue must cover all the opportunity costs, both explicit and implicit.

Quick Quiz Richard Collishaw is a dairy farmer who is also a skilled metal worker. He makes unique garden sculptures that could earn him €40 an hour. One day, he spends 10 hours milking his dairy herd. The cost of operating the machinery used in the milking process is €200. What opportunity cost has he incurred? What cost would his accountant measure? If the milk produced will yield €400 in revenue, does Richard earn an accounting profit? Does he earn an economic profit? Would you advise Richard to continue as a farmer or switch to metal working?

PRODUCTION AND COSTS

We saw in Chapter 2 how the transformation process was characterized by firms buying inputs to carry out production. There is clearly a relationship between the amount of inputs used, the cost of production and the amount produced. In this section we examine the link between a firm's production process and its total cost using Primo's Pizza Factory as an example.

The Production Function

production function the relationship between quantity of inputs used to make a good and the quantity of output of that good

The relationship between factor inputs can be expressed as a mathematical relationship called a production function. The **production function** shows the amount of output which can be produced given different combinations of factor inputs, land, labour and capital. This should be fairly intuitive – if Primo increases the number of people working in his factory then the amount of pizzas produced would be likely to rise. However, it could also

be possible that Primo might decide to cut the number of workers he employs and buy a machine which can do their job and still increase the amount of pizzas produced.

In reality, firms face complex decisions over how to organize production and a whole discipline has grown up around this called operations management which in turn is closely linked to organizational behaviour and organizational design. Some firms rely on large amounts of labour compared to other factors for production and are referred to as *labour intensive*, whereas other firms have relatively small amounts of labour but very large amounts of capital and are referred to as *capital intensive*.

Let us assume there are two factor inputs, labour (L) and capital (K). The production function can be expressed as follows:

$$Q = f(L_1, K_1)$$

This states that the level of output (Q) is dependent upon the amount of labour and capital employed. More complex production functions are developed which include more specific dependent variables which can have different values. For example, we can specify a particular amount of labour, a particular amount of capital, as well as a certain quantity of land. Using this as a model, we can then vary some factor inputs whilst holding the others constant and analyze the effect on the level of output.

Obviously, this is a very simple introduction to the production function and as we will see later, changing the amount of factor inputs can have different consequences.



what if... Primo's factory had a total floor area of 2000 square metres, three-quarters of which was taken up with machinery. Primo employs 50 workers. If he employs a further ten workers, will output of pizzas increase? If he employs another ten will output continue to increase? What do you think would happen to output if Primo continued to increase the number of workers he employs?

The Short Run and the Long Run

In business, the distinction between the short run and the long run is of considerable importance. The **short run** is defined as the period of time in which some factors of production cannot be altered. The **long run** is that period of time when all factors of production can be altered. The distinction is really a conceptual one rather than a specific one. Many businesses will not be able to calculate the short run or the long run but they will know that the distinction will be an important consideration in their decision making. The long run for a market trader in a local street market may be weeks or months but for an energy supply company could be 20 years.

short run the period of time in which some factors of production cannot be changed

long run the period of time in which all factors of production can be altered



For a market trader increasing the amount of labour employed, securing a new pitch and buying a new stall might only take three months; for an energy supply company, building a new power station could take many years as it goes through the process of designing the station, securing planning permission, signing contracts with construction companies and then the actual construction process.

Short-Run Production Relationships

A firm's costs of producing a specific output depend on the prices of the needed resources and the quantities of resources (inputs) needed to produce that output. Resource supply and demand determine resource prices. The technological aspects of production, specifically the relationships between inputs and output, determine the quantities of resources needed. Our focus will be on the *labor*-output relationship, given a fixed plant capacity. But before examining that relationship, we need to define three terms:

- **Total product (TP)** is the total quantity, or total output, of a particular good or service produced.
- **Marginal product (MP)** is the extra output or added product associated with adding a unit of a variable resource, in this case labor, to the production process. Thus,

$$\text{Marginal product} = \frac{\text{change in total product}}{\text{change in labor input}}$$

- **Average product (AP)**, also called labor productivity, is output per unit of labor input:

$$\text{Average product} = \frac{\text{total product}}{\text{units of labor}}$$

In the short run, a firm can for a time increase its output by adding units of labor to its fixed plant. But by how much will output rise when it adds more labor? Why do we say “for a time”?

Law of Diminishing Returns

The answers are provided in general terms by the **law of diminishing returns**. This law assumes that technology is

ORIGIN OF THE IDEA

○ 7.1

Law of diminishing returns

fixed and thus the techniques of production do not change. It states that as successive units of a variable resource (say, labor) are added to a fixed resource (say, capital or land), beyond some point the extra, or marginal, product that can be attributed to each additional unit of the variable resource will decline. For example, if additional workers are hired to work with a constant amount of capital equipment, output will eventually rise by smaller and smaller amounts as more workers are hired.

Rationale Suppose a farmer has a fixed resource—80 acres of land—planted in corn. If the farmer does not cultivate the cornfields (clear the weeds) at all, the yield will be

CONSIDER THIS . . .



Diminishing Returns from Study

Here is a noneconomic example of a relationship between “inputs” and “output” that may help you better understand the idea of

diminishing returns. Suppose for an individual that

$$\text{Total course learning} = f(\text{intelligence, quality of course materials, instructor effectiveness, class time, and study time})$$

where f means “function of” or “depends on.” So this relationship supposes that total course learning depends on intelligence (however defined), quality of course materials such as the textbook, the effectiveness of the instructor, the amount of class time, and the amount of personal study time outside the class.

For analytical purposes, let's assume that one's intelligence, the quality of course materials, the effectiveness of the instructor, and the amount of class time are *fixed*—meaning they do not change over the length of the course. Now let's add units of study time per day over the length of the course to “produce” greater course learning. The first hour of study time per day increases total course learning. Will the second hour enhance course learning by as much as the first? By how much will the third, fourth, fifth, . . . fifteenth hour of study per day contribute to total course learning relative to the *immediate previous hour*?

We think you will agree that eventually diminishing returns to course learning will set in as successive hours of study are added each day. At some point the marginal product of an extra hour of study time will decline and, at some further point, become zero.

This is also true of production relationships within firms. As successive units of a variable input (say, labor) are added to a fixed input (say, capital), the marginal product of the variable input eventually declines. In short, diminishing returns will occur sooner or later. Total product eventually will rise at a diminishing rate, reach a maximum, and then decline.

40 bushels per acre. If he cultivates the land once, output may rise to 50 bushels per acre. A second cultivation may increase output to 57 bushels per acre, a third to 61, and a fourth to 63. Succeeding cultivations will add less and less to the land's yield. If this were not so, the world's needs for corn could be fulfilled by extremely intense cultivation of this single 80-acre plot of land. Indeed, if diminishing returns did not occur, the world could be fed out of a flowerpot. Why not? Just keep adding more seed, fertilizer, and harvesters!

TABLE 7.1 Total, Marginal, and Average Product: The Law of Diminishing Returns

(1) Units of the Variable Resource (Labor)	(2) Total Product (TP)	(3) Marginal Product (MP), Change in (2)/ Change in (1)	(4) Average Product (AP), (2)/(1)
0	0		—
1	10	10	10.00
2	25	15	12.50
3	45	20	15.00
4	60	15	15.00
5	70	10	14.00
6	75	5	12.50
7	75	0	10.71
8	70	−5	8.75

The law of diminishing returns also holds true in non-agricultural industries. Assume a wood shop is manufacturing furniture frames. It has a specific amount of equipment such as lathes, planes, saws, and sanders. If this shop hired just one or two workers, total output and productivity (output per worker) would be very low. The workers would have to perform many different jobs, and the advantages of specialization would not be realized. Time would be lost in switching from one job to another, and machines would stand idle much of the time. In short, the plant would be understaffed, and production would be inefficient because there would be too much capital relative to the amount of labor.

The shop could eliminate those difficulties by hiring more workers. Then the equipment would be more fully used, and workers could specialize on doing a single job. Time would no longer be lost switching from job to job. As more workers were added, production would become more efficient and the marginal product of each succeeding worker would rise.

But the rise could not go on indefinitely. Beyond a certain point, adding more workers would cause overcrowding. Since workers would then have to wait in line to use the machinery, they would be underused. Total output would increase at a diminishing rate because, given the fixed size of the plant, each worker would have less capital equipment to work with as more and more labor was hired. The marginal product of additional workers would decline because there would be more labor in proportion to the fixed amount of capital. Eventually, adding still more workers would cause so much congestion that marginal product would become negative and total product would decline. At the extreme, the addition of more and more labor would exhaust all the standing room, and total product would fall to zero.

Note that the law of diminishing returns assumes that all units of labor are of equal quality. Each successive worker is presumed to have the same innate ability, motor coordination, education, training, and work experience. Marginal product ultimately diminishes, but not because successive workers are less skilled or less energetic. It declines because the firm is using more workers relative to the amount of plant and equipment available.

Tabular Example Table 7.1 is a numerical illustration of the law of diminishing returns. Column 2 shows the total product, or total output, resulting from combining each level of a variable input (labor) in column 1 with a fixed amount of capital.

Column 3 shows the marginal product (MP), the change in total product associated with each additional unit of labor. Note that with no labor input, total product is zero;

WORKED PROBLEMS

W 7.2

Total, marginal, and average product

a plant with no workers will produce no output. The first three units of labor reflect increasing marginal returns, with

marginal products of 10, 15, and 20 units, respectively. But beginning with the fourth unit of labor, marginal product diminishes continuously, becoming zero with the seventh unit of labor and negative with the eighth.

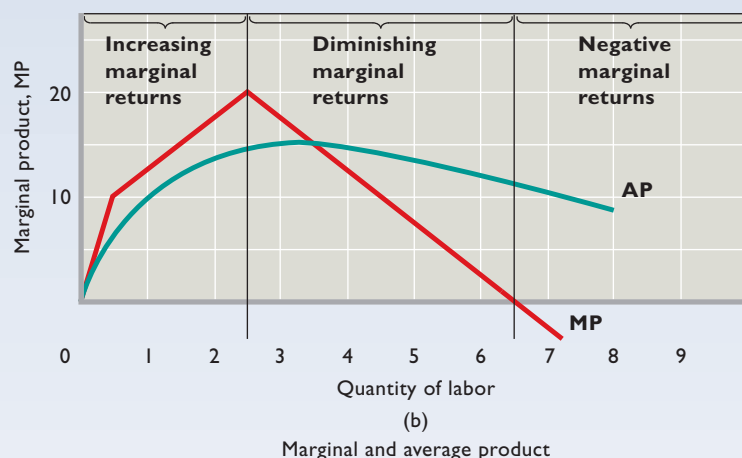
Average product, or output per labor unit, is shown in column 4. It is calculated by dividing total product (column 2) by the number of labor units needed to produce it (column 1). At 5 units of labor, for example, AP is 14 (= 70/5).

Graphical Portrayal Figure 7.2 (Key Graph) shows the diminishing-returns data in Table 7.1 graphically and further clarifies the relationships between total, marginal,

key graph



FIGURE 7.2 The law of diminishing returns. (a) As a variable resource (labor) is added to fixed amounts of other resources (land or capital), the total product that results will eventually increase by diminishing amounts, reach a maximum, and then decline. (b) Marginal product is the change in total product associated with each new unit of labor. Average product is simply output per labor unit. Note that marginal product intersects average product at the maximum average product.



QUICK QUIZ FOR FIGURE 7.2

- Which of the following is an assumption underlying these figures?
 - Firms first hire “better” workers and then hire “poorer” workers.
 - Capital and labor are both variable, but labor increases more rapidly than capital.
 - Consumers will buy all the output (total product) produced.
 - Workers are of equal quality.
- Marginal product is:
 - the change in total product divided by the change in the quantity of labor.
 - total product divided by the quantity of labor.
 - always positive.
 - unrelated to total product.
- Marginal product in graph (b) is zero when:
 - average product in graph (b) stops rising.
 - the slope of the marginal-product curve in graph (b) is zero.
 - total product in graph (a) begins to rise at a diminishing rate.
 - the slope of the total-product curve in graph (a) is zero.
- Average product in graph (b):
 - rises when it is less than marginal product.
 - is the change in total product divided by the change in the quantity of labor.
 - can never exceed marginal product.
 - falls whenever total product in graph (a) rises at a diminishing rate.

Answers: 1. d; 2. a; 3. d; 4. a

and average products. (Marginal product in Figure 7.2b is plotted halfway between the units of labor since it applies to the addition of each labor unit.)

Note first in Figure 7.2a that total product, TP, goes through three phases: It rises initially at an increasing rate; then it increases, but at a diminishing rate; finally, after reaching a maximum, it declines.

Geometrically, marginal product—shown by the MP curve in Figure 7.2b—is the slope of the total-product curve. Marginal product measures the change in total product associated with each succeeding unit of labor. Thus, the three phases of total product are also reflected in marginal product. Where total product is increasing at an increasing rate, marginal product is rising. Here, extra units of labor are adding larger and larger amounts to total product. Similarly, where total product is increasing but at a decreasing rate, marginal product is positive but falling. Each additional unit of labor adds less to total product than did the previous unit. When total product is at a maximum, marginal product is zero. When total product declines, marginal product becomes negative.

Average product, AP (Figure 7.2b), displays the same tendencies as marginal product. It increases, reaches a maximum, and then decreases as more and more units of labor are added to the fixed plant. But note the relationship between marginal product and average product: Where marginal product exceeds average product, average product rises. And where marginal product is less than average product, average product declines. It follows that marginal product intersects average product where average product is at a maximum.

This relationship is a mathematical necessity. If you add a larger number to a total than the current average of that total, the average must rise. And if you add a smaller number to a total than the current average of that total, the average must fall. You raise your average examination grade only when your score on an additional (marginal) examination is greater than the average of all your past scores. You lower your average when your grade on an additional exam is below your current average. In our production example, when

the amount an extra worker adds to total product exceeds the average product of all

workers currently employed, average product will rise. Conversely, when the amount an extra worker adds to total product is less than the current average product, average product will decrease.

The law of diminishing returns is embodied in the shapes of all three curves. But, as our definition of the law

of diminishing returns indicates, economists are most concerned with its effects on marginal product. The regions of increasing, diminishing, and negative marginal product (returns) are shown in Figure 7.2b.

Short-Run Production Costs

Production information such as that provided in Table 7.1 and Figures 7.2a and 7.2b must be coupled with resource prices to determine the total and per-unit costs of producing various levels of output. We know that in the short run, resources associated with the firm's plant are fixed. Other resources, however, are variable in the short run. As a result, short-run costs can be either fixed or variable.

Fixed, Variable, and Total Costs

Let's see what distinguishes fixed costs, variable costs, and total costs from one another.

Fixed Costs **Fixed costs** are those costs that do not vary with changes in output. Fixed costs are associated with the very existence of a firm's plant and therefore must be paid even if its output is zero. Such costs as rental payments, interest on a firm's debts, a portion of depreciation on equipment and buildings, and insurance premiums are generally fixed costs; they are fixed and do not change even if a firm produces more. In column 2 of Table 7.2 we assume that the firm's total fixed cost is \$100. By definition, this fixed cost is incurred at all levels of output, including zero. The firm cannot avoid paying fixed costs in the short run.

Variable Costs **Variable costs** are those costs that change with the level of output. They include payments for materials, fuel, power, transportation services, most labor, and similar variable resources. In column 3 of Table 7.2 we find that the total of variable costs changes directly with output. But note that the increases in variable cost associated with succeeding one-unit increases in output are not equal. As production begins, variable cost will for a time increase by a decreasing amount; this is true through the fourth unit of output in Table 7.2. Beyond the fourth unit, however, variable cost rises by increasing amounts for succeeding units of output.

The reason lies in the shape of the marginal-product curve. At first, as in Figure 7.2b, marginal product is increasing, so smaller and smaller increases in the amounts of variable resources are needed to produce successive units of output. Hence the variable cost of successive units of output decreases. But when, as diminishing returns are encountered, marginal product begins to decline, larger and larger additional amounts of variable resources are

ORIGIN OF THE IDEA

7.2

Production relationship

TABLE 7.2 Total-, Average-, and Marginal-Cost Schedules for an Individual Firm in the Short Run

Total-Cost Data				Average-Cost Data			Marginal Cost
(1) Total Product (Q)	(2) Total Fixed Cost (TFC)	(3) Total Variable Cost (TVC)	(4) Total Cost (TC) TC = TFC + TVC	(5) Average Fixed Cost (AFC) $AFC = \frac{TFC}{Q}$	(6) Average Variable Cost (AVC) $AVC = \frac{TVC}{Q}$	(7) Average Total Cost (ATC) $ATC = \frac{TC}{Q}$	(8) Marginal Cost (MC) $MC = \frac{\text{change in TC}}{\text{change in Q}}$
0	\$100	\$ 0	\$ 100				\$ 90
1	100	90	190	\$100.00	\$90.00	\$190.00	80
2	100	170	270	50.00	85.00	135.00	70
3	100	240	340	33.33	80.00	113.33	60
4	100	300	400	25.00	75.00	100.00	70
5	100	370	470	20.00	74.00	94.00	80
6	100	450	550	16.67	75.00	91.67	90
7	100	540	640	14.29	77.14	91.43	110
8	100	650	750	12.50	81.25	93.75	130
9	100	780	880	11.11	86.67	97.78	150
10	100	930	1030	10.00	93.00	103.00	

needed to produce successive units of output. Total variable cost therefore increases by increasing amounts.

Total Cost Total cost is the sum of fixed cost and variable cost at each level of output:

$$TC = TFC + TVC$$

TC is shown in column 4 of Table 7.2. At zero units of output, total cost is equal to the firm's fixed cost. Then for each unit of the 10 units of production, total cost increases by the same amount as variable cost.

Figure 7.3 shows graphically the fixed-, variable-, and total-cost data given in Table 7.2. Observe that total variable cost, TVC, is measured vertically from the horizontal axis at each level of output. The amount of fixed cost, shown as TFC, is added vertically to the total-variable-cost curve to obtain the points on the total-cost curve TC.

The distinction between fixed and variable costs is significant to the business manager. Variable costs can be controlled or altered in the short run by changing production levels. Fixed costs are beyond the business manager's current control; they are incurred in the short run and must be paid regardless of output level.

Per-Unit, or Average, Costs

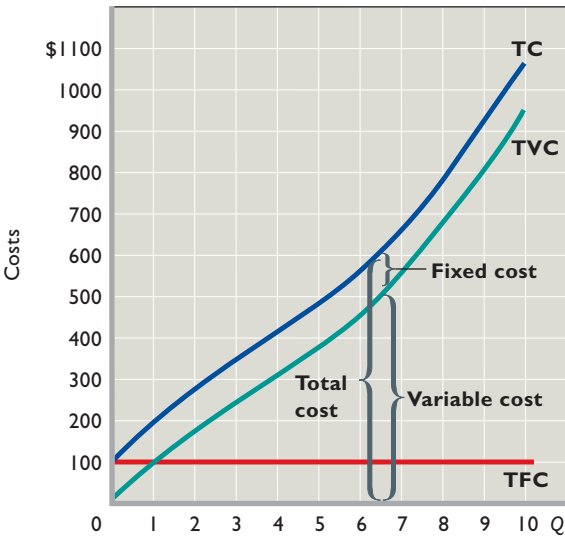
Producers are certainly interested in their total costs, but they are equally concerned with per-unit, or average, costs. In particular, average-cost data are more meaningful for making comparisons with product price, which is always stated on a per-unit basis. Average fixed cost, average variable

cost, and average total cost are shown in columns 5 to 7, Table 7.2.

AFC Average fixed cost (AFC) for any output level is found by dividing total fixed cost (TFC) by that amount of output (Q). That is,

$$AFC = \frac{TFC}{Q}$$

FIGURE 7.3 Total cost is the sum of fixed cost and variable cost. Total variable cost (TVC) changes with output. Total fixed cost (TFC) is independent of the level of output. The total cost (TC) at any output is the vertical sum of the fixed cost and variable cost at that output.



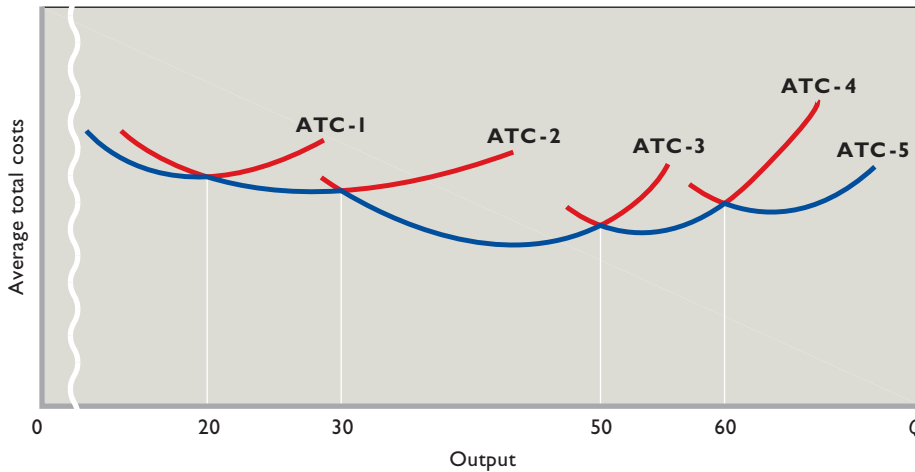


FIGURE 7.7 The long-run average-total-cost curve: five possible plant sizes. The long-run average-total-cost curve is made up of segments of the short-run cost curves (ATC-1, ATC-2, etc.) of the various-size plants from which the firm might choose. Each point on the bumpy planning curve shows the lowest unit cost attainable for any output when the firm has had time to make all desired changes in its plant size.

The Long-Run Cost Curve

The vertical lines perpendicular to the output axis in Figure 7.7 indicate the outputs at which the firm should change plant size to realize the lowest attainable average total costs of production. These are the outputs at which the per-unit costs for a larger plant drop below those for the current, smaller plant. For all outputs up to 20 units, the lowest average total costs are attainable with plant size 1. However, if the firm's volume of sales expands beyond 20 units but less than 30, it can achieve lower per-unit costs by constructing a larger plant, size 2. Although total cost will be higher at the expanded levels of production, the cost per unit of output will be less. For any output between 30 and 50 units, plant size 3 will yield the lowest average total costs. From 50 to 60 units of output, the firm must build the size-4 plant to achieve the lowest unit costs. Lowest average total costs for any output over 60 units require construction of the still larger plant, size 5.

Tracing these adjustments, we find that the long-run ATC curve for the enterprise is made up of segments of the short-run ATC curves for the various plant sizes that can be constructed. The long-run ATC curve shows the lowest average total cost at which *any output level* can be produced after the firm has had time to make all appropriate adjustments in its plant size. In Figure 7.7 the blue, bumpy curve is the firm's long-run ATC curve or, as it is often called, the firm's *planning curve*.

In most lines of production the choice of plant size is much wider than in our illustration. In many industries the number of possible plant sizes is virtually unlimited, and in time quite small changes in the volume of output will lead to changes in plant size. Graphically, this implies

an unlimited number of short-run ATC curves, one for each output level, as suggested by **Figure 7.8 (Key Graph)**. Then, rather than being made up of segments of short-run ATC curves as in Figure 7.7, the long-run ATC curve is made up of all the points of tangency of the unlimited number of short-run ATC curves from which the long-run ATC curve is derived. Therefore, the planning curve is smooth rather than bumpy. Each point on it tells us the minimum ATC of producing the corresponding level of output.

Economies and Diseconomies of Scale

We have assumed that, for a time, larger and larger plant sizes will lead to lower unit costs but that, beyond some point, successively larger plants will mean higher average total costs. That is, we have assumed the long-run ATC curve is U-shaped. But why should this be? It turns out that the U shape is caused by economies and diseconomies of large-scale production, as we explain in a moment. But before we do, please understand that the U shape of the long-run average-total-cost curve *cannot* be the result of rising resource prices or the law of diminishing returns. First, our discussion assumes that resource prices are constant. Second, the law of diminishing returns does not apply to production in the long run. This is true because the law of diminishing returns only deals with situations in which a productive resource or input is held constant. Under our definition of "long run," all resources and inputs are variable.

Economies of Scale **Economies of scale**, or economies of mass production, explain the downsloping part of

key graph

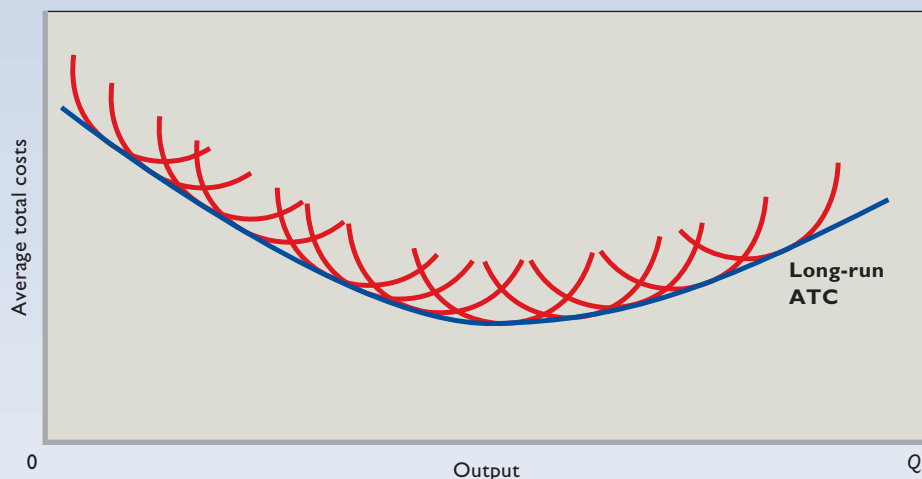


FIGURE 7.8 The long-run average-total-cost curve: unlimited number of plant sizes. If the number of possible plant sizes is very large, the long-run average-total-cost curve approximates a smooth curve. Economies of scale, followed by diseconomies of scale, cause the curve to be U-shaped.

QUICK QUIZ FOR FIGURE 7.8

- The unlabeled red curves in this figure illustrate the:
 - long-run average-total-cost curves of various firms constituting the industry.
 - short-run average-total-cost curves of various firms constituting the industry.
 - short-run average-total-cost curves of various plant sizes available to a particular firm.
 - short-run marginal-cost curves of various plant sizes available to a particular firm.
- The unlabeled red curves in this figure derive their shapes from:
 - decreasing, then increasing, short-run returns.
 - increasing, then decreasing, short-run returns.
 - economies, then diseconomies, of scale.
 - diseconomies, then economies, of scale.
- The long-run ATC curve in this figure derives its shape from:
 - decreasing, then increasing, short-run returns.
 - increasing, then decreasing, short-run returns.
 - economies, then diseconomies, of scale.
 - diseconomies, then economies, of scale.
- The long-run ATC curve is often called the firm's:
 - planning curve.
 - capital-expansion path.
 - total-product curve.
 - production possibilities curve.

Answers: 1. c; 2. b; 3. c; 4. a

the long-run ATC curve, as indicated in Figure 7.9, graphs (a), (b), and (c). As plant size increases, a number of factors will for a time lead to lower average costs of production.

Labor Specialization Increased specialization in the use of labor becomes more achievable as a plant increases in size. Hiring more workers means jobs can be divided and subdivided. Each worker may now have just one task to perform instead of five or six. Workers can work full-time on the tasks for which they have special skills. By contrast, skilled machinists in a small plant may spend half their time performing unskilled tasks, leading to higher production costs.

Further, by working at fewer tasks, workers become even more proficient at those tasks. The jack-of-all-trades

doing five or six jobs is not likely to be efficient in any of them. Concentrating on one task, the same worker may become highly efficient.

Finally, greater labor specialization eliminates the loss of time that occurs whenever a worker shifts from one task to another.

Managerial Specialization Large-scale production also means better use of, and greater specialization in, management. A supervisor who can handle 20 workers is underused in a small plant that employs only 10 people. The production staff could be doubled with no increase in supervisory costs.

Small firms cannot use management specialists to best advantage. For example, a marketing specialist working in

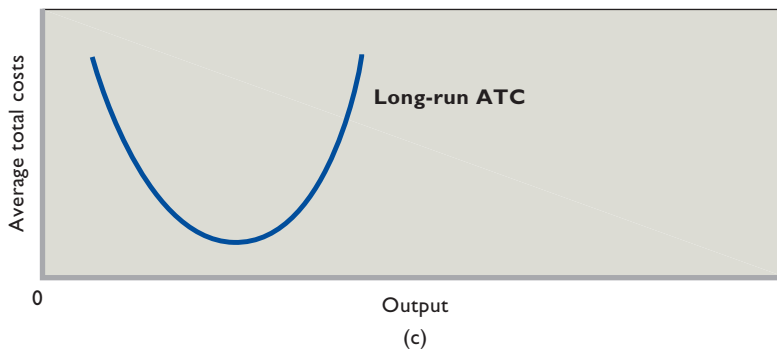
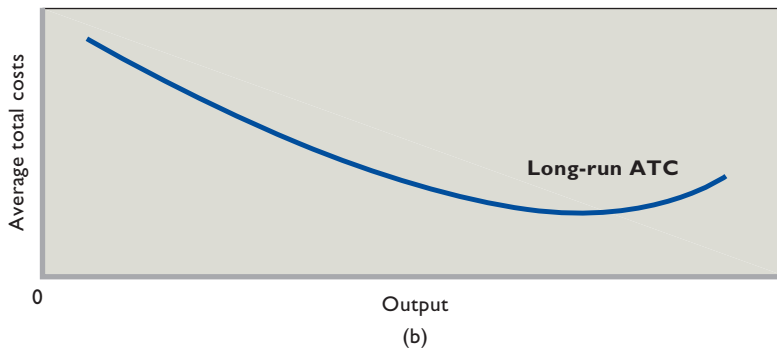
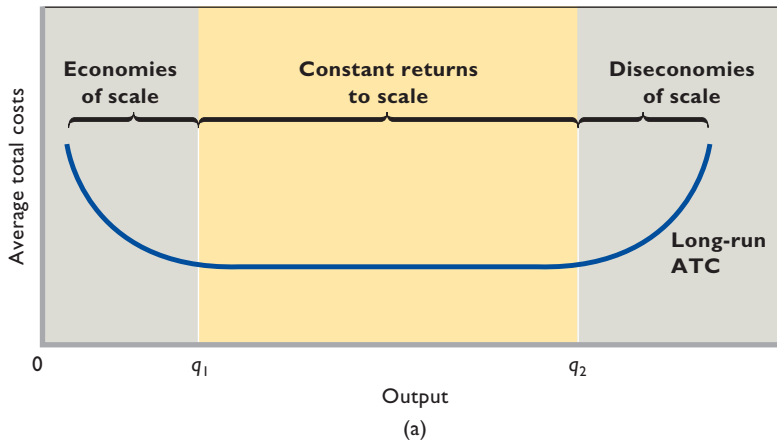


FIGURE 7.9 Various possible long-run average-total-cost curves. In (a), economies of scale are rather rapidly obtained as plant size rises, and diseconomies of scale are not encountered until a considerably large scale of output has been achieved. Thus, long-run average total cost is constant over a wide range of output. In (b), economies of scale are extensive, and diseconomies of scale occur only at very large outputs. Average total cost therefore declines over a broad range of output. In (c), economies of scale are exhausted quickly, followed immediately by diseconomies of scale. Minimum ATC thus occurs at a relatively low output.

a small plant may have to spend some of her time on functions outside of her area of expertise—for example, accounting, personnel, and finance. A larger scale of operations would allow her to supervise marketing full-time, while other specialists perform other managerial functions. Greater productivity and efficiency, along with lower unit costs, would be the net result.

Efficient Capital Small firms often cannot afford the most efficient equipment. In many lines of production such machinery is available only in very large and extremely expensive units. Furthermore, effective use of

the equipment demands a high volume of production, and that again requires large-scale producers.

In the automobile industry the most efficient fabrication method employs robotics and elaborate assembly-line equipment. Effective use of this equipment demands an annual output of several hundred thousand automobiles. Only very large-scale producers can afford to purchase and use this equipment efficiently. The small-scale producer is faced with a dilemma. To fabricate automobiles using other equipment is inefficient and therefore more costly per unit. But so, too, is buying and underutilizing the equipment used by the large manufacturers. Because it

cannot spread the high equipment cost over very many units of output, the small-scale producer will be stuck with high costs per unit of output.

Other Factors Many products entail design and development costs, as well as other “start-up” costs, which must be incurred regardless of projected sales. These costs decline per unit as output is increased. Similarly, advertising costs decline per auto, per computer, per stereo system, and per box of detergent as more units are produced and sold. Also, the firm’s production and marketing expertise usually rises as it produces and sells more output. This *learning by doing* is a further source of economies of scale.

All these factors contribute to lower average total costs for the firm that is able to expand its scale of operations. Where economies of scale are possible, an increase in all resources of, say, 10 percent will cause a more-than-proportionate increase in output of, say, 20 percent. The result will be a decline in ATC.

In many U.S. manufacturing industries, economies of scale have been of great significance. Firms that have expanded their scale of operations to obtain economies of mass production have survived and flourished. Those unable to expand have become relatively high-cost producers, doomed to struggle to survive.

Diseconomies of Scale In time the expansion of a firm may lead to diseconomies and therefore higher average total costs.

The main factor causing **diseconomies of scale** is the difficulty of efficiently controlling and coordinating a firm’s operations as it becomes a large-scale producer. In a small plant a single key executive may make all the basic decisions for the plant’s operation. Because of the firm’s small size, the executive is close to the production line, understands the firm’s operations, and can make efficient decisions because the small plant size requires only a relatively small amount of information to be examined and understood in optimizing production.

This neat picture changes as a firm grows. One person cannot assemble, digest, and understand all the information essential to decision making on a large scale. Authority must be delegated to many vice presidents, second vice presidents, and so forth. This expansion of the management hierarchy leads to problems of communication and cooperation, bureaucratic red tape, and the possibility that decisions will not be coordinated. At the same time, each new manager must be paid a salary. Thus, declining efficiency in making and executing decisions goes hand-in-hand with rising average total costs as bureaucracy expands beyond a certain point.

Also, in massive production facilities workers may feel alienated from their employers and care little about working efficiently. Opportunities to shirk, by avoiding work in favor of on-the-job leisure, may be greater in large plants than in small ones. Countering worker alienation and shirking may require additional worker supervision, which increases costs.

Where diseconomies of scale are operative, an increase in all inputs of, say, 10 percent will cause a less-than-proportionate increase in output of, say, 5 percent. As a consequence, ATC will increase. The rising portion of the long-run cost curves in Figure 7.9 illustrates diseconomies of scale.

Constant Returns to Scale In some industries a rather wide range of output may exist between the output at which economies of scale end and the output at which diseconomies of scale begin. That is, there may be a range of **constant returns to scale** over which long-run average cost does not change. The q_1q_2 output range of Figure 7.9a is an example. Here a given percentage increase in all inputs of, say, 10 percent will cause a proportionate 10 percent increase in output. Thus, in this range ATC is constant.

Minimum Efficient Scale and Industry Structure

Economies and diseconomies of scale are an important determinant of an industry’s structure. Here we introduce the concept of **minimum efficient scale (MES)**, which is the lowest level of output at which a firm can minimize long-run average costs. In Figure 7.9a that level occurs at q_1 units of output. Because of the extended range of constant returns to scale, firms producing substantially greater outputs could also realize the minimum attainable long-run average costs. Specifically, firms within the q_1 to q_2 range would be equally efficient. So we would not be surprised to find an industry with such cost conditions to be populated by firms of quite different sizes. The apparel, food processing, furniture, wood products, snowboard, banking, and small-appliance industries are examples. With an extended range of constant returns to scale, relatively large and relatively small firms can coexist in an industry and be equally successful.

Compare this with Figure 7.9b, where economies of scale continue over a wide range of outputs and diseconomies of scale appear only at very high levels of output. This pattern of declining long-run average total cost occurs in the automobile, aluminum, steel, and other heavy industries. The same pattern holds in several of the new industries related to information technology, for example, computer microchips, operating system software, and Internet service provision.

total revenue minus total cost, would increase. Hence, if marginal revenue is greater than marginal cost, as it is at Q_1 , the firm can increase profit by increasing production.

A similar argument applies when output is at Q_2 . In this case, marginal cost is greater than marginal revenue. If the firm reduced production by 1 unit, the costs saved (MC_2) would exceed the revenue lost (MR_2). Therefore, if marginal revenue is less than marginal cost, as it is at Q_2 , the firm can increase profit by reducing production. Where do these marginal adjustments to the level of production end? Regardless of whether the firm begins with production at a low level (such as Q_1) or at a high level (such as Q_2), the firm will eventually adjust production until the quantity produced reaches Q_{MAX} . This analysis confirms the general rule for profit maximization: at the profit-maximizing level of output, marginal revenue and marginal cost are exactly equal.

If we assume that the firm aims to maximize profit then it aims at an output level where $MC = MR$. In Figure 8.2, that quantity of output is Q_{MAX} .

FYI



The Mathematics of Profit Maximization

Assume that the equations for an inverse demand curve and total cost facing a firm are given as:

$$P = 230 - Q$$

and:

$$TC = 40 + Q^2$$

We know that we have to differentiate both equations to find the marginal revenue and marginal cost and that the power function rule is:

$$\frac{dy}{dx} = nx^{n-1}$$

$$\text{So, } MC = \frac{d(TC)}{dq} = -2q$$

$$TR = P \times Q$$

$$TR = (230 - Q)Q$$

$$TR = 230Q - Q^2$$

$$MR = \frac{d(TR)}{dq} = 230 - 2q$$

Setting the profit maximizing output at $MC = MR$ gives us:

$$-Q = 230 - 2Q$$

$$Q = 230$$

Profit would be maximized at an output level of 230. It should be noted that that the outcome figures are not always neat round numbers. The point, as has been made earlier, is that this is a model to help firms make decisions and as a result this firm might aim for an output level between two values and know that it is maximizing profit.

JEOPARDY PROBLEM

A firm finds itself in a position of seeing an increase in sales growth of 15 per cent over the previous year and productivity is up by 5 per cent. The firm's directors, however, have reported losses of €25 million compared to a profit of €15 million in the previous year. How could this have happened?

BREAK-EVEN ANALYSIS

break-even the level of output/sales at which total cost equals total revenue found by dividing the fixed costs by the contribution (selling price minus variable costs per unit)

When a new business starts up or if an existing business decides to develop a new product one goal may be to achieve break-even. **Break-even** refers to the output level at which the total costs of production are equal to the total revenue generated from selling that output. Many firms will look at break-even analysis as part of their planning tools.

They can look at the variables involved and how different figures plugged into these variables can affect decision making.

Figure 8.3 shows how we can represent break-even graphically. The vertical axis shows both costs and revenues and the horizontal axis shows output and sales. We know that a firm faces fixed costs which must be paid regardless of whether it produces any output. As a result the Fixed Cost curve FC is represented as a horizontal line at a cost of C_1 . Variable costs (VC) are zero when the firm produces no output but rise as output rises. The VC curve shows these costs rising in direct proportion to output. Total cost is the sum of variable costs and fixed cost ($TC = VC + FC$) and so the TC curve has a vertical intercept equal to the level of fixed costs (C_1) and then rises as output rises. The vertical distance between the TC and VC curve is constant because of the proportional relationship between output and VC assumed above.

The total revenue curve is dependent on the price that the firm chooses to charge for its product. The higher the price charged the steeper the TR curve. The break-even level of output in Figure 8.3 is the amount where $TC = TR$ which is depicted as Q_{BE} . At this output level the TC are C_1 and the TR is the same at R_1 shown by C_1/R_1 in Figure 8.3. At any output level below Q_{BE} the firm will find that its TC is greater than TR and so it will make a loss on that output. The loss made is represented by the area shown by the shaded triangle A .

At output levels above Q_{BE} , for example Q_1 , the firm's TR , shown by R_2 , is greater than its TC , shown by C_2 and as a result it makes profits on this output. The amount of profit made at output Q_1 is represented by the shaded triangle B .

If the firm were operating at Q_1 then it could experience a fall in sales and still be generating a profit provided sales did not fall below the break-even output Q_{BE} . The distance between the break-even output and current production where TR is greater than TC is called the **margin of safety**.

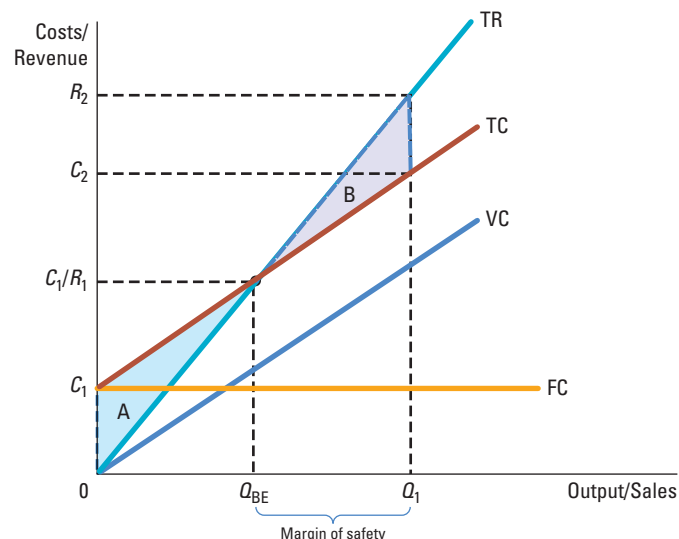
We can also look at the break-even point in a more mathematical way. A firm needs to aim at selling its output at a price greater than the variable cost of production.

margin of safety the distance between the break-even output and current production where total revenue is greater than total cost

FIGURE 8.3

Break-Even Analysis

A break-even chart shows a firm's total cost, made up of fixed and variable costs and its total revenue. The output level where $TR = TC$ is the break-even point shown by output Q_{BE} . Any output level below Q_{BE} would mean that the firm was producing at a loss because TC would be greater than TR but any output above Q_{BE} would mean the firm was operating at a profit. At output level Q_1 , the amount of profit is indicated by the shaded triangle B . The distance between Q_{BE} and Q_1 is called the margin of safety and denotes how far sales could fall before the firm starts to make losses.



contribution the difference between the selling price and the variable cost per unit

Imagine a firm producing chocolate bars. The ingredient costs are €0.30 per bar and the labour costs are estimated at €0.10 per bar. Total variable costs are €0.40 per bar. If the selling price of the chocolate bar is €0.60 then every bar sold covers the variable costs of production and leaves €0.20 which can be used to help pay off the fixed costs which also have to be paid. This sum is called the **contribution**. The contribution is the difference between the selling price and the variable cost per unit. Knowing the contribution per unit the break-even output can be given by:

$$\text{Break-even} = \frac{\text{Fixed costs}}{\text{Contribution per unit}}$$

The break-even point can be affected by the costs of production, therefore, any change in fixed costs, for example, will shift the FC line up or down and thus the TC curve. Equally, changes in raw materials costs can affect the variable costs and as a result the TC curve.

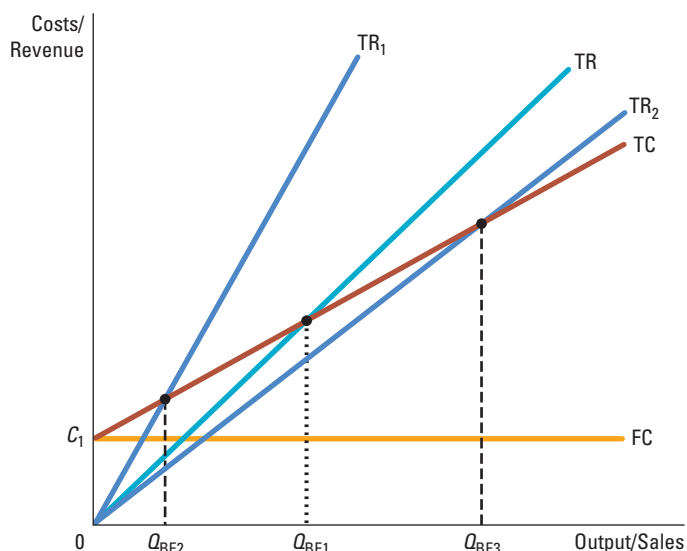
It may be that the firm has little control over changes to its costs but it can use break-even analysis to assess the impact of the changes and make decisions as a result. What many firms do have control over is the price they charge. We know that $TR = P \times Q$. If a firm changes its price the shape of the TR curve will alter. If a firm chooses to increase its price then the TR curve will pivot and become steeper as shown in Figure 8.4. With the TR curve now indicated as TR_1 , the firm will not have to sell as many products to cover its costs and so the break-even output will be lower at Q_{BE2} . A firm might choose to reduce its price in which case the TR curve will become flatter. In this case TR curve will now be represented as TR_2 and the firm will now have to sell more products in order to cover its costs and so the break-even output would rise to Q_{BE3} .

Limitations of break-even analysis As mentioned above, break-even is best seen as a planning tool and the outline given here is a simplistic one. There are a number of limitations with break-even analysis. Firstly there is an assumption that all output is sold. This is not the case for many firms where production takes place but products are not immediately sold and as such firms have stock (inventory).

FIGURE 8.4

The Effect on Break-Even Output of Changes in Price

Total revenue is found by multiplying price times the quantity sold. If the price is increased the TR curve will become steeper as shown by the curve TR_1 . As a result, the firm would now only need to produce output Q_{BE2} in order to break-even. If price was reduced the TR curve becomes flatter indicated by curve TR_2 and the firm would have to sell a higher output level, Q_{BE3} in order to break even.



Secondly, the shape of the TR and TC curves is unlikely to be smooth as increases in output can be associated with economies of scale and not every additional product a firm sells can be sold at the same price. Equally, given the dynamic nature of markets, any estimation of costs or revenues are not likely to be relevant for very long.

Thirdly, we know that if a firm changes price, demand will be affected. Break-even analysis says nothing about what the effect of a price change on actual sales will be, nor how long it will take a firm to sell the output level to break even. The effect on sales of a change in price will be dependent on the price elasticity of demand.

CASE STUDY

On 2 February 2011, Rupert Murdoch, the chairman and CEO of News Corporation, launched a new business product *The Daily*. *The Daily* is a newspaper designed exclusively for Apple's iPad. It has a mission to: 'provide the best news experience by combining world-class storytelling with the unique interactive capabilities of the iPad'. Customers can download the app and get two weeks free access before having to subscribe for US\$0.99 a week or \$39.99 a year.



NEWS INTERNATIONAL

Clearly there will have been a cost to setting the product up in the first place which has been estimated at \$30 million. The running costs have been reported to be \$500 000 a week or \$26 million a year. Given these figures we can get some idea of what the sales of subscriptions need to be for the product to break even. *The Daily* would need some 651 163 subscribers a year just to cover the running costs of the product, let alone the set up costs (which we can loosely refer to as the fixed costs) of \$30 million.

What are the chances that it will be able to recruit this number of subscribers? Well, the first thing to note is that the price being charged for subscriptions is relatively low, hence the high numbers needed to break even. To get some idea we might look at data on other subscriptions to online news services. French newspaper *Le Monde* is reported to have around 39 000 web-only subscribers, *The Times*, another of Murdoch's papers, reported subscribers to its digital version as rising to 110 000 in October 2011. A Long Island, New York daily newspaper, *Newsday*, was said to have received just 35 subscriptions after three months of launch. *The Wall Street Journal* is estimated to have around 400 000 online paid subscriptions and *Wired* magazine began its online life with 100 000 subscribers but this fell to 23 000 half a year later. Given these figures, some from established brands, the numbers for *The Daily* do not look promising. Even if subscribers can be encouraged to download the app and pay, Apple will take a cut from the subscription amount so the actual revenues would be lower than that estimated here.

However, a newspaper's revenue does not just come from subscriptions; advertising also brings in revenue. The next question is: is the amount of advertising spend sufficient to make up any gap between costs and subscriber revenue to help *The Daily* break even. Here again, things do not look that promising. Digital intelligence firm, eMarketer, estimates that the total spent on advertising in 2011 was around \$1 billion. *The Daily*, therefore, needs to get a share of that spending and given this spending covers all mobile advertising business, it has been

estimated that *The Daily* would have to capture around 2.5 per cent of this annual spending.

One other thing to consider is that *The Daily* is only available on iPad and so the number of iPad users represents a constraint on the potential market size. Estimates for total iPad sales up to the end of 2012 since its launch are around 62 million. *The Daily*, therefore, needs to get around 1.05 per cent of iPad users to subscribe to its product and not only to subscribe but to maintain the subscription over a period of time. Is that possible? Of course it is, but the challenges facing *The Daily* to even reach break-even level seem daunting, never mind being able to become profitable.

Revenue Maximization

In recent years there has been some criticism of executives in businesses because their reward packages seem to have grown disproportionately to the performance of the business. One of the problems with this is understanding the definition of ‘performance’. Executives might be persuaded to focus on goals which may give an impression of the business doing well such as targeting sales. Other things being equal, a rise in sales is a ‘good’ thing although of course this might not say anything about the cost involved in achieving any such increase in sales.

We know that total revenue is price multiplied by quantity sold. The goal of revenue maximization may also be referred to as sales revenue maximization. The principle is straight forward. There are three ways to increase sales: do something with price, something to influence how much is sold, or a combination of the two. We have seen how both reducing and increasing price can increase total revenue depending on the price elasticity of demand for the product. There are a number of pricing strategies that can be employed and we will look at these in more detail in Chapter 11. Equally, firms will have a number of strategies and tactics to try and increase sales which will include other elements of the marketing mix apart from price

- the product itself, how consumers are able to access the product (place), how consumers are made aware of the product (promotion), the processes involved which includes things like customer service commitments and the information customers can access, a focus on the people who are involved in the business (the employees and physical evidence), how the firm is viewed in the eyes of its customers, does the view conform with prior assumptions? For example, going into a car showroom and seeing new cars which are dirty and poorly presented in a shabby environment would not be an image most car dealers would want to present.

Graphical Representation of Sales Revenue Maximization We know from our look at the theory of demand that in order to increase demand, a firm needs to reduce its price. We also know that along a straight line demand curve the elasticity ranges from elastic through to inelastic.

The price elasticity of demand varies at every point along a straight line demand curve. The higher up the demand curve (towards the vertical axis) where price is relatively high but quantity demanded relatively low, *ped* will be elastic, whereas at the lower end of the demand curve (towards the horizontal axis) the price is relatively low but quantity demanded relatively high and as a result *ped* will be inelastic. There will also be a point midway between these two ranges where the *ped* is of unit elasticity.

This knowledge is important for when you look at market structure and imperfect competition such as oligopoly and monopoly where the firm does not face a horizontal

Forms of Business organization

One of the first decisions that you'll have to make as a business owner is how your business should be structured. You need to know the advantages and disadvantages of each of the different forms of business organization to make sure you're making the right decision for your new business.

All businesses must adopt some legal configuration that defines the rights and liabilities of participants in the business's ownership, control, personal liability, lifespan and financial structure. The form of business determines which income tax return form to file and the company's and owners' legal liabilities.

When you're forming your new business, you will want to take into account the following:

- Your (practical) vision regarding the size and nature of your business.
- The level of control you wish to have.
- The level of "structure" you are willing to deal with.
- The business's vulnerability to lawsuits.
- Tax implications of the different organizational structures.
- Expected profit (or loss) of the business.

Sole proprietorship

The vast majority of small businesses start out as sole proprietorships. These businesses are owned by one person, usually, the individual who has day-to-day responsibility for running the business. Sole proprietors can be independent contractors, freelancers or home-based businesses.

Features of a Sole Proprietorship

➤ Formation of Business and Closure

Such business organisations are formed by a sole proprietor at his discretion. Similarly, the owner has the right and the legal authority to determine when he will close the business. If, for example, the owner feels his fully-owned entity is facing a fiscal crunch, he may decide to shut shop. Since there are no hard and fast rules that govern sole proprietorship in India, there is no legal obligation for him to change his mind.

➤ **Liability**

In any sole proprietorship, the owner has “unlimited liability”. If he avails a business loan from a financial establishment, he will be liable to repay it. If he is unable to settle up on time, his private estates may be put on sale by the lender. The sole owner is liable to repay anyone who has helped him financially for his business purposes. This concept of unlimited liability is one of the significant disadvantages of a sole proprietorship. However, it remains a crucial characteristic of such ventures.

➤ **Risk Bearing and Profit-sharing**

The sole owner is the only individual who bears any business risks associated with his venture. No one else will bear any portion of that risk.

Conversely, the owner is also eligible to enjoy all the profits which his business delivers. If there are any losses, he will have to deal with them on his own.

➤ **Sole Control**

Because the owner controls a business, he alone controls all the rights and responsibilities. It is his prerogative to change the course of his business or to exit a particular sector altogether. It is one of the many characteristics of sole proprietorship and wholly-owned businesses. The owner’s judgements even if error-ridden, are final and no one else can interfere in them if not otherwise solicited.

If a business’ operations and methods of functioning are to be modified, only the owner can do so. No one else is authorised.

➤ **No separate Entity**

Unlike other types of business ventures, a sole proprietor and his business are the same entity. According to the existing laws, since the sole owner operates a business, he is the entity. In case the owner expires, the business wraps up automatically. However, if the business has been passed down to successors, the clauses are different. The continuity of business clause then kicks

in. The same concept also takes precedence when the proprietor is taken ill, imprisoned or otherwise aggravated.

Advantages of a Sole Proprietorship

There are several advantages. The most basic ones are enumerated here

➤ **Prompt decision-Making Process:**

Since these are necessarily a one-man show, there are minimal hassles when a particular particular decision has to be reached at. The proprietor does not need anyone's permission or consent, or dissent, to carry on exactly as he wills. This lack of such agility in decision-making in bigger businesses with partners is a significant hurdle at times.

➤ **Full Confidentiality:**

Secrecy is paramount in modern businesses given the ferocious competition. There is no leak of inside information in this business model, making it arguably the safest type of business.

➤ **Direct Incentive:**

The owner alone gets to enjoy all profits- the fruits of labour. He will also retain any benefits apart from revenues that a business may generate.

➤ **Ease of formation and Closure:**

Should there be losses in operations and the business proves untenable, a sole proprietor is free to close the rm down. He can always use the leftover capital or profits hitherto generated for other purposes.

Disadvantages of a Sole Proprietorship

The following are some major limitations of self-proprietorship.

➤ **Resources Limited:**

A single owner does not have access to a large corpus initially. He may have to take a bank loan or ask relatives or friends for loans. If his business fails to take off, he will have difficulty repaying.

➤ **Lack of Continuity (Limited life of a business concern):**

Yet another disadvantage of sole proprietorships is a lack of continuity of a business' life-cycle. If the owner dies, his business ceases to exist. While it is true that

he can decree that his successor will run the business on his demise or ailment, there is no guarantee that the business will be run as efficiently.

➤ **Unlimited Liability:**

Speaking on merits and demerits of a sole proprietorship, this is a major demerit. A sole owner has to carry the burden of unlimited liability. For example, in direct relation to point A of this section, creditors may also seek compensation from the owner's private estate if he is unable to repay loans. It is a significant deal-buster for many would businesspersons are aware of this risk.

➤ **Limited Expertise:**

Running a business requires a lot of varying expertise in different elds. There has to be clarity on operations, client relationships, margin setting, accounts, and so on. Technological know-how is also crucial. For a single owner, these elements present unique challenges.

Partnership

In India, we have a definite law that covers all aspects and functioning of a partnership, The Indian Partnership Act 1932. The act also defines a partnership as “the relation between two or more persons who have agreed to share the profits from a business carried on by either all of them or any of them on behalf of/acting for all”

So in such a case two or more persons come together as a unit to achieve some common objective. And the profits earned in pursuit of this objective will be shared amongst themselves. The entity is collectively called a “Partnership Firm” and all the individual members are the “Partners”.

Features of a Partnership

➤ **Formation/Partnership Agreement**

A partnership firm is not a separate legal entity. But according to the act, a firm must be formed via a legal agreement between all the partners. So a contract must be entered into to form a partnership firm. Its business activity must be lawful, and the motive should be one of profit.

➤ **Unlimited Liability**

In a unique feature, all partners have unlimited liability in the business. The partners are all individually and jointly liable for the firm and the payment of all debts. This means that even personal assets of a partner can be liquidated to meet the debts of the firm.

If the money is recovered from a single partner, he can, in turn, sue the other partners for their share of the debt as per the contract of the partnership.

➤ **Risk bearing/Sharing of Profit and Loss:**

Another significant component of the partnership is, the accord between partners has to share gains and losses of a trading concern. However, the definition held in the Partnership Act elucidates – partnership as an association between people who have consented to share the gains of a business, the sharing of loss is implicit. Hence, sharing of gains and losses is vital.

➤ **Decision making and control:**

In sole proprietorship one person was taking all the decision but here in the partnership firm all the decision related to business are made with the consultation from each partner in democratic way (mutually agree). All the efforts and research activities are done by all the partners.

➤ **Continuity**

A partnership cannot carry out in perpetuity. The death or retirement or bankruptcy or insolvency or insanity of a partner will dissolve the firm. The remaining partners may continue the partnership if they so choose, but a new contract must be drawn up. Also, the partnership of a father cannot be inherited by his son. If all the other partners agree, he can be added on as a new partner.

➤ **Number of Members**

As we know that there should be a minimum of two members. However, the maximum number will vary according to a few conditions. The Partnership Act itself is silent on this issue, but the Companies Act, 2013 provides clarity.

For a banking business, the number of partners must not exceed ten. For a business of any other nature, the maximum number is twenty. If the number of partners increases it will become an illegal entity or association.

➤ **Mutual Agency**

In this type of organisation, the business must be carried out by all the partners together. Or alternatively, it can be carried out by any of the partners (one or several) acting for all of them or on behalf of all of them. So this means every partner is an agent as well as the principal of the partnership. He represents the other partners in some cases so he is their agent.

Advantages of a business partnership

The business partnership offers a lot of advantages to those who choose to use it.

➤ **Ease of doing business**

The partners can agree to create the partnership verbally or in writing. There's no need to register with Companies House and registering the business partnership for taxation with HMRC is quite simple. The partners will also individually need to register for self-assessment, which they can do online.

Although it will take longer and incur additional cost, it's usually sensible to put in place a partnership agreement. This documents how the partnership will work, the rights and responsibilities of partners and what would happen in various possible situations, including if the partners fundamentally disagree or someone wants to leave.

➤ **Better decision-making**

Compared with operating on your own, in a partnership the business benefits from the unique perspective brought by each partner. In business, very often two heads really are better than one, with the combined conclusion of debating a situation far better than what each partner could have achieved individually.

➤ **More partners, more capital**

The more partners there are, the more money there may be available from their combined resources to invest into the business, which can help to fuel growth. Together, their borrowing capacity is also likely to be greater.

➤ **Privacy**

Compared to a limited company, the affairs of a partnership business can be kept confidential by the partners. By contrast, in a limited company certain documents are available for public inspection at Companies House and a company's shareholders can choose to inspect various registers and other documents the company is required to keep.

➤ **Sharing Risk –**

All loss incurred by the firm is equally distributed amongst each partner.

Disadvantages of a business partnership

➤ **Unlimited liability**

Again because the business does not have a separate legal personality, the partners are personally liable for debts and losses incurred. So if the business runs into trouble your personal assets may be at risk of being seized by creditors, which would generally not be the case if the business was a limited company.

The partners are jointly and severally liable. As one partner can bind the partnership, you can effectively find yourself paying for the actions of the other partners. If your partners are unable to settle debts, you'll be responsible for doing so. In an extreme example where you only own 10% of the partnership, if your partners have no assets you might end up having to settle 100% of the debts of the partnership and need to sell your possessions in order to do so.

➤ **Limited Capital:**

As the partnership business is established and managed by a few partners, it has less chance of accumulating a large amount of capital. In comparison to the joint stock company, the partnership has less capital.

➤ **Problem of dispute (Conflict):**

Even though the partnership business is formed by the agreement of partners, the partners may not agree all the time. The partners may disagree regarding the profit and use of authority. This dispute between partners may create a problem in the existence of a business.

➤ **Uncertain existence (Lack of continuity):**

A partnership business may face dissolution in case of death, insolvency or mental or physical illness

of active partners. The partnership of business could be shut down by the partner after making the agreement between them. Therefore, it has an uncertain existence.

➤ **Lack of public faith:**

Since the partnership business has limited sizes, non-existence in the eye of the law, it has less public faith. The public doesn't believe in partnership business as much as the joint stock company because it has difficulty in both expansion and growth.

Cooperative Society

'Cooperative Society' is intended to help each other. Thus, cooperatives are those institutions which are formed for mutual assistance of its members. The cooperative movement is mainly the movement of the poor. Cooperative organisations are formed to serve their members. The weaker sections of the society can create such organisations to protect themselves from exploitation through modern means of production and distributions. A cooperative society is a voluntary organisation of individuals who are mostly workers and small producers. They are organised under joint management on democratic methods to improve their domestic and business conditions and capital collection.

Features of a Cooperative Society

➤ **Voluntary Membership:**

As it is a voluntary association, the membership is also voluntary. A person is free to join a cooperative society, and can also leave anytime as per his desire. Irrespective of their religion, gender & caste, membership is open to all.

➤ **Legal status:**

It is compulsory for the co-operative society to get registration. The co-operative society is a separate legal identity to the society.

It does not get affected by the entry or exit of its members.

➤ **Limited liability**

There is limited liability of the members of co-operative society. Liability is limited to the extent of the amount contributed by members as capital.

➤ **Control:**

An elected managing committee has the powers to take decisions. Members have the right to vote, by which they elect the members who will constitute the managing committee.

➤ **Service motive**

The cooperative society works on the principle of mutual help & welfare. Hence, the principal of service dominates its working. If any surplus is generated, it is distributed amongst the members as a dividend in conformity with the bye-laws of the society.

Cooperative Society- Advantages:

➤ **Democratic Management-**

A cooperative society is managed in a democratic manner. It is based on the principle of 'one man one vote'. All members have equal rights and can have a voice in its management.

➤ **Limited Liability-**

The liability of the members of a co-operative society is limited to the extent of capital contributed by them. They do not have to bear personal liability for the debts of the society.

➤ **Stability-**

A co-operative society has a separate legal existence. It is not affected by the death, insolvency, lunacy or permanent incapacity of any of its members. It has a fairly stable life and continues to exist for a long period.

➤ **Economical Operations-**

The operation of a cooperative society is quite economical due to elimination of middlemen and the voluntary services provided by its members.

➤ **Government Patronage-**

Government gives all kinds of help to co-operatives, such as loans at lower rates of interest and relief in taxation.

➤ **Easy to Form-**

A cooperative society is a voluntary association and may be formed with a minimum of ten adult members. Its registration is very simple and can be done without much legal formalities.

Cooperative Society- Disadvantages:

➤ **Limited Capital-**

Cooperatives are usually at a disadvantage in raising capital because of the low rate of return on capital invested by the members.

Inefficient Management-

The management of a co-operative society is generally inefficient because the managing committee consists of part-time and inexperienced people. Qualified managers are not attracted towards a cooperative on account of its limited capacity to pay adequate remuneration.

➤ **Lack of Secrecy-**

The affairs of a co-operative society are openly discussed in the meetings of the members. Every member is free to inspect the books and records of the society. Therefore, it becomes difficult to keep the secrets of business.

➤ **Government Intervention:**

The daily operations of a cooperative society are subject to government rules and regulations. Regular book-keeping, auditing and inspection of accounts by the government officials are

mandatory aspects of the organisation. The reports have to be submitted to the registrar. All these legal formalities take a lot of time and consequently inhibit efficiency.

➤ **Conflict among Members:**

The members of cooperative society come from different walks of life. Often, their views on important issues may differ from each other leading to strong resentment and disharmony among them. Some ambitious members also want to control the functions of the organisation. Blinded by the society, thus, crippling its efficiency.

Company

A company is a natural legal entity formed by the association and group of people to work together towards achieving a common objective.

According to the definition of a company by the Indian Act 2013;

“A registered association which is an artificial legal person, having an independent legal, entity with perpetual succession, a common seal for its signatures, a common capital comprised of transferable shares and carrying limited liability.”

Features of Company

➤ **Separate Legal Entity**

When we say legal entity, what it means that it's completely independent of its people who control its operations. In other words, the company won't be responsible if its members don't pay their debt. The same goes for the company as well; that the members don't have to pay for the debt of the company, if it's unable to pay to its creditors.

➤ **Common Seal**

As we know that a company acts as an artificial legal individual, therefore, it has a stamp or seal with the name and address engraved on it. This stamp would be like the signature of the company. The stamp and company's seal is used for the verification and authorization of various documents.

➤ **Separation of Ownership and Management**

A company may have multiple shareholders and, in several cases, the number of shareholders may be large enough to be responsible for managerial affairs. So, shareholders usually hire directors who take the responsibility of running all the daily operations of a company.

➤ **Artificial person**

The law treats the company as a legal artificial person because it has its name and bank accounts. It can also own property under its name, file a lawsuit against other companies or personals, or be partnered up with other companies. It performs all of the activities that a person can legally do; a company can do it well. Therefore, it acts as an artificial individual.

➤ **Perpetual Existence**

Unlike proprietorship, partnership or any other type of business, a company doesn't depend upon its owners, board of directors, shareholders, or employees. Many people come and go in the company, but it stays. Therefore, the existence of the company is much stable than

➤ **Transferability of Shares:**

Shareholders of a public limited company can transfer their shares as per the rules laid down in the articles of association. However, in case of a private limited company, there might be some restrictions on the transfer of shares.

➤ **Limited Liability**

The liability of shareholders is limited to their share price only; it is in the limited companies by share. On the other hand, in the case of limited companies by guarantee, where the share of contributors is like an asset in the company; if the company goes bankrupt, then the shareholders have to pay a small amount to cover up the loss of the company.

Company-Advantages

➤ Limited Liability:

The liability of shareholders, unless and otherwise stated, is limited to the face value of shares held by them or guarantee given by them.

➤ Perpetual Existence:

Deaths, insanity, insolvency of shareholders or directors do not affect the company's existence. A company has a separate legal entity with perpetual succession.

➤ Expansion Potential:

As there is no limit to the maximum number of shareholders in a public limited company, expansion of business is easy by issuing new shares and debentures. Companies normally use their reserves for expansion purposes.

➤ Transferability of Shares:

If the shareholders of a company are displeased with the progress of the business, they can sell their shares any time. During all this change of ownership, the business continues to operate.

The shareholder can withdraw the profitable amount also from the existing company anytime and invest it in any other company.

➤ Professional Management:

In company business, the management is in the hands of the directors who are elected by the shareholders and are well experienced persons. In order to manage the day-to-day activities, salaried professional managers are appointed. Thus, the company business offers professional management.

➤ Diffusion of Risk:

As the membership is very large, the whole business risk is divided among the several members of the company. This is an advantage particularly for small investors.

Disadvantages of Company:

➤ Complexity of a formality:

Compared to proprietorship and partnership, a company has to comply with more legal requirements. It consumes considerable time and effort.

➤ Lack of Secrecy:

As per the legal provisions, a company has to make various statements available to the Registrar of the Companies, Financial Institutions; the secrecy of business comes down. It is further reduced when the company provides its annual report to the shareholders as the competitors do also find out the details of all financial data.

➤ Delay in Decision-making:

In company form of organization no single individual can make a policy decision. All important decisions are taken either by the Board of Directors or are referred to general house. Decision-taking process is time consuming. If some business opportunity arises and a quick decision is needed, it will not be possible to arrange meetings all of a sudden. So many opportunities may be lost because of a delay in decision-making.